TERAFLASH pro

High Power Version

Time-Domain Terahertz Spectroscopy Platform

Manual

Manual: M-127 Version 01
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(February 2026 Subject to change without notice)

Dear Customer,

Welcome to the TOPTICA community!

We have designed this product to be easy to use and reliable so that you can focus on your work. If you have questions or need advice on how to integrate it into your setup, please contact us immediately so we can walk you through the process. We will provide you with quick and competent help through our service staff and product managers.

You can always contact TOPTICA service on the Internet by

- Email: service@toptica.com
- Web form: service.toptica.com

The service department at our headquarters in Munich can be reached by phone

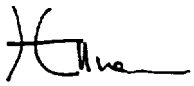
- Tel: +49 89 85837150

For up-to-date phone numbers and addresses of our regional offices please refer to our web site: www.toptica.com/contact-us/

Please have your product -ID and serial number ready when contacting us so we can quickly retrieve all relevant information.

As we are constantly improving our products, we greatly value all customer feedback. We encourage you to tell us what you like about our products as well as any suggestions for improvement.

Best regards,



Harald Ellmann
Vice President Service
TOPTICA Photonics SE

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1 Introduction

1.1 Technology

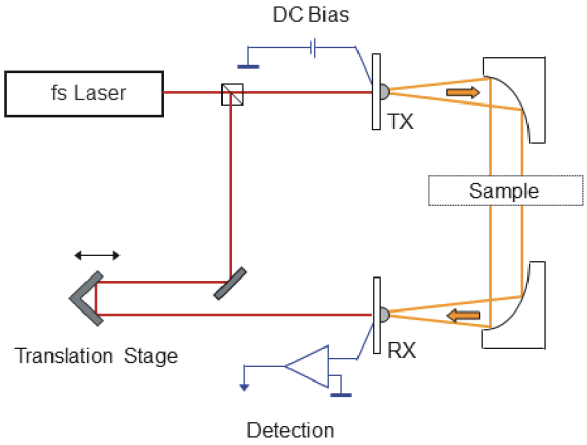


Figure 1 Pulsed terahertz generation with photoconductive antennas.
TX = terahertz emitter, RX = terahertz receiver.

Pulsed terahertz radiation is generated with femtosecond lasers. In a typical time-domain setup (see Figure 1), the laser pulse is split in two, with one part traveling to the terahertz emitter (or transmitter - TX), and the second part traveling to the terahertz receiver (RX), after interacting with a sample. The ultrashort laser pulses produce charge carriers in the emitter which are accelerated by an externally applied bias field. The resulting current transient gives rise to an electromagnetic pulse - the terahertz pulse that is radiated into free space[1].

The receiver works in a "pump and probe" fashion: the incident terahertz pulse generates a voltage across the receiver antenna, which accelerates the charge carriers produced simultaneously by the "probe" laser pulse. The resulting photocurrent is directly proportional to the terahertz electric field.

A characteristic of any THz-TDS system is that the width of the generated terahertz pulse (~ 1 ps) is much longer than the width of the probing laser pulse (< 80 fs). Thus, only a fraction of the terahertz pulse is sampled each time the laser pulse hits the receiver antenna. To retrieve the full pulse shape, a variable delay stage scans the terahertz wave packet with the much shorter "probe" pulse. A fast Fourier transform (FFT) of the terahertz amplitude then produces the terahertz spectrum.

1. D.H. Auston et al., *Picosecond photoconducting Hertzian dipoles;* Appl. Phys. Lett. **45** (1984) 284.

1.2 System Design

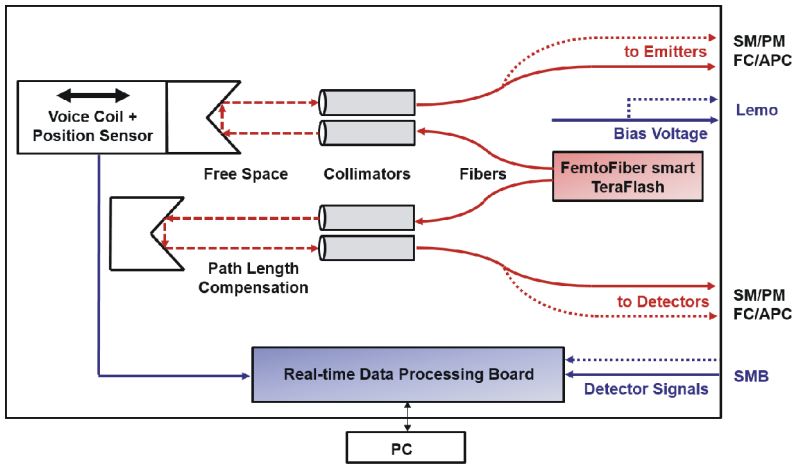


Figure 2 Schematic representation. Blue lines denote electric connections, red lines the optical signal paths, dotted lines correspond to the dual option.

TOPTICA's TERAFLASH pro system combines established femtosecond laser technology and state-of-the-art InGaAs photoconductive antennas into a table-top terahertz platform. The system achieves a bandwidth greater than 6 THz and - owing to a precise mechanical delay stage[2] - a peak dynamic range of up to 120 dB. The entire system is based on mature 1.5 µm technology and the antenna modules are equipped with single-mode, polarization-maintaining (SM/PM) fiber pigtails.

Figure 2 shows a schematic representation of the TERAFLASH pro system. The unit comprises a 1.5 µm femtosecond fiber laser with dispersion compensation. A 50/50 fiber splitter divides the output beam into an emitter and a receiver branch. Both optical paths feature highly precise, fiber-coupled mechanical delay stages: the emitter path includes a fast, scanning delay and the receiver path a slow, long-travel delay. Laser, fiber splitter, delay stages, driver electronics, power supply as well as a microcomputer for system control and data acquisition are all housed in one 19" box. The system is equipped with an Ethernet interface for software control.

The optical pulses in both arms are guided to fiber-coupled, InGaAs-based photoconductive antennas, which serve as terahertz emitter and receiver. The fiber-pigtailed design permits a flexible arrangement in either transmission or reflection geometry.

The TERAFLASH pro system offers the option to operate two emitter-receiver pairs simultaneously. To this end, both the emitter and the receiver branch are divided into two fiber output ports each, by using additional 50/50 fiber splitters after the delay stages (dotted lines in Figure 2).

2. N. Vieweg et al., *Terahertz-time domain spectrometer with 90 dB peak dynamic range; J Infrared Milli. Terahz. Waves* **35:10** (2014) 823.

1.3 Femtosecond Laser

The TERAFLASH pro uses a robust, all-fiber-based femtosecond laser system (TOPTICA FemtoFiber smart TeraFlash). The laser radiation is centered at 1560 nm, the repetition rate is 80 MHz and the pulse width is below 80 fs. Carefully designed fiber optics compensate for dispersion effects and guide the optical pulses to the delay stages and photoconductive antennas. The entire assembly uses polarization-maintaining fibers.

1.4 Delay Stages

1.4.1 Fast and Highly Precise Delay

A fast and highly precise delay provides the time variation between the emitter and receiver branches, required for sampling the terahertz pulse. It consists of a voice-coil driven corner-cube mirror with a digital, high-precision position sensor. This combination enables reproduceable time delays with a resolution of 1.3 fs. The high accuracy of the time scale results in superior dynamic-range values when multiple pulse traces are averaged.

Data acquisition is accomplished both during the forward and the backward movement of the mirror, which minimizes the "dead time" of the system. The acquisition time of one pulse trace varies with the chosen delay range, since the swing speed of the voice-coil is maintained constant. Consequently, a long travel distance requires more time: The system samples 166 traces/s at a scan range of 5 ps, 95 traces/s at 20 ps, 60 traces/s at 50 ps, and 6 traces/s at 200 ps. A range of intermediate settings can be selected in the control software.

1.4.2 Long-Travel Delay Line

A second, long-travel delay line introduces a constant timing offset between the emitter and receiver branch and thus compensates any changes in the length of the terahertz path. It comprises a motor-driven corner-cube mirror with a 4-fold optical path, and produces a delay of up to 4000 ps. The value of the time delay is also measured with a resolution of 1.3 fs.

The "wide-scan" option combines the movement of the fast delay with discrete steps of the long-travel delay. By stitching several time traces together, the scan range can be extended up to 3000 ps, attaining a frequency resolution of 0.33 GHz in the spectrum.

1.5 Photoconductive Antennas

The TERAFLASH pro system provides four fiber connector ports, so that up to four photoconductive antennas (i.e. one or two emitter-receiver pairs) can be connected. Each emitter module employs a high-mobility InAlAs/InGaAs multilayer heterostructure (MLHS)[3, 4, 5]. The receiver is based on an LT-grown Beryllium doped InAlAs/InGaAs MLHS[6, 7, 8] with short carrier lifetime. A strip-line antenna geometry is chosen for the emitter, and a dipole geometry for the receiver. Both emitter and receiver feature a mesa-structured gap region. The antennas are packaged into compact housings of 25 mm diameter, and are equipped with SM/PM fiber pigtails.

New high-power emitters[9] produce a typical output power of 300 .. 400 µW. In conjunction with a suitable photoconductive receiver, this translates into a dynamic range of more than 120 dB.

A hyperhemispherical silicon lens generates a divergent terahertz beam, with a divergence angle of about 25° (FWHM). The radiation is linearly polarized; the polarization axis is marked with two grooves at the front of the modules.

1.6 Key Features of the TERAFLASH pro Systems

- Table-top terahertz time-domain spectrometer.
- InGaAs-based photoconductive switches.
- Robust, all-fiber-based femtosecond laser system (TOPTICA FemtoFiber smart TeraFlash).
- Excellent terahertz performance: peak dynamic range: up to 120 dB, spectral bandwidth: > 6 THz.
- High measurement speed: 60 traces/second with 50 ps scan length.
- Flexible setup due to 2.5 m long fiber delivery from main unit to terahertz antennas.
- Variable terahertz path length between 15 cm and 110 cm.
- Dual option for 2 emitter/receiver pairs.

The TERAFLASH pro is a powerful terahertz platform for a wide range of applications, both in scientific research and industrial quality control. The excellent performance – in particular, a broad spectral bandwidth combined with an outstanding dynamic range - makes the TERAFLASH pro a versatile laboratory tool for terahertz spectroscopy and imaging.

Due to its robust design and high measurement speed, the TERAFLASH pro system also lends itself for industrial usage, e.g. as part of a production and quality control system. Applications range from fundamental research to terahertz-based thickness gauging, non-destructive testing and characterization of semiconductors.

3. R.J.B. Dietz et al., *THz generation at 1.55 µm excitation: six-fold increase in THz conversion efficiency by separated photoconductive and trapping regions*; Opt. Express **19** (2011) 25911.
4. R.J.B. Dietz et al., *64 µW pulsed terahertz emission from growth optimized InGaAs/InAlAs heterostructures with separated photoconductive and trapping regions*; Appl. Phys. Lett. **103** (2013) 61103.
5. B. Globisch et al.: Iron doped InGaAs: Competitive THz emitters and detectors fabricated from the same photoconductor; J. Appl. Phys. **121** (2017) 053102.
6. B. Globisch et al., *Carrier dynamics in Beryllium doped low-temperature-grown InGaAs/InAlAs*; Appl. Phys. Lett. **104** (2014) 172103.
7. R.J.B. Dietz et al., *Influence and adjustment of carrier lifetimes in InGaAs/InAlAs photoconductive pulsed terahertz detectors: 6 THz bandwidth and 90 dB dynamic range*; Opt. Express **22** (2014) 19411.
8. R.B. Kohlhaas et al., *Improving the dynamic range of InGaAs-based THz detectors by localized beryllium doping: up to 70 dB at 3 THz*; Opt. Lett. **43**, 5423-5426 (2018)
9. A. Dohms et al., *Fiber-coupled THz TDS system with mW-level THz power and up to 137 dB dynamic range*; IEEE Transact. Terahertz Science Technol. (2024)

1.7 System Extensions and Options

1.7.1 Additional Antenna Pair

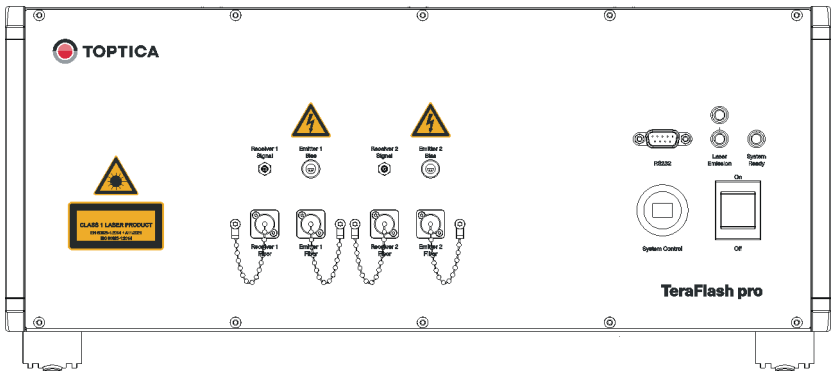


Figure 3 TERAFLASH pro with outputs for two emitter-receiver pairs

TOPTICA's TERAFLASH pro system enables simultaneous measurements with one or two emitters, and one or two receivers. Adding a second antenna pair therefore permits synchronous measurements in transmission and reflection, or the addition of a reference channel to monitor environmental changes.

1.7.2 Imaging Extensions

NOTE !	Please refer to section 6 for detailed information on the Imaging Extensions.

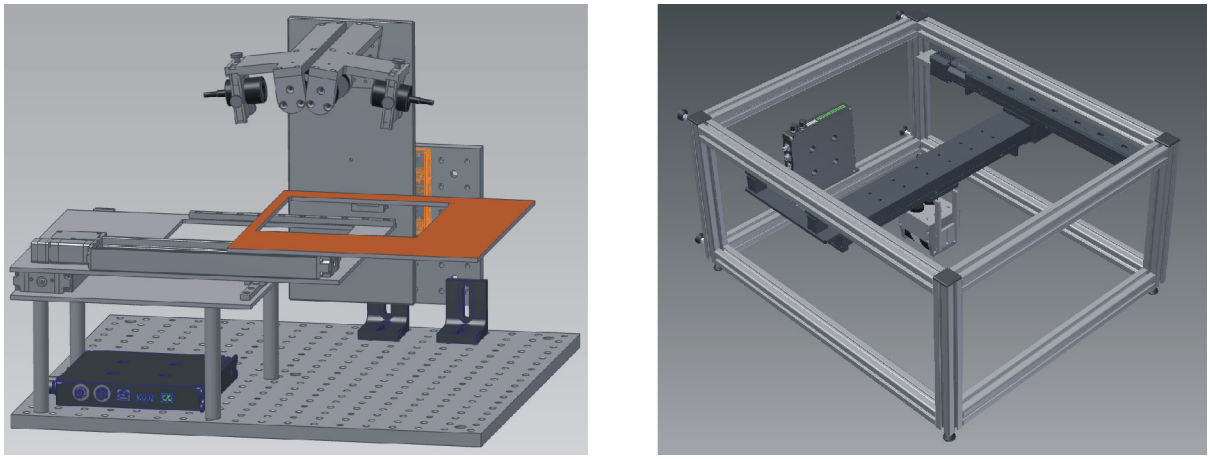


Figure 4 Schematic of the Imaging Extension options for the TERAFLASH pro system (left: Complete Imaging Extension, right: Flexible Imaging Extension)

TOPTICA offers two Imaging Extensions as accessories to the TERAFLASH pro system.

The Complete Imaging Extension employs two precise linear stages to scan a sample through the terahertz beam, and four parabolic mirrors for beam shaping and focusing. Users can quickly reconfigure the optics from a transmission setup to a reflection geometry and vice versa, with the help of alignment pins.

The Flexible Imaging Extension includes a compact reflection head that scans across a static sample. A rigid aluminum frame allows the assembly to be positioned horizontally, vertically or at any oblique angle. The "Flexible" version is therefore well suited to study heavy objects, including paintings and murals.

For all setups, the translational movement is synchronized with the voice-coil delay of the TERAFLASH pro, resulting in a high measurement speed of up to 40 pixels per second. The positioning accuracy is better than 200 µm over a field of view of 15 x 15 cm (Complete Imaging Extension) or 30 x 30 cm (Flexible Imaging Extension).

The Imaging Extension comes with a software package that enables customers to select different contrast parameters for the image, including pulse amplitude, pulse position and layer thickness. The software permits time gating and frequency filtering of the raw data, which is a powerful method to increase image resolution. In addition, users can generate amplitude and height profiles along arbitrary cross-sections of the sample. Detailed specifications of the Imaging Extensions are listed in section 8.2.

1.7.3 Optomechanics

TOPTICA offers different sets of optomechanics, designed for the most common beam-path configurations. For transmission-mode experiments, three rail-based assemblies produce a collimated terahertz beam (THZ OPTICS/TR COMP, compact 2-mirror setup, and THZ OPTICS/TR 2M, finely adjustable 2-mirror setup), or an additional focus (THZ OPTICS/TR 4M, 4-mirror setup). For applications that require a reflection geometry, a compact module (THZ OPTICS/REF 4M) generates a focus at the location of the sample. Customized optomechanical assemblies are available upon request.

Specifications:

- THZ OPTICS/TR COMP: Compact rail system with 2 off-axis parabolic mirrors, for transmission-mode experiments with collimated beam.
- THZ OPTICS/TR 2M: Finely adjustable rail system with 2 off-axis parabolic mirrors, for transmission-mode experiments with collimated beam.
- THZ OPTICS/TR 4M: Finely adjustable rail system with 4 off-axis parabolic mirrors, for transmission-mode experiments with focussed or collimated beam. Focus size approx. 1 mm.
- THZ OPTICS/REF 4M: Compact reflection head with 4 off-axis parabolic mirrors. Focus size approx. 1 mm.

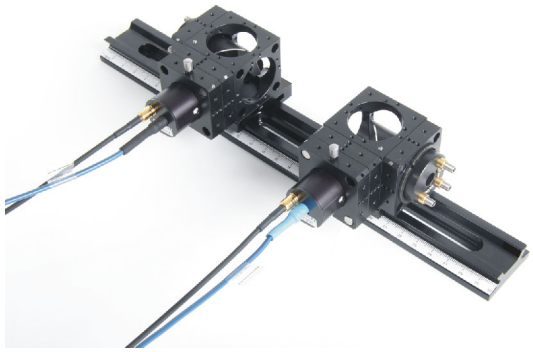


Figure 5 Optomechanics THZ OPTICS/TR COMP (shown with photoconductive antennas)

Package Contents THZ OPTICS/TR COMP

- 2 off-axis parabolic mirrors, diameter 1", focal length 2"
- 2 photoconductive antenna mounts (not adjustable)
- 1 optical rail, 25 cm length

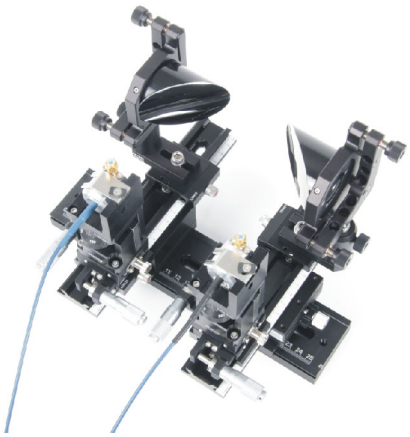


Figure 6 Optomechanics THZ OPTICS/TR 2M (shown with photoconductive antennas)

Package Contents THZ OPTICS/TR 2M

- 2 off-axis parabolic mirrors, diameter 2", focal length 3",
- 2 kinematic mirror mounts,
- 2 xyz stages for photoconductive antennas,
- Manual delay stage,
- Assembly mounted on optical rails.

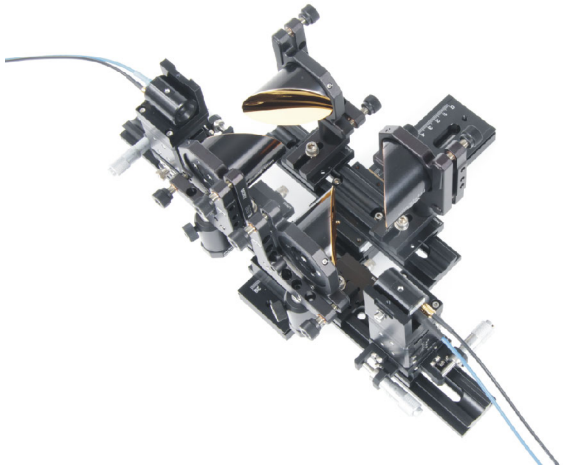


Figure 7 Optomechanics THZ OPTICS/TR 4M (shown with photoconductive antennas)

Package Contents THZ OPTICS/TR 4M

- 4 off-axis parabolic mirrors, diameter 2", focal length 3" (collimating mirrors) and 2" (focussing mirrors),
- 4 kinematic mirror mounts,
- 2 xyz stages for photoconductive antennas,
- Manual delay stage,
- Assembly mounted on optical rails.

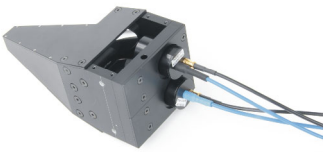


Figure 8 Optomechanics THZ OPTICS/REF 4M (shown with photoconductive antennas)

Package Contents THZ OPTICS/REF 4M

Reflection head with:

- 4 off-axis parabolic mirrors, diameter 1", focal length 2" (collimating mirrors), and 4" (focussing mirrors),
- Connector for tubes (FESTO - CK-M5-PK-4 - outer tube diameter 6 mm) for dry air or nitrogen purging

2 Safety Instructions and Warnings

Before using the TERAFLASH pro system please read and follow the Safety Instructions and Warnings noted in this manual.

The following safety terms are used in this manual:
The **DANGER !** heading in this manual explains hazards that could result in personal injury or death.
The **CAUTION !** heading in this manual explains hazards that could damage the instrument.
In addition, a **NOTE !** heading gives information to the user that may be beneficial when using the device.

2.1 General Safety Terms

The TERAFLASH pro system is manufactured according to the Laser Safety Standards EN 60825-1:2014+A11:2021, IEC 60825-1:2014 and complies with US laws 21 CFR §1040.10 and §1040.11.

DANGER !
CAUTION ! Before operating the TERAFLASH pro system please read this manual carefully to prevent damage to the device, connected photoconductive antennas and injury to persons. The following safety instructions must be followed at all times.

DANGER !
CAUTION !



Possibility of electrical shock ! Wherever this symbol is attached, the possibility of an electrical shock may appear. Use only equipment and accessories supplied by TOPTICA.



Caution ! Wherever this symbol is attached read and understand the manual before operating the device. The manual must be consulted in order to find out the nature of the potential HAZARDS and any actions which have to be taken to avoid them.

DANGER !
CAUTION ! The TERAFLASH pro Main Unit is equipped with two LEDs which indicate laser emission. Please refer to section 4.1 in this manual for detailed information. Be aware of laser emission when these LEDs light up.

DANGER ! **Laser light is emitted from the fibers when the photoconductive antennas are optically disconnected**. Never disconnect/connect any fibers while laser emission is switched on ! Do not look into any fiber connector ports or fiber facets. At the fiber connector ports a **divergent** laser beam is emitted (power up to 50 mW at each fiber output port, Class 1 laser product).

DANGER ! The photoconductive antenna modules feature a silicon lens with a black protective coating, which suppresses residual laser radiation. The antenna modules must not be used if the black protective coating on the silicon lens is scratched or damaged.

DANGER ! The equipment must be positioned so that the disconnecting device can be easily reached by the operator.

DANGER ! Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

DANGER ! Do not look into the laser beam under conditions which exceed the limits for Class 1 specified by the United States Food and Drug Administration, Department of Health and Human Services, Center for Devices and Radiological Health, 21 CFR 1040.10 and 2 CFR 1040.11.

To be continued on the next page.

DANGER ! For safe operation and for proper grounding please use the supplied mains cable only. Improper or missing grounding can lead to serious injury.

DANGER ! The electrical units should not be operated in a hazardous environment.

DANGER ! If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

DANGER ! The equipment must be positioned so that the disconnecting device can be easily reached by the operator.

CAUTION ! **Photoconductive antennas are ESD-sensitive devices. We recommend wearing a high-impedance grounding strap around the wrist when handling photoconductive antennas. Take care not to touch the inner pins of the connectors.**

CAUTION ! When handling the photoconductive antennas, it is necessary to protect them from any inadvertent electrical short-circuits and to avoid contact with current-carrying parts. The antennas are particularly sensitive to voltage peaks. These might considerably affect the service life of the antennas, even if a disturbance is not immediately noticeable.

CAUTION ! **Do not attempt to open the photoconductive antenna modules.**

2.2 Safety Labels on the TERAFLASH pro System

2.2.1 Laser Beam

The TERAFLASH pro system emits invisible pulsed laser radiation of up to 50 mW average power at the optical connector ports. It is classified as Class 1 product.

DANGER !	The TERAFLASH pro system emits invisible pulsed laser radiation of up to 50 mW average power at the optical connector ports (Class 1 laser product). Avoid exposing eyes and skin to the laser beam, including any laser stray light.

The following label is affixed to the housing of the TERAFLASH pro Main Unit according to EN 60825-1:2014+A11:2021, IEC 60825-1:2014:



Size: 52 mm x 26 mm
Color: yellow/black
Location: At the front panel of the TERAFLASH pro Main Unit

Specifications of the Laser Beam Output:

P_{ave}	up to 50 mW / Output
t_{pulse}	< 80 fs
Center Wavelength	1560 nm

2.2.2 Laser Beam Output

During operation, the laser beam is emitted from the four FC/APC fiber connector ports on the front panel of the TERAFLASH pro Main Unit. (For location of the laser beam outputs please refer to section 8.10).

DANGER !	At the optical connector ports, the TERAFLASH pro Main Unit emits invisible pulsed laser radiation of up to 50 mW power per output (Class 1 laser product). Avoid exposing eyes and skin to the laser beam, including any laser stray light.

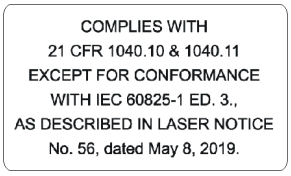
The following label is affixed on the TERAFLASH pro Main Unit according to EN 60825-1:2014+A11:2021, IEC 60825-1:2014:



Size: 24 mm x 21 mm
Color: Yellow/black
Location: At the front panel of the TERAFLASH pro Main Unit

2.2.3 CFR Compliance

Compliance with US laws 21 CFR §1040.10 and §1040.11 is declared by the following label:



Size: 38 mm x 19 mm
Color: silver/black
Location: At the side of the TERAFLASH pro Main Unit

2.2.4 ESD-Sensitive Devices



Size: 13 mm x 13 mm
Color: yellow/black
Location: At the photoconductive antennas

2.3 Identification of Manufacturer

Manufacturer (name and address), production date, product ID number and compliance with CE standard are noted on the identification label. The crossed out wheeled dust bin symbol indicates that products must be collected and disposed of separately from household waste (please refer to section 8.13 for details).



Sizes: 38 mm x 19 mm
Colors: Silver/black
Locations: At the side of the TERAFLASH pro Main Unit

3 Installation

3.1 Handling during Transport

As long as the TERAFLASH pro system is inside the shipping box, keep the box in a horizontal position. Do not turn it upright or upside down during transport and storage.

Always use the supplied shipping box to transport the TERAFLASH pro system.

3.2 Inspection after Delivery

DANGER !	Heavy equipment – personal injury or equipment damage might result if mishandled.
CAUTION !	Use safe practices when lifting. At least two persons are required to lift and install the system. Always use the installed handles located at the TERAFLASH pro Main Unit.

The TERAFLASH pro system (typical package contents shown in Figure 9) and the Imaging Extension are packed in wooden boxes designed to give maximum protection during shipment. If the outside of a shipping box is damaged, notify your shipping department immediately. The shipping agent may wish to notify the carrier at this point.

If the shipping box is undamaged externally, the parts of the TERAFLASH pro system should be removed from the box. **Keep the TERAFLASH pro Main Unit in a horizontal position, do not turn it upright or upside down**. If any damage is evident visually, notify TOPTICA SE and your shipping agent. It is recommended to save the box for future storage or transportation.



Figure 9 Basic package contents of the TERAFLASH pro system

TERAFLASH pro

(1) TERAFLASH pro Main Unit
(2) Emitter photoconductive antenna - TX
(3) Mains supply cable
(4) USB-Ethernet adapter
(5) Ethernet cross-link network cable
(6) Receiver photoconductive antenna - RX
USB flash drive with operating software and TERAFLASH pro manual

Depending on the requested specification, your TERAFLASH pro system may be delivered with additional options not shown here, e.g. an additional photoconductive antenna pair, optomechanics assemblies, or Imaging Extensions (see below).

Complete Imaging Extension

2 stages with motion controller, mounted on breadboard
4 parabolic mirrors with mounts
RS 232 cable for scanner control
Power supply (48 V DC, 3.75 A) with mains supply cable
Structured aluminum plate (adjustment aid)

Flexible Imaging Extension

2 stages with motion controller, mounted on aluminum frame
Reflection head (THz Optics/Ref 4M)
RS 232 cable for scanner control
Power supply (48 V DC, 3.75 A) with mains supply cable
Structured aluminum plate (adjustment aid)

3.3 Installation Instructions

When installing the TERAFLASH pro system, the following instructions have to be observed:

- The TERAFLASH pro Main Unit must be installed horizontally. The voice-coil delay will not function properly if it has to act against gravity.
- Do not cover the ventilation grilles at the bottom side and at the rear side of the TERAFLASH pro Main Unit.
- For environmental conditions and weight, please refer to section 8.1.
- Never use the unit near water, for example near a wash basin, a sink or other damp environment.
- The laser beam is emitted from the FC/APC fiber connector ports on the front panel of the TERAFLASH pro Main Unit. Locations are shown in section 8.10.

DANGER ! The photoconductive antenna modules feature a silicon lens with a black protective coating, which suppresses residual laser radiation. The antenna modules must not be used if the black protective coating on the silicon lens is scratched or damaged.

CAUTION ! Always clean the fiber facets as described in section 3.3.1 before connecting a fiber to the connector port.

CAUTION ! Photoconductive antennas are ESD-sensitive devices. We recommend wearing a high-impedance grounding strap around the wrist when handling photoconductive antennas. Take care not to touch the inner pins of the connectors.

CAUTION ! When handling the photoconductive antennas, it is necessary to protect them from any inadvertent electrical short-circuits and to avoid contact with current-carrying parts. The antennas are particularly sensitive to voltage peaks. These might considerably affect the service life of the antennas, even if a disturbance is not immediately noticeable.

CAUTION ! Do not attempt to open the photoconductive antenna modules.

3.3.1 Handling of Optical Fibers

DANGER !	**Before checking and/or cleaning a fiber facet, switch off the TERAFLASH pro system and disconnect it from mains. Never check or clean a fiber facet when laser emission is switched on !**

CAUTION !	Dust particles on a fiber facet will burn when illuminated by a laser beam, and damage the fiber core. Unless an optical inspection (e.g. with the help of a fiber microscope) proves the fiber facets to be dust-free, it is strongly recommended to always clean the facets before connecting a fiber to a connector port, or to another fiber.

CAUTION !	For the fiber pigtails at the antennas, the bend radius should not be smaller than 1.5 cm. If possible, usa a bend radius larger than 2.5 cm. Do not drop any sharp or heavy objects on the fiber.

Checking a Fiber Facet:

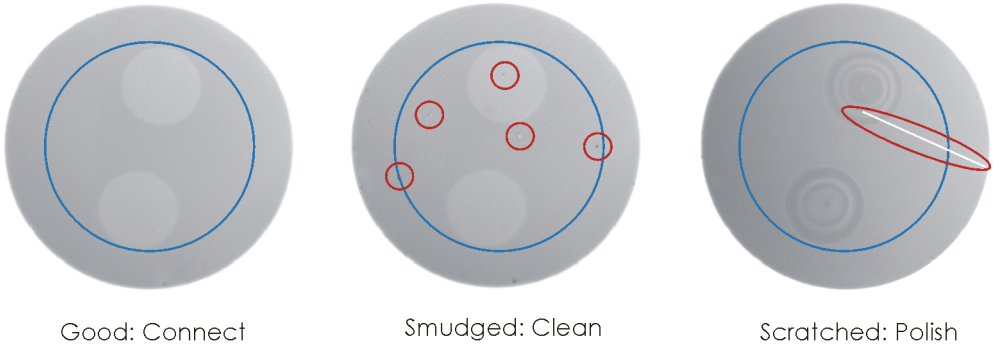


Figure 10 Appearance of fiber facets (examples)

Cleaning a Fiber Facet:

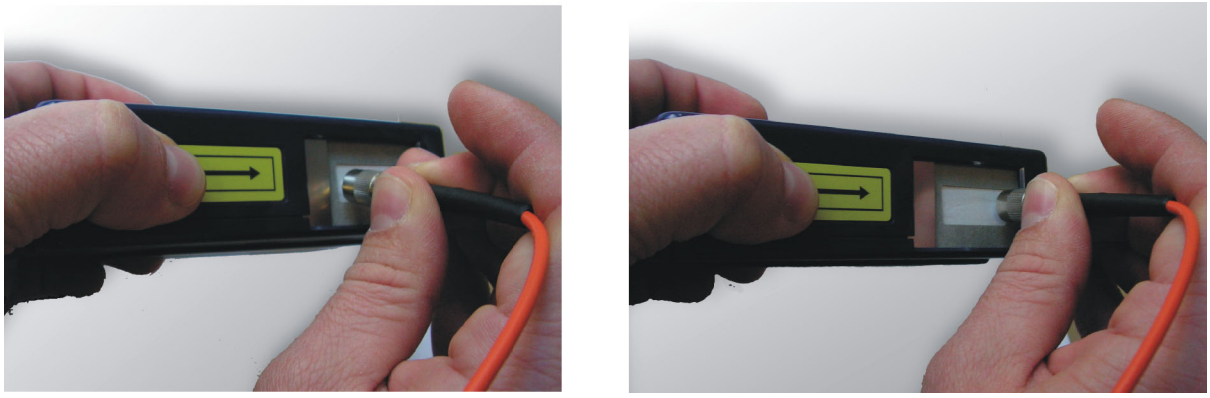


Figure 11 Cleaning the fiber.
Hold the fiber as shown on the left side and move it twice across the cleaning surface as shown on the right side

Before connecting an optical fiber, its facet should be cleaned with a commercially available 'reel cleaner' or a similar device that incorporates a dry woven polyester cloth as cleaning medium (see Figure 11).

Protecting a Fiber Connector when not in use:

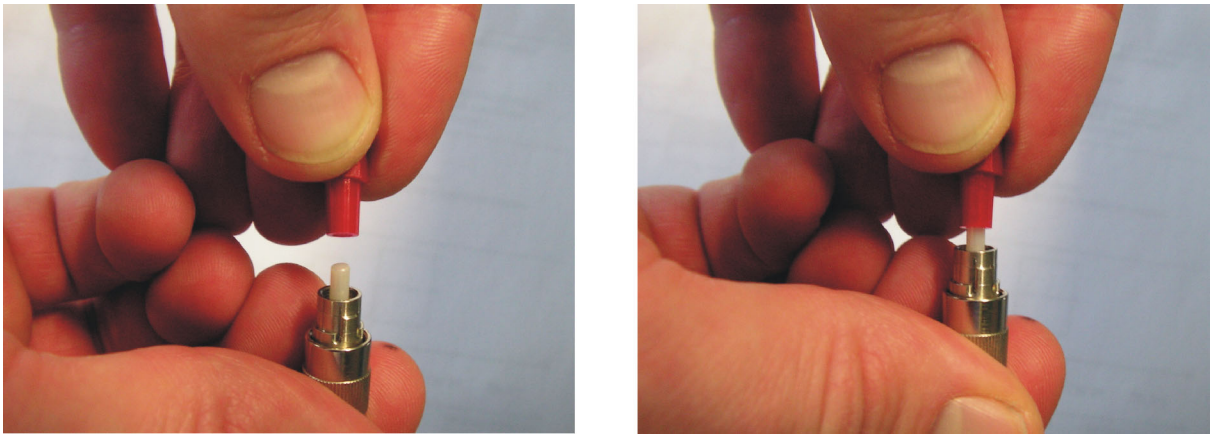


Figure 12 Protecting a fiber

When the fiber is not used, carefully place the protective cap on the fiber end as shown in Figure 12.

4 Operation

4.1 Operator Controls of the TERAFLASH pro Main Unit

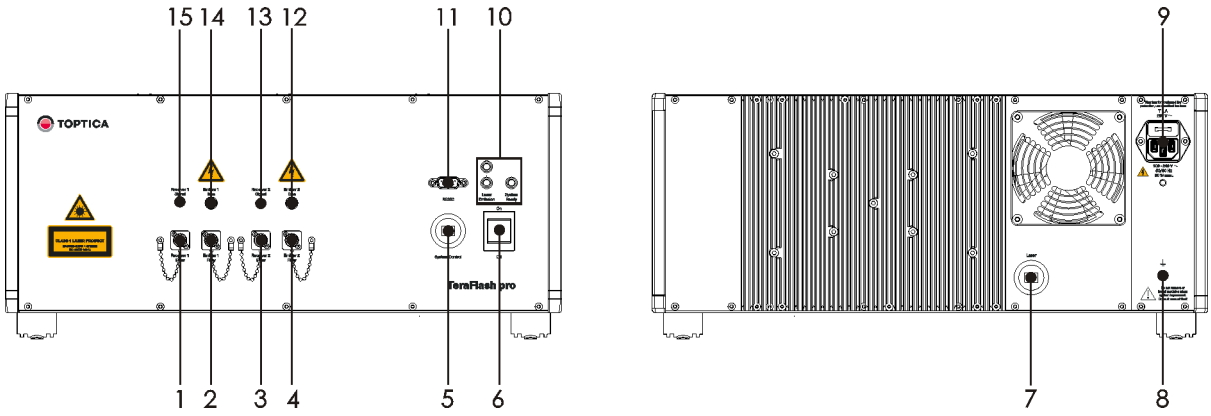


Figure 13 Front panel (left) and rear panel (right) of TERAFLASH pro Main Unit

1 Receiver 1 Fiber Connector Port

2 Emitter 1 Fiber Connector Port

3 Receiver 2 Fiber Connector Port

4 Emitter 2 Fiber Connector Port

5 System Control Ethernet Connector

6 Power Switch

7 Laser Control Ethernet Connector

8 Ground pin with M5 thread

9 Mains Connector

10 Control LEDs:
System Status and
Laser Radiation Emission Warning LEDs

11 RS 232 connector (not used)

12 Emitter 2 Bias Output Port

13 Receiver 2 Signal Input Port

14 Emitter 1 Bias Output Port

15 Receiver 1 Signal Input Port

4.2 Description of Operator Controls

1 Receiver 1 Fiber Connector Port	Connector port for the receiver 1 fiber, with protective cover.
2 Emitter 1 Fiber Connector Port	Connector port for the emitter 1 fiber, with protective cover.
3 Receiver 2 Fiber Connector Port	Connector port for the receiver 2 fiber, with protective cover.
4 Emitter 2 Fiber Connector Port	Connector port for the emitter 2 fiber, with protective cover.
5 System Control Ethernet Connector	Connector port for computer control of the TERAFLASH pro Main Unit.
6 Power Switch	Power switch for the TERAFLASH pro Main Unit.
7 Laser Control Ethernet Connector	**Do not use** this connector unless advised by TOPTICA service.
8 Ground Pin	Ground pin with M5 thread.
9 Mains Connector	Mains cable connector.
10 Control LEDs **Laser Radiation Emission Warning LED**	**System Ready** While the TERAFLASH pro system is booting, the green LED flashes. Once booting is completed, the LED lights up continuously. **Laser Emission** The two yellow LEDs light up when the built-in femtosecond laser is switched on and laser light is emitted. Two LEDs are installed to guarantee redundancy. **DANGER !** When the yellow Laser Radiation Emission Warning LEDs light up, invisible laser radiation is emitted from all of the fiber connectors. Avoid eye-contact with plugs or sockets of optical fibers ! Do not connect or disconnect optical fibers while the laser is operating.
11 RS 232 connector	Not used.
12 Emitter 2 Bias Output Port • LEMO-connector	Output port for the emitter 2 bias voltage supply. **DANGER !** Check the LEMO-connector for damage prior to connecting a cable ! Only use a connector free of defects. Do not use any tools to touch the inner pins of the connector, as a voltage of up to 200 V DC is present. **CAUTION !** Only use emitters that withstand a bias voltage of 200 V ! Other emitters may easily be destroyed.
13 Receiver 2 Signal Input Port • SMB-connector	Input Port for the receiver 2 signal.

14 Emitter 1 Bias Output Port • LEMO-connector	Output port for the emitter 1 bias voltage supply. **DANGER !** Check the LEMO-connector for damage prior to connecting a cable ! Only use a connector free of defects. Do not use any tools to touch the inner pins of the connector, as a voltage of up to 200 V DC is present. **CAUTION !** Only use emitters that withstand a bias voltage of 200 V ! Other emitters may easily be destroyed.
15 Receiver 1 Signal Input Port • SMB-connector	Input Port for the receiver 1 signal.

4.3 Connections

CAUTION ! **Photoconductive antennas are ESD-sensitive devices. We recommend wearing a high-impedance grounding strap around the wrist when handling photoconductive antennas. Take care not to touch the inner pins of the connectors.**

CAUTION ! When handling the photoconductive antennas, it is necessary to protect them from any inadvertent electrical short-circuits and to avoid contact with current-carrying parts. The antennas are particularly sensitive to voltage peaks. These might considerably affect the service life of the antennas, even if a disturbance is not immediately noticeable.

NOTE ! **Before connecting the TERAFLASH pro system as described below**, please install the driver for the USB-Ethernet adapter and configure its IP address (see section 5.1.1), then install the driver (see section 5.1.2), and the TERAFLASH pro Control Software (see section 5.1.3). Modify the firewall settings of your computer (see section 5.1.4). Make sure that the USB-Ethernet adapter remains connected to your PC.

1. Connect the TERAFLASH pro Main Unit to mains (100 - 240 V~, 50/60 Hz).

2. Connect the supplied Ethernet cross link network cable to the System Control Ethernet Port at the front panel of the TERAFLASH pro Main Unit, and to the USB-Ethernet adapter.

CAUTION ! Use the supplied cables only !

3. Connect the photoconductive antennas optically to channel 1.
Clean the fiber facets as described in section 3.3.1, using e.g. a Reel Cleaner.
Insert the fiber connectors of the antenna modules into the corresponding FC/APC sockets, at the front of the TERAFLASH pro Main Unit (Emitter 1 Fiber and Receiver 1 Fiber).

NOTE ! Make sure that the polarization maintaining fibers are oriented correctly. The fiber connectors feature a small key that has to be inserted in the corresponding slit in the fiber receptacle.

4. Connect the photoconductive antennas electrically to channel 1.
-Terahertz emitter: Connect the TX to the Emitter Bias 1 connector of the TERAFLASH pro Main Unit.
-Terahertz receiver: Connect the RX to the Receiver 1 Signal connector of the TERAFLASH pro Main Unit.

DANGER ! Check the LEMO-connector for damage prior to connecting a cable ! Only use a connector free of defects. Do not use any tools to touch the inner pins of the connector, as a voltage of up to 200 V DC is present.

CAUTION ! Only use emitters that withstand a bias voltage of 200 V ! Other emitters may easily be destroyed.

NOTE ! For safety reasons, emitter and receiver have different connectors and thus cannot be interchanged.

5. If available: Connect the second pair of photoconductive antennas to channel 2 (repeat steps 3. and 4. for the channel 2 connectors).

6. Mount the terahertz antennas in a suitable optomechanical setup.

NOTE !	The antennas are sensitive to the polarization of the terahertz beam. Take care that both antennas are fixed in the same angular orientation. The axis of polarization is indicated by two grooves on the front side of the antenna modules (see Figure 14).



Figure 14 Photoconductive antennas with SM/PM fiber pigtail. The grooves indicate the axis of polarization.

4.4 Power Up

NOTE !	An automatic input voltage detection circuit adjusts the TERAFLASH pro Main Unit to your local mains supply voltage.

For safe operation of the TERAFLASH pro system follow the steps below.

1. Switch on the PC that you use to control the TERAFLASH pro system.

2. Switch on the TERAFLASH pro Main Unit at the red Power Switch at the front panel. Wait until the green System Ready LED lights up continuously.

4. Start the TERAFLASH pro Control Software. The Main window as shown in Figure 16 opens.

5. In the Control tab of the System Control Panel (see section 5.2.1), click the **Connect** button to establish a connection to the TERAFLASH pro Main Unit. Wait until the status message "Connected" is displayed next to the button.

6. Wait until the TERAFLASH pro system has warmed up and the status changes to "Ready".

7. Click the **Laser** button to switch on the laser.

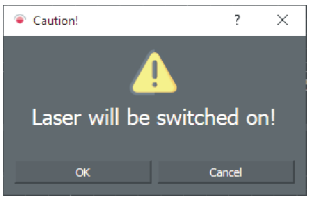


Figure 15 Laser Emission Alert

Confirm by selecting **OK** once the Laser Emission Alert window appears. Wait until the status message "On" is displayed next to the **Laser** button.

8. Once the status message "Ready" is displayed next to the **Emitter 1/2** buttons, click on **Emitter 1** and/or **Emitter 2** to activate the bias voltage of the respective emitter(s).

9. Click **Measurement** to start the scanning of the voice-coil delay. You can select the scan length for the terahertz pulse trace via the **Shaker Range** drop-down menu (see section 5.2.1). A useful value to start with is 50 ps.

10. If no terahertz pulse is displayed in the **Pulse Trace Display** (time-domain trace, upper graph in Figure 16), adjust the value for Delay (ps) (see section 5.2.1), i.e. the position of the stationary delay stage. You can type a numerical value in the spinbox, or use the arrow buttons next to it, or click on the **+50**/**-50** buttons to change the delay position by 50 ps.

NOTE !	The antennas are sensitive to the polarization of the terahertz beam. Take care that both antennas are fixed in the same angular orientation. The axis of polarization is indicated by two grooves at the front side of the antenna modules (see Figure 14).

11. Select the number of averages for the time-domain trace in the Acquisition Section of the System Control Panel (see section 5.2.2). A low number enables a faster acquisition time, a high number provides a higher dynamic range (see Figure 16).

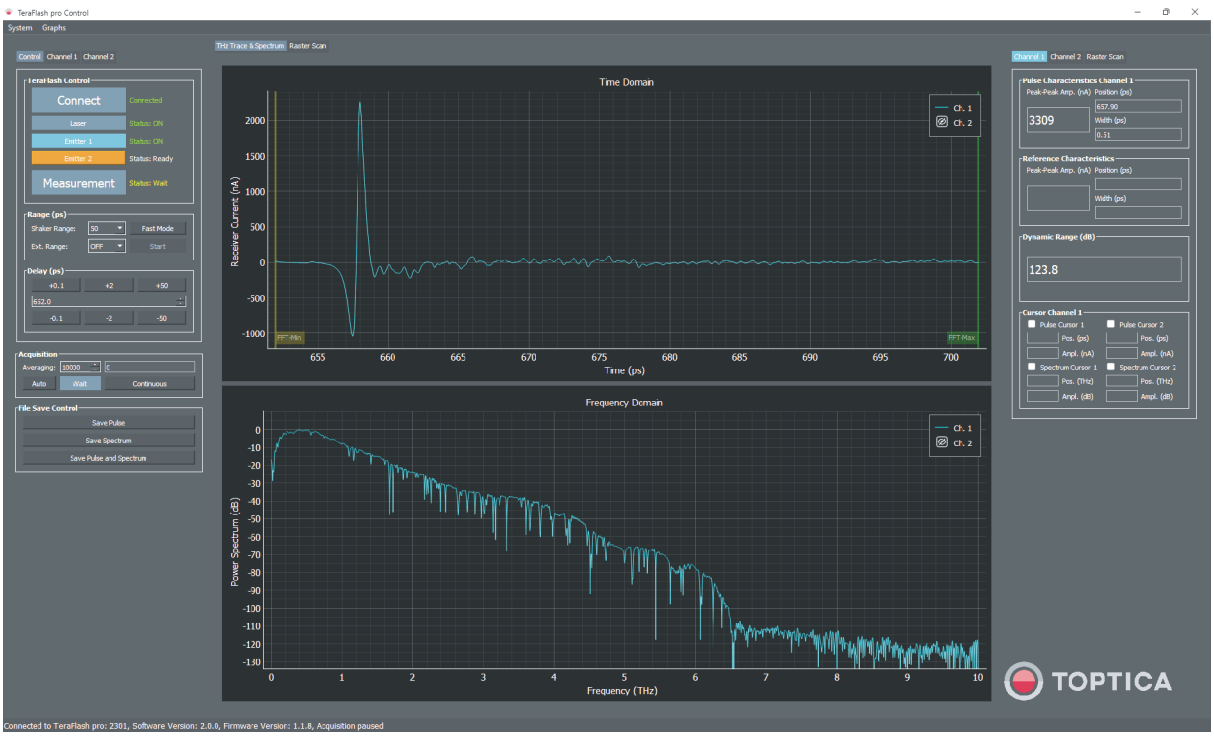


Figure 16 Main window of the TERAFLASH pro Control Software

4.5 Power Down

1. Click the **Measurement** button to stop the scanning of the voice-coil delay and the data acquisition.

2. Click the Emitter 1/2 button(s) to switch off the bias supply of the emitter antennas.

3. Click the **Laser** button to switch off the laser.

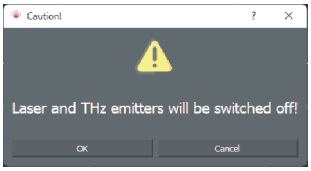


Figure 17 Laser Emission OFF Alert

Confirm by selecting **OK** in the Alert window.

4. Click the **Connect** button to terminate the connection to the TERAFLASH pro Main Unit.

5. Close the TERAFLASH pro Control Software.

6. Switch off the TERAFLASH pro Main Unit at the red Power Switch at the front panel.

5 TERAFLASH pro Control Software

5.1 Software Installation

PC Requirements

- PC with Intel CPU of generation 14 or higher, with one free USB interface (for TERAFLASH pro systems with Imaging Extensions two USB ports are required).
- Windows 10 (or higher) operating system

You need to have administrator rights in order to install new software and drivers, and permission to change IP addresses and firewall settings.

5.1.1 USB-Ethernet Adapter

1. Switch on the control PC. Insert the supplied USB flash drive and copy the content to the hard disk.

2. Remove the USB flash drive and insert the supplied USB-Ethernet adapter. Wait until the computer detects the device. Do not yet connect any Ethernet cable to the adapter.

3. Depending on the computer settings, the driver for the USB-Ethernet adapter is installed automatically. If this is not the case, then use Windows Explorer to navigate to the folders copied from the USB flash drive. Right-click on Thinkpad_USB3_Ethernet_adapter.exe and select "Run as administrator".

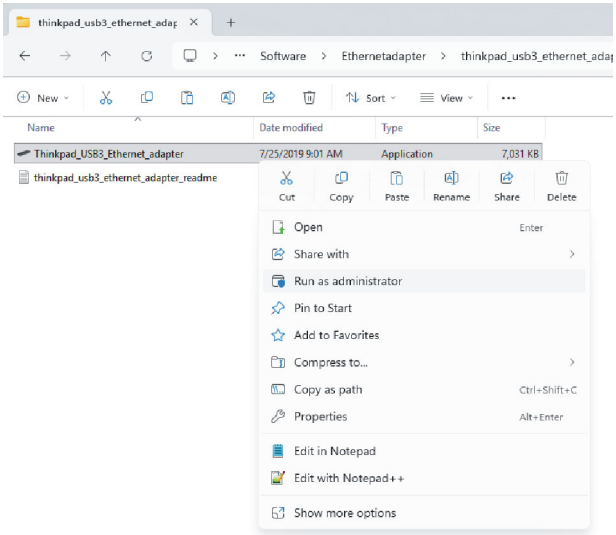


Figure 18 Installing the driver for the USB-Ethernet adapter

4. A "Setup Wizard" window opens. Accept the license agreement and select the installation directory. Always click **Next** to continue and follow the instructions.

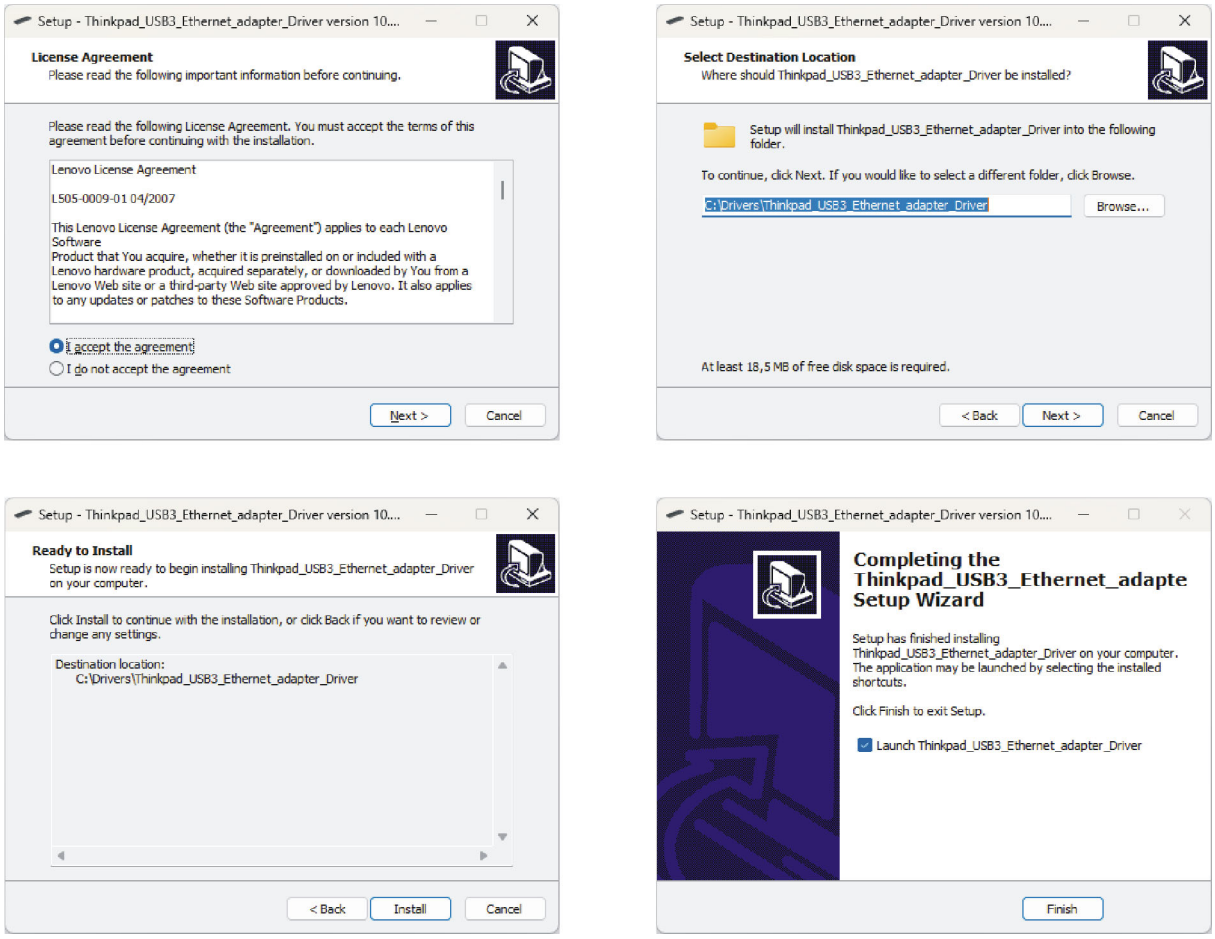


Figure 19 Setup Wizard for the driver installation

5. Configure the IP address of the USB-Ethernet adapter: Navigate to Control Panel > Network and Sharing Center > Change Adapter Settings. Select the new adapter and right-click on **Properties**.

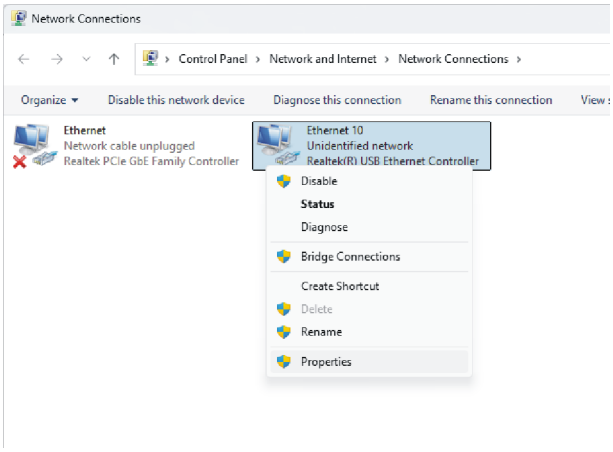


Figure 20 Selecting the properties of the USB-Ethernet adapter

6. Navigate to IPv4 > Properties. Set the IP address to 10.8.8.10 and the Subnet mask to 255.255.255.0, as shown in Figure 21. Confirm with **OK**.

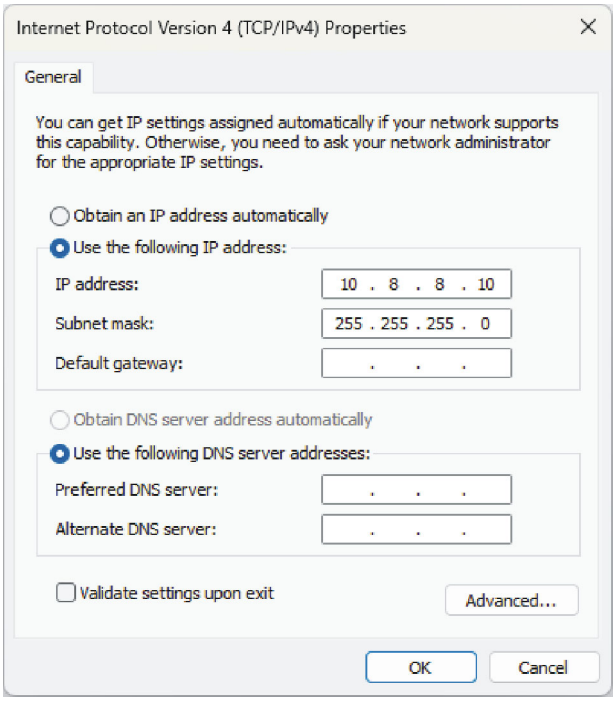


Figure 21 IP address configuration

CAUTION ! Only one device in the connected network is allowed to have this IP-Address.

5.1.2 NI-VISA Driver

The NI-VISA driver is required for the communication of the control PC with the TERAFLASH pro Main Unit.

1. Right-click on ni-visa_24.0_online.exe and select "Run as administrator".

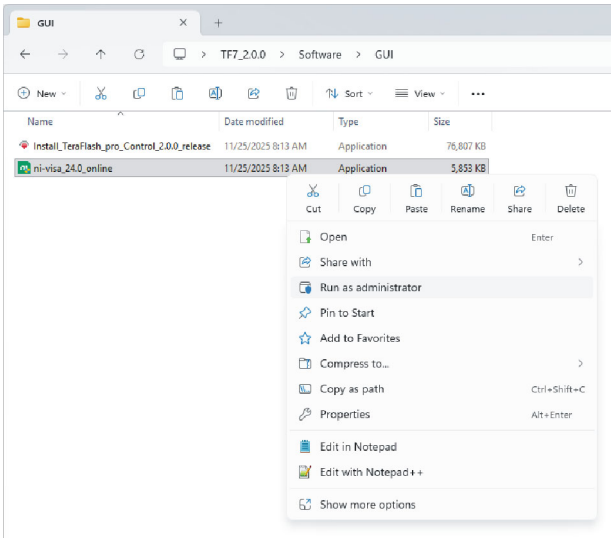
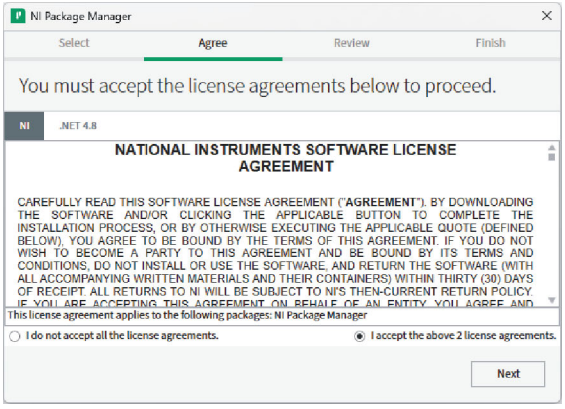


Figure 22 Initiating the installation of the NI Package Manager and the NI-VISA Driver

2. Accept the license agreements and continue by clicking **Next** in the following windows to start the installation of the NI Package Manager. The installation process is indicated by green progress bars.



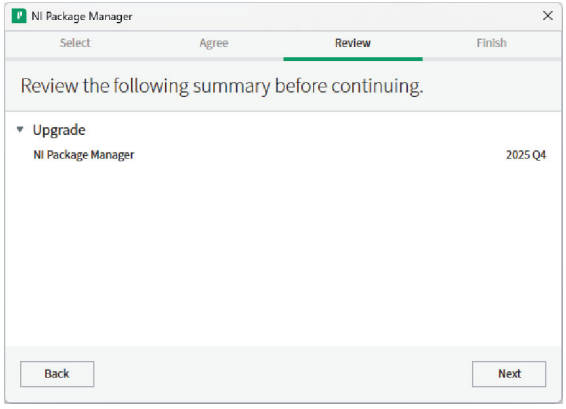


Figure 23 Installing the NI Package Manager

3. To install the NI-VISA driver, click **Next** in the following window. Accept the license agreements and continue by clicking **Next** in the following windows to start the installation of the NI-VISA driver. The installation process is indicated by green progress bars.

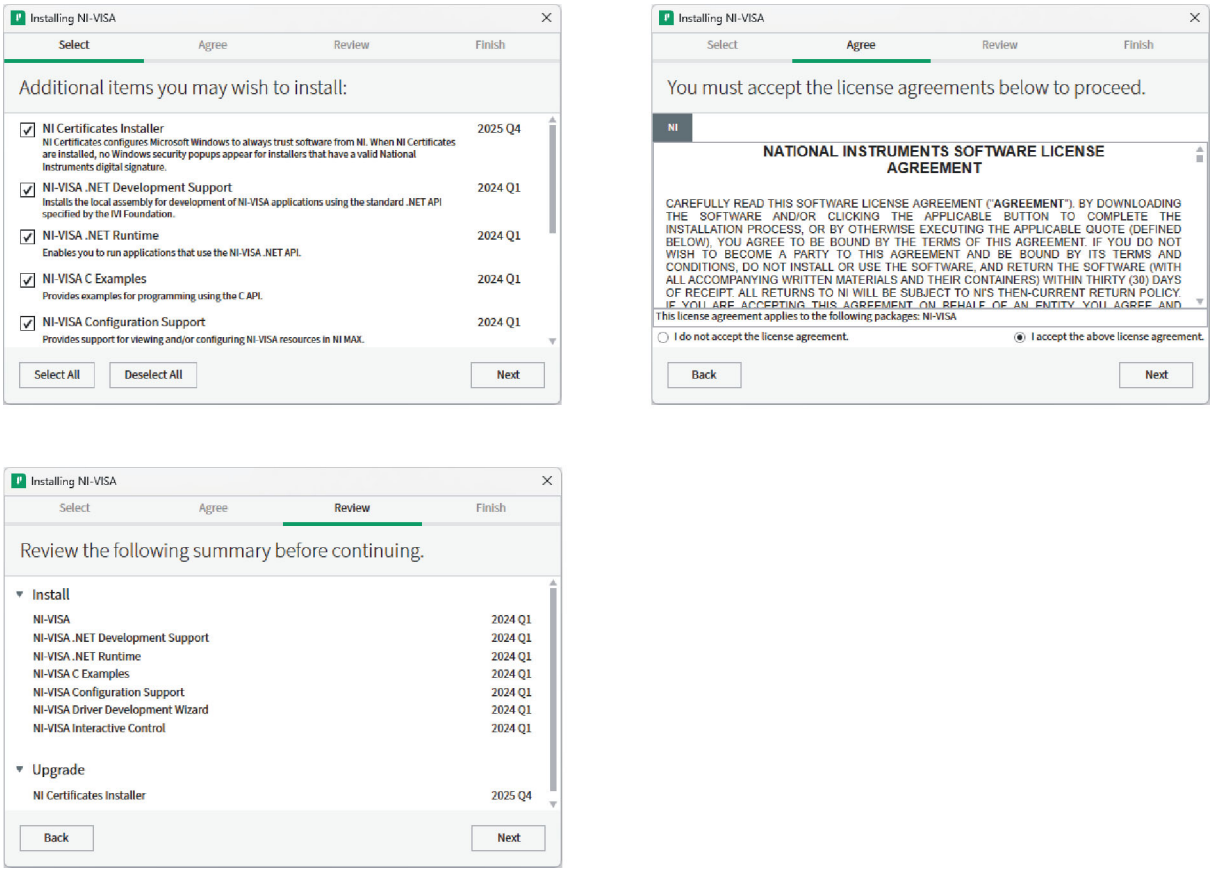


Figure 24 Installation of the NI-VISA Driver

4. When prompted, restart your computer to complete the installation by clicking **Reboot Now**.

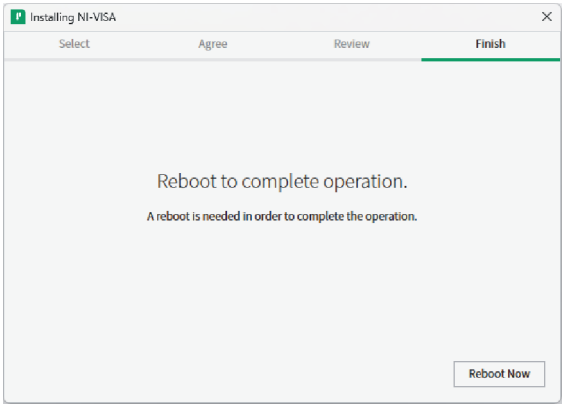


Figure 25 Completion of installation and restart prompt

5.1.3 TERAFLASH pro Control Software

1. Use Windows Explorer to navigate to the directory on the hard disk to which you copied the TERAFLASH pro installation files: Software > Install_TeraFlash_pro_Control_xxx (additional letters and numbers can indicate the system and software version).

   Right-click on Install_TeraFlash_pro_Control_xxx.exe (with "xxx" denoting the software version) and select "Run as administrator".

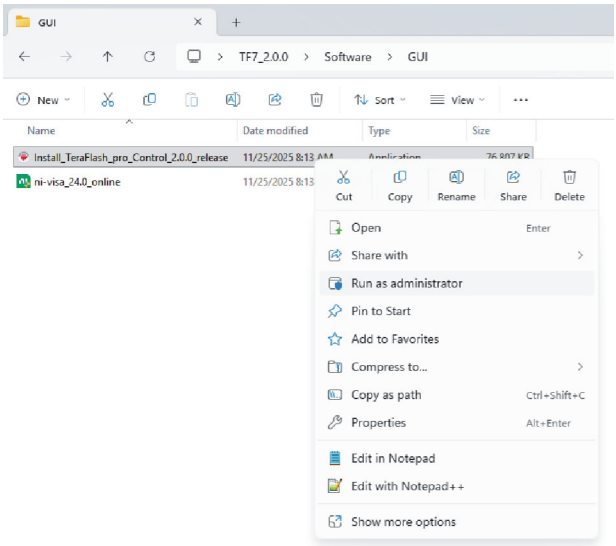


Figure 26 Initiating the installation of the TERAFLASH pro Control Software.

2. Select components to install and click **Next** to continue (Figure 27 left). Select the installation directory and start the installation by clicking **Install** (Figure 27 right).

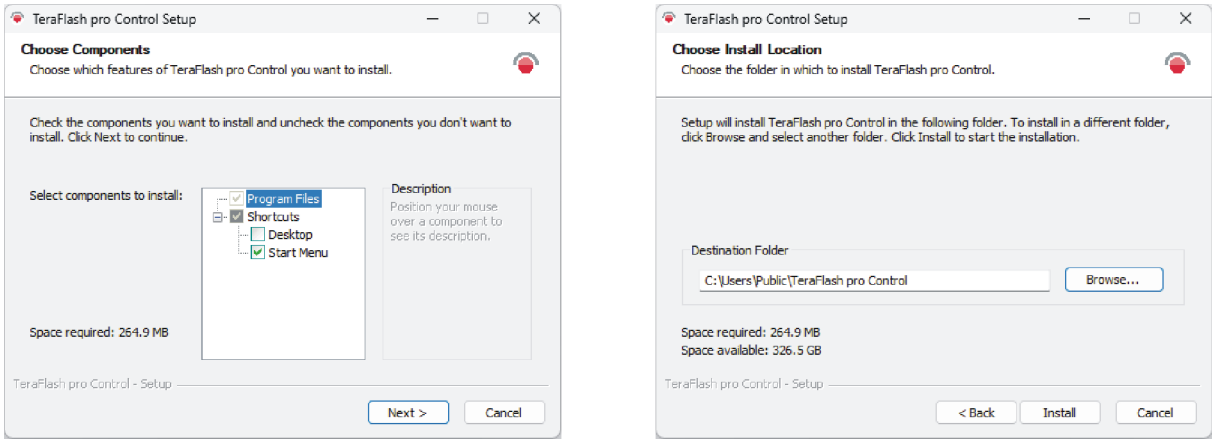


Figure 27 TERAFLASH pro Control Software installation

4. Please wait while the TERAFLASH pro Control Software is being installed. Continue by clicking **Next** (Figure 28 left). Click **Finish** to complete the installation (Figure 28 right).

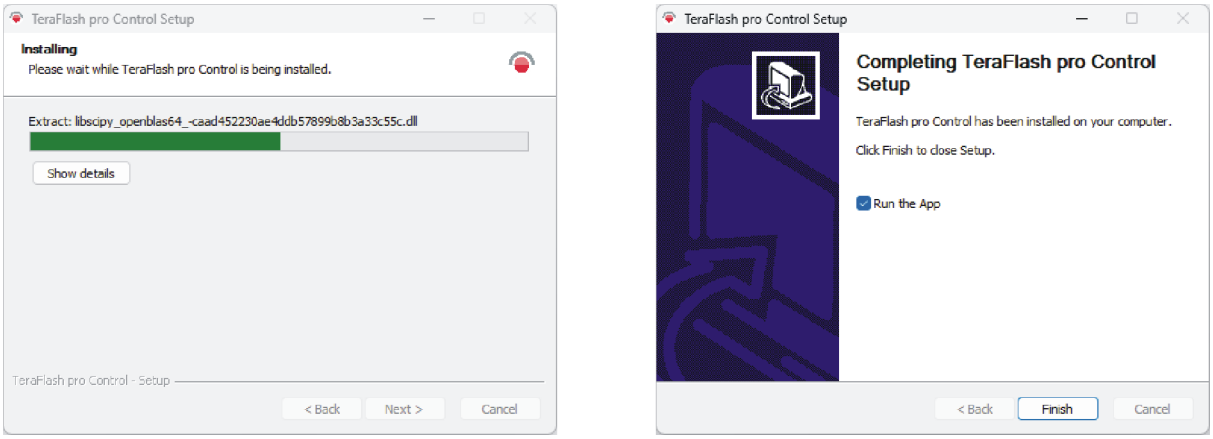


Figure 28 TERAFLASH pro Control Software installation

5.1.4 Firewall Settings

1. Connect the TERAFLASH pro to the USB-Ethernet adapter, using the provided Ethernet cross-link network cable.

2. Allow the TERAFLASH pro Control Software to operate through Windows Firewall:
Navigate to Control Panel > Windows Defender Firewall > Allow an app or feature through Windows Defender Firewall. Click **Allow another app**.

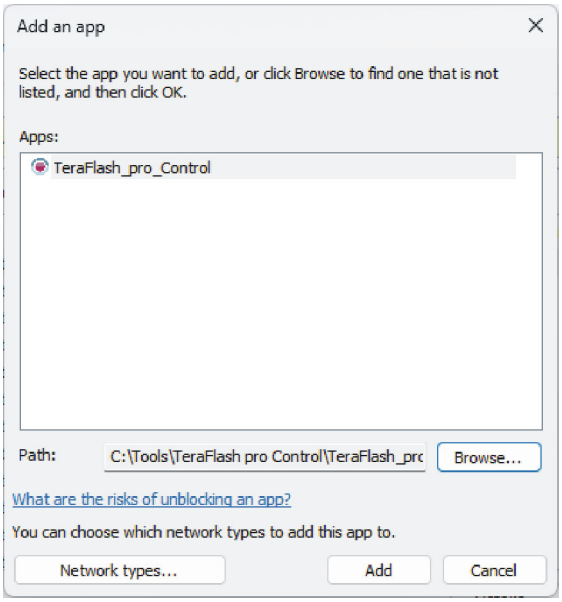


Figure 29 Adding the TERAFLASH pro Control Software to the list of allowed programs in the firewall.

3. In the dialog window that opens, click **Browse** and navigate to the location of the TERAFLASH pro Control Software in your file system. Select the TeraFlash_pro_Control.exe file (additional letters and numbers indicate the system and software version). Confirm by clicking on **Add**.

4. In the dialog box which opens, make sure that private and public access is granted to TeraFlash_pro_Control. Please confirm with **OK**.

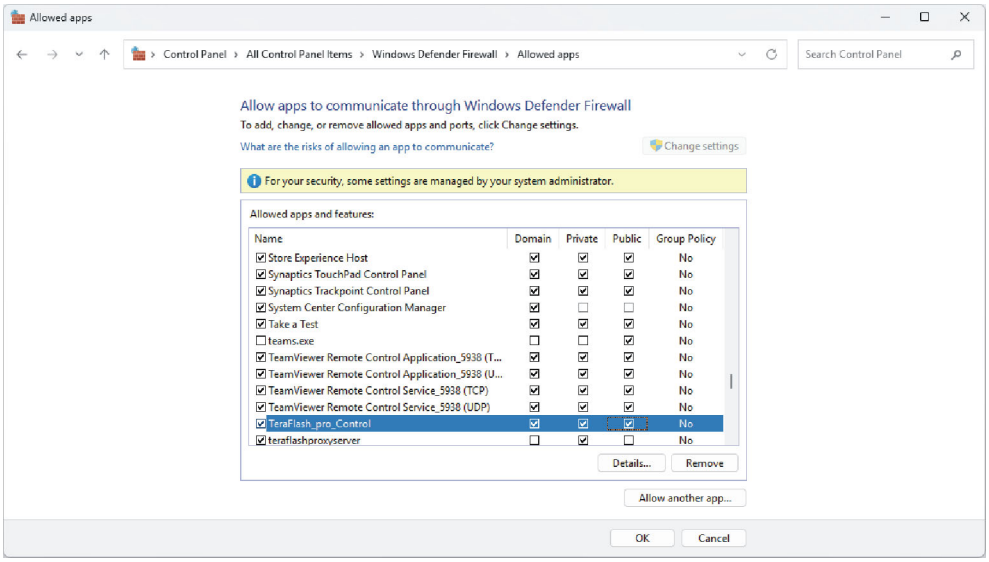


Figure 30 Granting access to TeraFlash_pro_Control

5. Start the TERAFLASH pro Control Software. When asked for granting private and public access during the first start of the software, click on **Allow**. In the Control tab of the System Control Panel (see section 5.2.1), click the **Connect** button to establish a connection to the TERAFLASH pro Main Unit. Wait until the status message "Connected" is displayed next to the button.

6. Should the Connection failed window appear (see Figure 31), verify that the green System Ready LED on the front panel of the TERAFLASH pro Main Unit is continuously illuminated. Further, check the Ethernet cable connection and ensure that the TCP/IPv4 settings (see Figure 21) are configured correctly. Also check the IP address settings of the TERAFLASH pro (see Figure 21 and Figure 48), as well as the Firewall Settings.

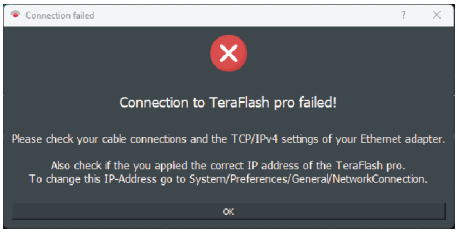


Figure 31 Connection failed window

5.2 User Interface and Software Functions

The TERAFLASH pro Control Software enables convenient system operation via a graphical user interface (see Figure 32).

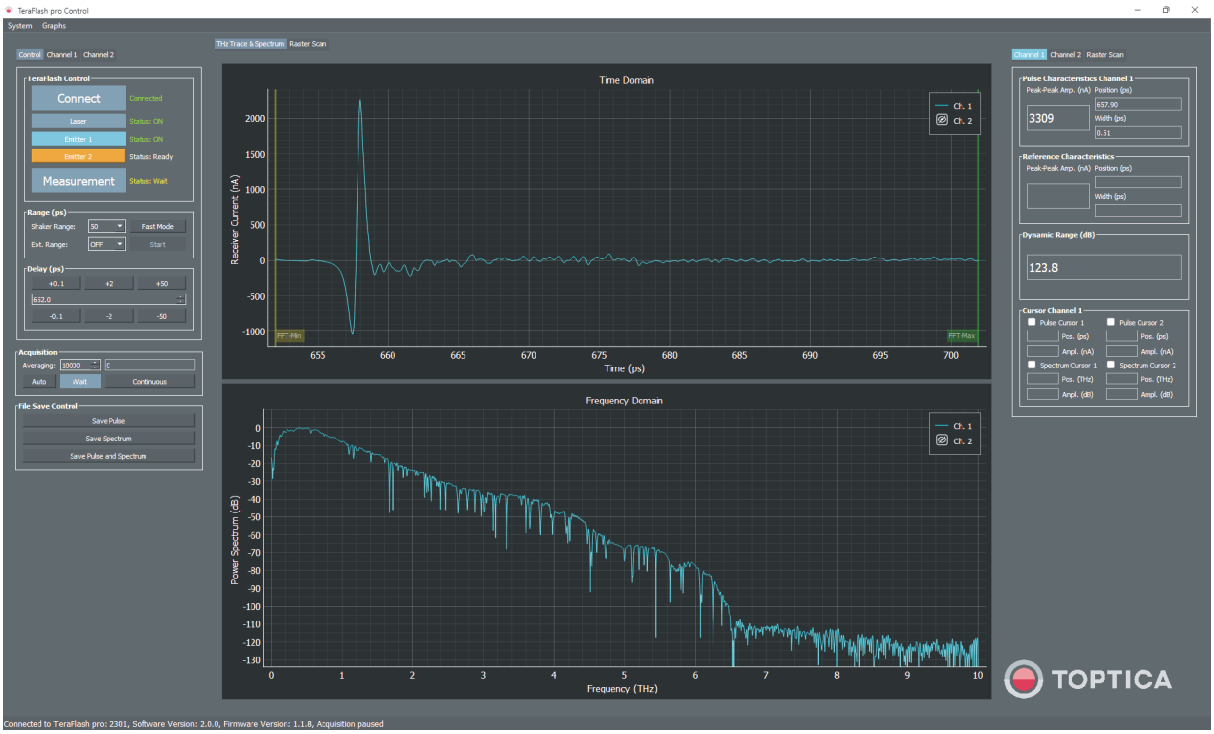


Figure 32 Main window of the TERAFLASH pro Control Software

On the left-hand side, the System Control panel (tabs "Control", "Channel 1", "Channel 2" + boxes "Acquisition" and "File Save Control") comprise all the parameters and settings to acquire and save terahertz data.

At the center, two panels depict the pulse trace and spectrum, respectively. The Raster Scan tab displays a terahertz imaging measurement if one of TOPTICA's Terahertz Imaging Extensions is used.

On the right-hand side, the Analysis panel (tabs "Channel 1", "Channel 2" and "Raster Scan") provides information on time-domain and frequency-domain properties of a measurement, such as pulse amplitude, pulse position, spectral peak dynamic range, and Cursor settings. The Raster Scan tab allows to access the control parameters of the Terahertz Imaging Extensions.

In the top-left corner, two menus "System" and "Graphs" include general settings for the system configuration and data display.

5.2.1 System Control Panel, Control Tab

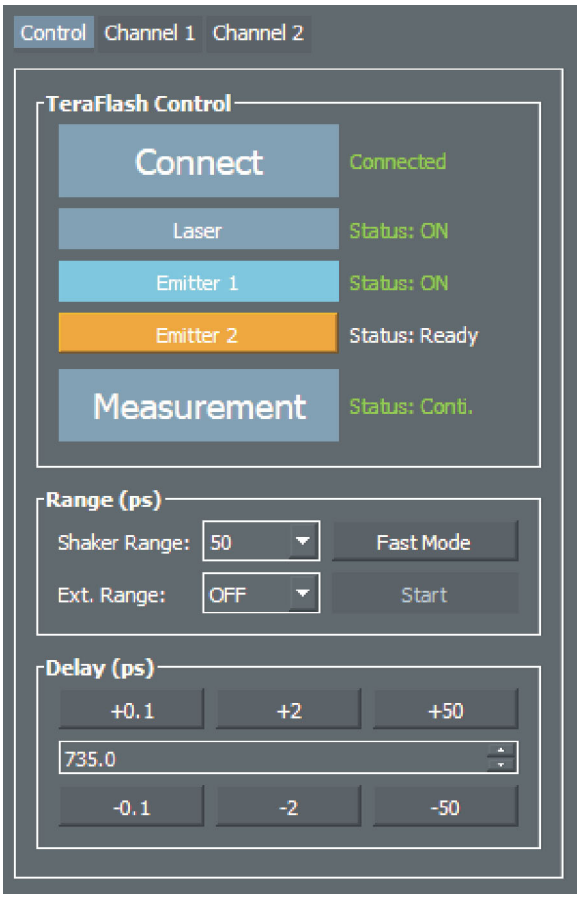


Figure 33 System Control panel, Control tab

TeraFlash Control Section

Connect — Click the **Connect** button to establish a connection to the TERAFLASH pro. If successful, the status message briefly changes from *Disconnected* to *Connecting* and then to *Connected*.
At this stage, the status of the **Laser** changes from *OFF* to *Warming up* and then to *Ready*.

NOTE !	When powering-down the system, you can click **Connect** once again in order to close the connection. This is not mandatory, you may as well just close the TERAFLASH pro Control Software.

Laser Click the **Laser** button to activate laser emission.

CAUTION !	After switching the TERAFLASH pro on, please wait for at least 70 seconds before activating laser emission.

Confirm by selecting **OK** once the Laser Emission Alert window appears (see Figure 15). When the laser is switched on, the **Laser** status message changes from *Ready* to *ON*, and the Laser Emission Warning LEDs at the TERAFLASH pro Main Unit front panel light up. At this stage, the **Emitter 1 and 2** status messages change from *OFF* to *Wait* and then to *Ready*. Clicking the **Laser** button again disables laser emission.

Emitter 1 Click the **Emitter 1** button to apply a bias voltage to the emitter connected to channel 1. The **Emitter 1** status message changes from *Ready* to *ON*. Clicking the **Emitter 1** button again switches the bias voltage off.

Emitter 2 Same function as above, but for channel 2.

NOTE !	If you have two emitters connected, you can choose to operate either one at a time, or both emitters simultaneously.

Measurement Click the **Measurement** button to start the movement of the voice-coil delay stage and initiate a terahertz measurement. The **Measurement** status changes from *OFF* to *Busy* and then to *Cont*. (= Continuous).
The audible movement indicates the delay line positioning itself at the desired position. After a short moment, real-time data is displayed on the central graphs. Clicking the **Measurement** button again stops the measurement.

Range (ps) Section

Shaker Range Defines the scan length of the shaker (i.e., voice-coil delay stage) in picoseconds. The drop-down menu offers settings from 5 ps to 200 ps.

NOTE !	After changing the **Shaker Range** setting, the **Measurement** status (see above) might change to *Busy* for several seconds.

Extended Range This drop-down menu enables the acquisition of scans of up to 3000 ps length. The maximum useable scan length is calculated automatically, taking the position of the delay stage into account:
Max. scan range = 4000 ps - (Position of delay stage).
Once a scan range is selected, the **Start** button on the right-hand side is enabled.

Start Click **Start** to initiate the **Extended Range** measurement. A progress bar displays the status of the measurement. While a measurement is ongoing, the **Start** button changes to **Abort**. Clicking **Abort** terminates the measurement.

Fast Mode — The shaker of the TERAFLASH pro operates in two different modes, which can be selected via the **Fast Mode** button:
Fast Mode inactive (button not clicked): Standard operation mode; the shaker moves at normal speed, enabling the use of all Range options.
Fast Mode active (button clicked): Fast operation mode; the shaker moves at a higher speed.

NOTE !	The fast operation mode is only compatible with **Shaker Range** settings up to 70 ps.

The following table lists the measurement speed for different **Shaker Range** settings.

Shaker Range (ps)	**Fast Mode** deactivated (regular speed): traces/s	**Fast Mode** activated: traces/s
5	98	166
10	89	135
15	64	104
20	39	95
35	25	68
50	21	59
70	16	37
100	11	
120	10	
150	8	
200	6	

Table 1 Measurement speed in traces per second, for different **Shaker Range** settings.

Delay (ps) Section

The stationary delay stage compensates changes in the length of the terahertz beam path (see section 1.4.2). The selected value (in ps) indicates the starting position of the acquired pulse trace.

+0.1 and **-0.1** — Changes the delay position in steps of 0.1 ps.

+2 and **-2** — Changes the delay position in steps of 2 ps.

+50 and **-50** — Changes the delay position in steps of 50 ps.

Numerical Control (spinbox) — Permits a direct, numerical input of the desired delay position.

5.2.2 System Control Panel, Acquisition Section

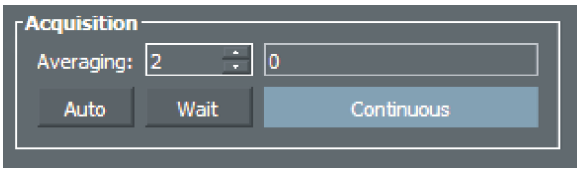


Figure 34 System Control panel, Acquisition section

Up to 599,999 averages can be applied to the pulse trace (see Figure 34). Once the desired number of averages has been completed, the software uses a "sliding mean" where old pulse traces are replaced by new ones.

Averaging (spinbox) — Number of pulse traces to be averaged. To start the integration, simply type the number of measurements to be averaged in the input field and confirm by pressing the Enter button of the PC. Alternatively, you can use the arrow keys on the right-hand side of the input field. The display on the right shows a "countdown" that indicates the remaining number of averages to be completed.

Auto — Clicking the **Auto** button resets the pulse trace averaging and initiates a new "countdown" of the number of averages. The data acquisition stops automatically, activating **Wait**, once pulse trace averaging has been completed. This mode is recommended for an individual measurement with a pre-defined number of averages.

Wait — Clicking the **Wait** button temporarily pauses the continuous data stream, preserving the data that is momentarily displayed. The **Save** buttons are activated (see File Save Control section below).

Continuous — When activated, continuous live data is displayed in the central plots (**Pulse Trace Display** and **Spectrum Display**).

5.2.3 System Control Panel, File Save Control Section

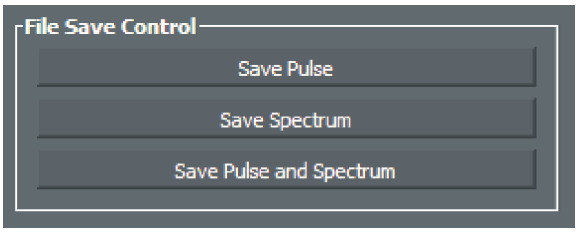


Figure 35 System Control panel, File Save Control section

Once data acquisition is paused, you can save the displayed data to hard disk. After clicking any of the **Save** buttons, a window opens where path and file name can be selected. File name extensions are added automatically.

Save Pulse	Saves pulse trace data (upper panel of the central plots).
Save Spectrum	Saves the displayed spectrum (lower panel of the central plots).
Save Pulse and Spectrum	Option to save both the pulse trace and the spectrum data.

For details on the saved data, please refer to section 8.6.

Once a file is saved, a Save Complete dialog window opens (see Figure 36).

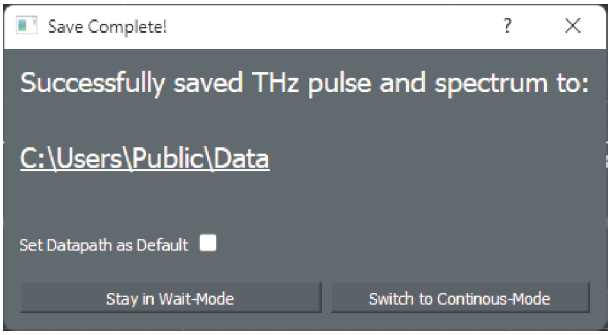


Figure 36 Save Complete dialog window

Set Datapath as Default	Check the box to use the data path for future files.
Stay in Wait-Mode	Click to return to the main window, measurement remains paused.
Switch to Continuous-Mode	Click to return to the main window, measurement continues.

5.2.4 System Control Panel, Channel 1 Tab

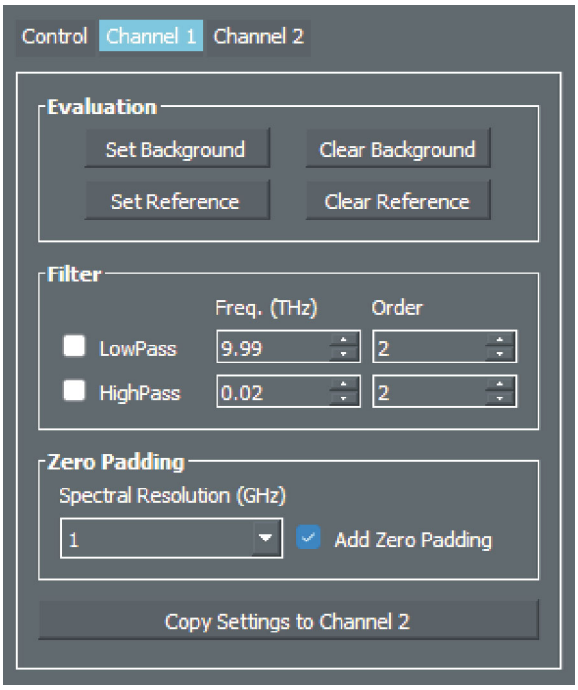


Figure 37 System Control panel, Channel 1 tab

Evaluation Section Click the respective button to execute the desired function.

Set Background Captures the currently displayed pulse trace as background signal, which is subtracted from all subsequently acquired pulse traces. This is useful in case of a disturbing background, e.g. due to cross-talk in a reflection setup. The button turns grey when clicked.

Clear Background Clears the background signal and stops background subtraction.

Set Reference Captures the currently displayed pulse traces together with the displayed terahertz spectrum as a reference dataset, along with the values for intensity, peak position and pulse width. For the display of reference curves and the ratio of signal and reference spectrum, please refer to section 5.2.12. The button turns grey when clicked.

Clear Reference Deletes the reference dataset.

Filter Section

Applies a filter to the displayed data. There are three filter options: low-pass, high-pass and band pass. A bandpass filter is obtained by activating both the HighPass and LowPass checkboxes. Alternatively, you can choose to use solely a low-pass or a high-pass filter, or no filter at all.

LowPass, HighPass

Check the respective box to activate a low-pass filter or a high-pass filter, respectively.

Freq. (THz)

Defines the cutoff frequencies of the low-pass and high-pass filters. You can either enter a value here, or drag the cursors in the spectrum display by clicking on the respective lines.
For the low-pass filter, frequencies below the cutoff are passed and frequencies above this limit are rejected.
For the high-pass filter, frequencies above the cutoff are passed and frequencies below this limit are rejected.

NOTE !	Applying a filter to the spectrum will also change the appearance of the terahertz pulse trace.

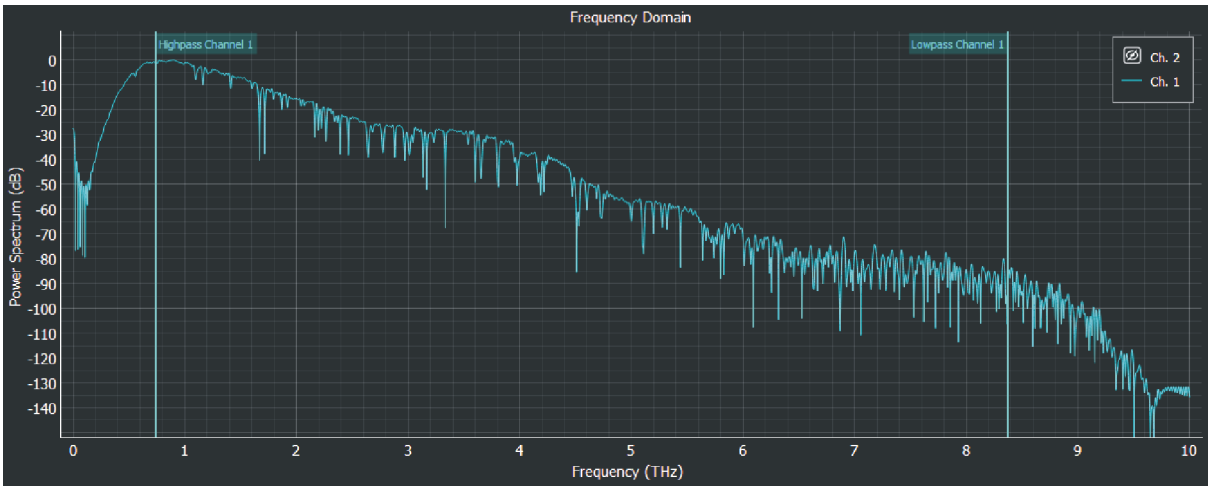


Figure 38 Spectrum display with both high-pass and low-pass filters activated.

Order

Order of the filter. A higher order filter means a steeper roll-off. The maximum order is 16.

Zero Padding Section

Sets the spectral resolution in GHz by applying zero padding to the time-trace signal.

Spectral Resolution (GHz)

Frequency resolution of the spectrum, in GHz. A frequency resolution of 1 GHz, 2 GHz or 5 GHz can be selected via the drop-down menu. Zero-filling is applied accordingly in the FFT analysis.

Add Zero Padding

Check/uncheck the box to toggle zero-padding of the time-trace data. If no zero padding is applied, the FFT analysis uses the "natural" resolution of the selected scan range.

Copy Settings to Channel 2

Click to apply the settings made in the Channel 1 tab to channel 2.

5.2.5 System Control Panel, Channel 2 Tab

The control options for Channel 2 are identical to those of Channel 1. Please refer to the descriptions in section 5.2.4.

5.2.6 Central Panels, THz Trace & Spectrum Display

The two central panels of this display depict the pulse trace (time-domain representation, upper plot) and the corresponding spectrum (frequency-domain representation, lower plot). The spectrum is obtained by applying a fast Fourier Transform (FFT) to the pulse trace.

Pulse Trace Display

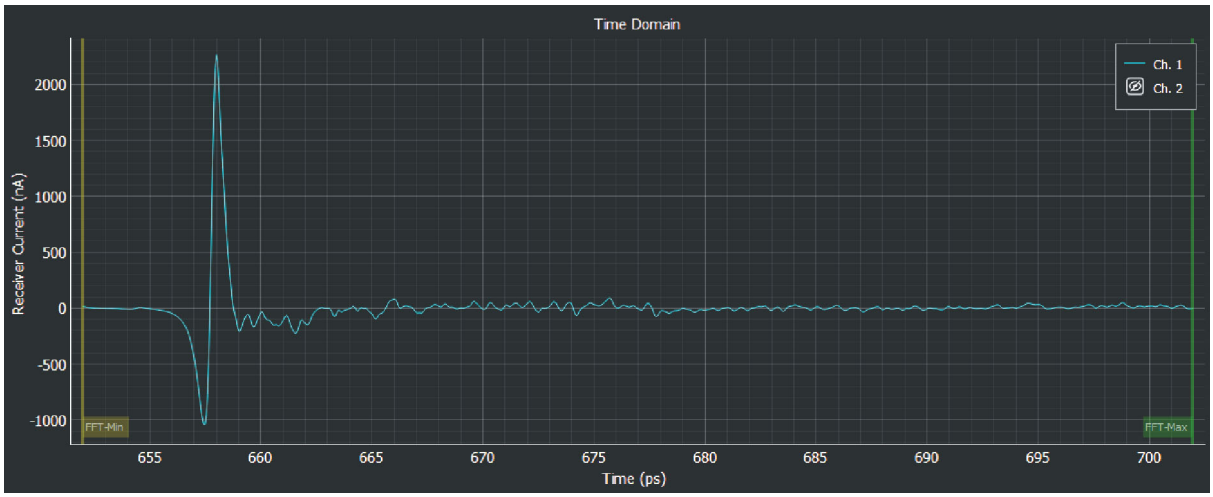


Figure 39 Central panel, pulse trace display

Ch. 1, Ch. 2
You can activate or deactivate the pulse data of Channel 1 (Ch. 1) and Channel 2 (Ch. 2) by clicking on the respective symbol on the left side of the legend, as shown in Figure 39. This feature allows you to view either only one or a few selected traces at a time.
Further display options are accessible in the Graphs menu (see section 5.2.12). The spectrum is obtained by applying a fast Fourier Transform (FFT) to the pulse trace.

FFT-Min (yellow)
Cursor to select the beginning of the time window for the FFT analysis.

FFT-Max (green)
Cursor to select the end of the time window for the FFT analysis.

Spectrum Display

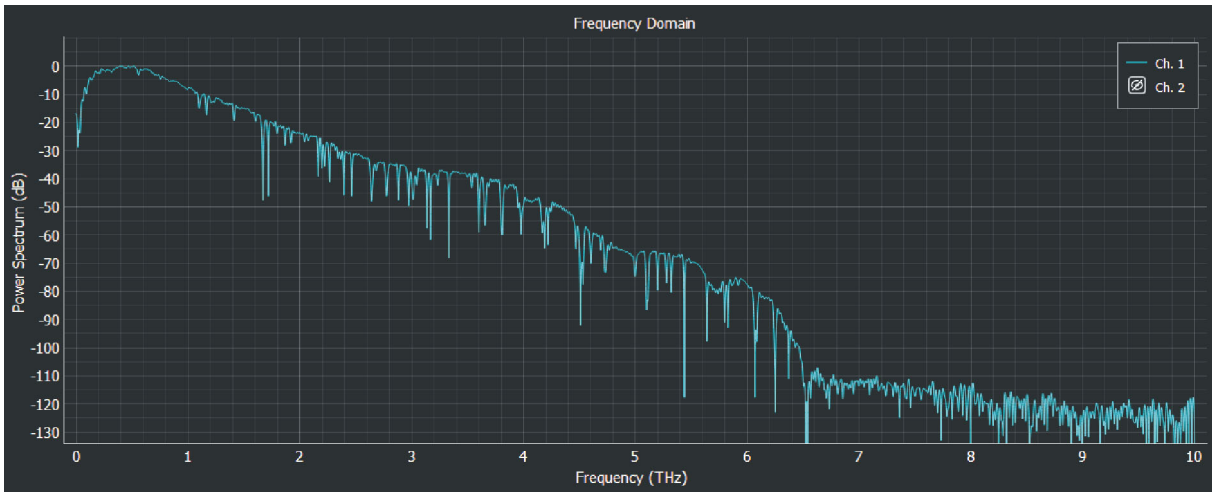


Figure 40 Central panel, spectrum display

Ch. 1, Ch. 2 As for the pulse trace display, you can activate or deactivate the display of the spectra of Channel 1 (Ch. 1) and Channel 2 (Ch. 2) by clicking on the respective symbol on the left side of the legend.
Further display options are accessible in the Graphs menu (see section 5.2.12).

Mouse-Based Display Controls for Pulse Trace and Spectrum Display

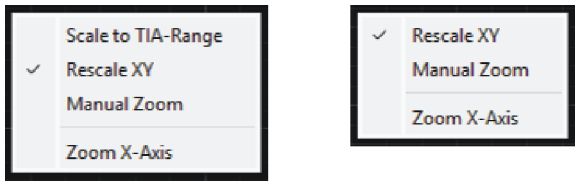


Figure 41 Display controls for pulse trace graph (left) and spectrum graph (right)

The following mouse interactions are accessible via a right mouse click the respective panel:

Scale to TIA-Range When activated, the y axis of the pulse trace graph is rescaled to match the selected range setting of the transimpedance amplifier (TIA, see section 5.2.11).

Rescale XY Rescales x and y axis when activated.

Manual Zoom Disables auto-scaling of the plots. The Manual Zoom Mode is automatically activated when you zoom into the plot.

Zoom X-Axis Zooms into the x axis of the respective plot (zoom factor is 3.33).

Once you have zoomed into a section of a graph, you can change (drag) the view by holding the left mouse button pressed down.

NOTE ! In order to zoom, you can
- Use the mouse wheel,
- Select a rectangular zoom box by holding the central mouse button pressed down,
- "Drag-zoom": Hold down the right mouse button, then drag the mouse to define an area.

5.2.7 Central Panels, Raster Scan Display

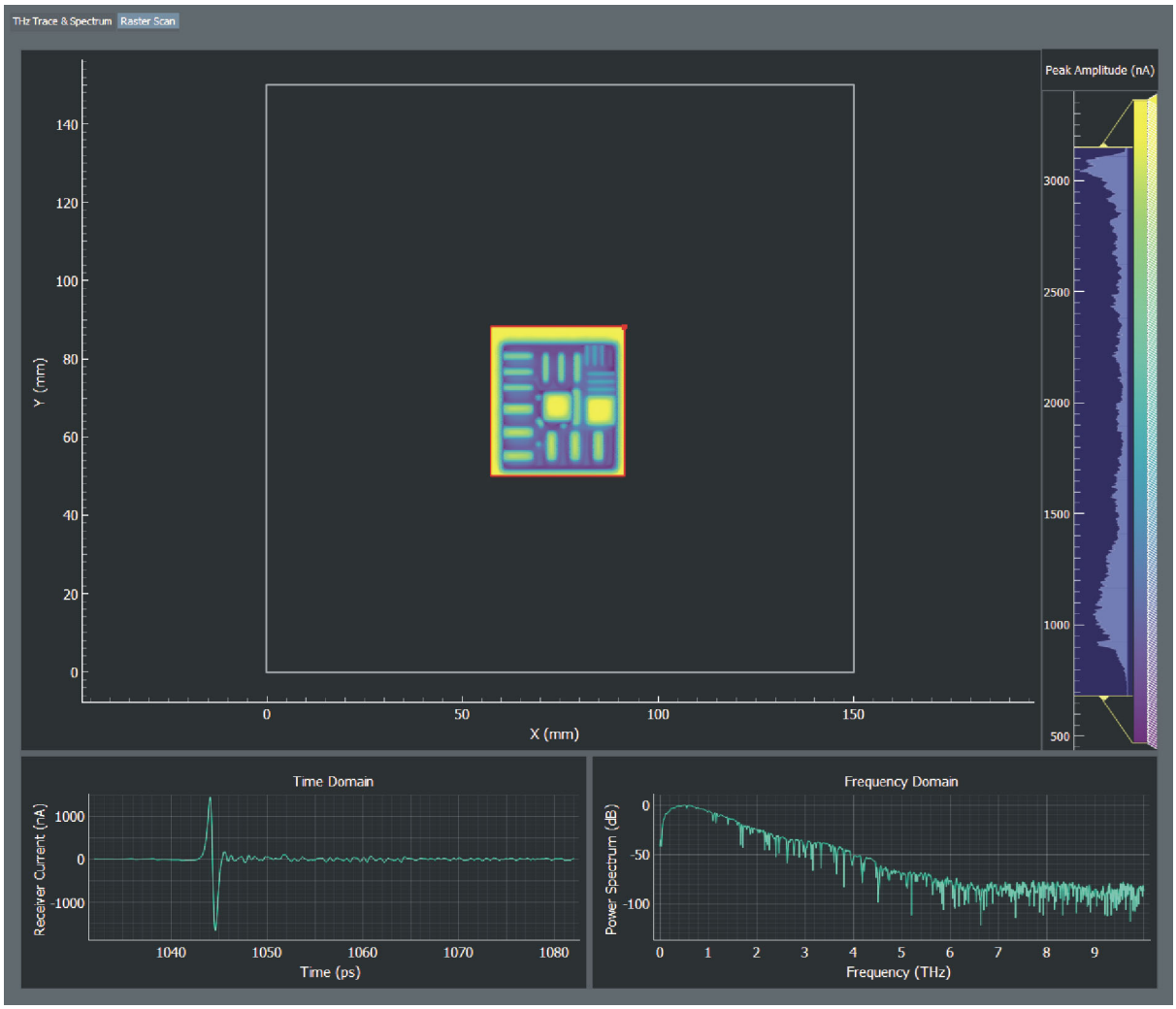


Figure 42 Central panel, Raster Scan display

This display visualizes a THz image while it is being recorded. The control parameters are accessible in the Raster Scan tab on the right-hand side (see section 5.2.10).

The upper panel shows an xy map of the THz image, with dimensions given in millimeters. Either the peak amplitude or the peak position can be selected as contrast parameters (see section 5.2.10). The color scale with corresponding units is displayed on the right-hand side of the plot. Once the scan is completed, the upper and lower limits of the color scale can be changed by dragging the yellow cursors next to the color bar up or down. A histogram next to the color scale visualizes the distribution of the numerical values displayed in the THz image.

The two lower panels show the pulse trace and spectrum of the pixel that is being imaged.

NOTE !	If no Imaging Extension is connected, this tab displays a dummy image along with pre-recorded data of pulse trace and spectrum.

5.2.8 Analysis Panel, Channel 1 Tab

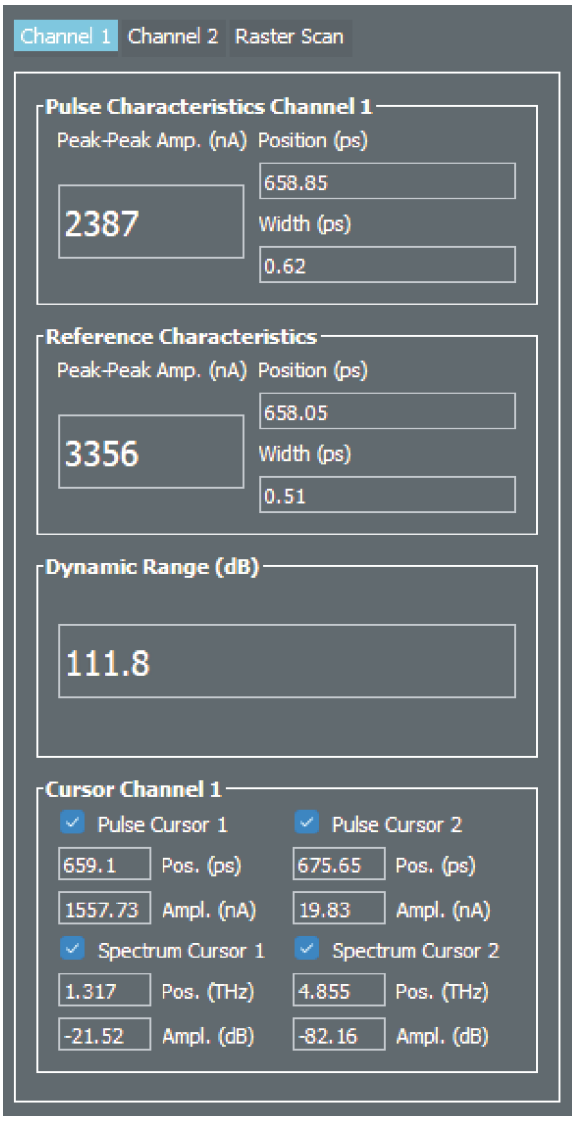


Figure 43 Analysis Panel, Channel 1 tab

Pulse Characteristics Channel 1 Section

Peak-Peak Amp. (nA) Indicates the peak-to-peak amplitude value of the main terahertz pulse, in nanoampere (nA). Observing this value is useful for the alignment of the terahertz beam path.

Position (ps) Position of the main terahertz pulse in the signal pulse trace, in picoseconds (ps).

Width (ps) Halfwidth of the main terahertz pulse in the signal pulse trace, in picoseconds (ps).

Reference Characteristics Section When a reference dataset is captured (see section 5.2.4), its properties are displayed in the fields Peak-Peak Amp. (nA), Position (ps) and Width (ps). Please refer to the previous section "Pulse Characteristics Channel 1" for descriptions.

Dynamic Range (dB) Section Displays the peak dynamic range of the spectrum, i.e. the ratio of the maximum signal in the spectrum to the noise floor. The noise floor is calculated from the RMS value of the signal between 9 THz and 10 THz.

NOTE !	The display of the peak dynamic range is deactivated when a filter is applied to the spectrum (see section 5.2.4).

Cursor Channel 1 Section Up to two cursors can be activated in each of the panels, i.e. in the pulse trace and spectrum, respectively. They facilitate the analysis of features observed in either panel.

Pulse Cursor 1, Pulse Cursor 2 Check/uncheck the boxes to activate/deactivate the first and second cursor displayed in the pulse trace.

Pos. (ps) Momentary position of the first (second) cursor on the abscissa, in ps.

Ampl. (nA) Momentary position of the first (second) cursor on the ordinate, in nA.

Spectrum Cursor 1, Spectrum Cursor 2 Check/uncheck the boxes to activate/deactivate the first and second cursor displayed in the spectrum graph.

Pos. (THz) Momentary position of the first (second) cursor on the abscissa, in THz.

Ampl. (dB) Momentary position of the first (second) cursor on the abscissa, in dB.

NOTE !	If the scale of the abscissa is changed from logarithmic to linear, then the amplitude values are displayed in arbitrary units.

5.2.9 Analysis Panel, Channel 2 Tab

The control options for Channel 2 are identical to those of Channel 1. Please refer to the descriptions in section 5.2.8.

5.2.10 Analysis Panel, Raster Scan Tab

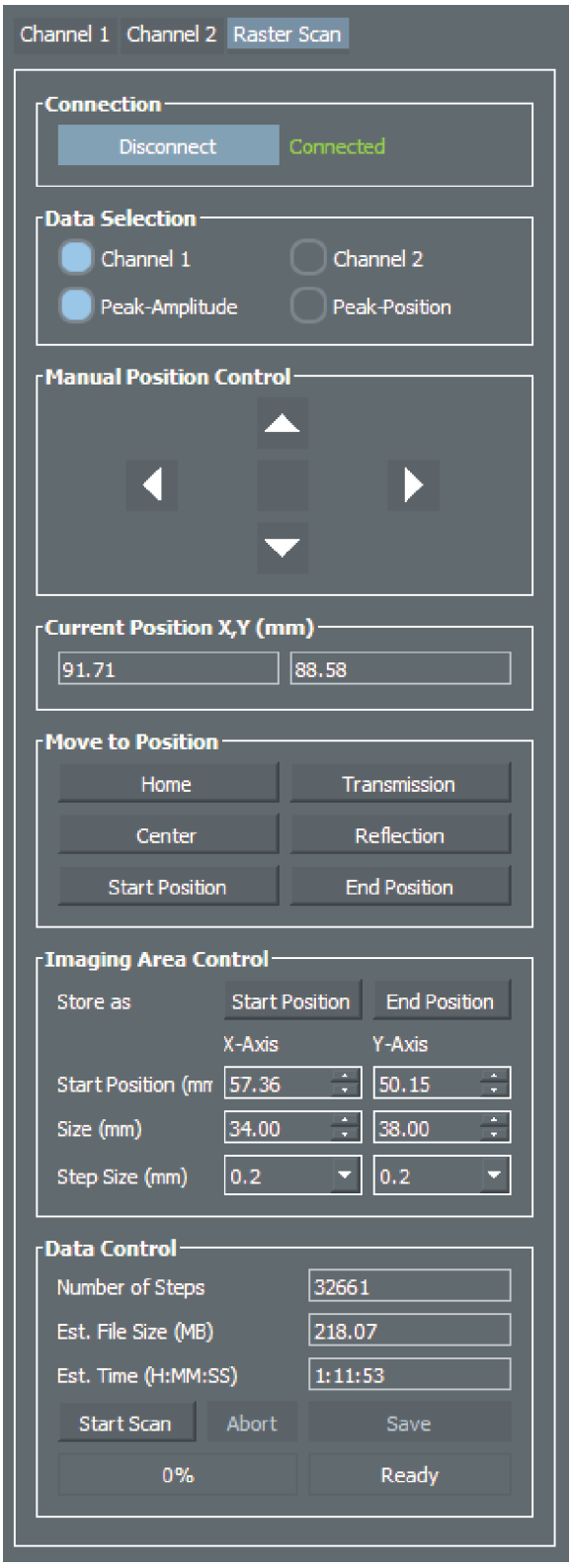


Figure 44 Analysis Panel, Raster Scan tab

This tab provides the control parameters for THz imaging.

NOTE !	The features of this tab become available once an Imaging Extension is connected.

Connection Section

Connect	Establishes a connection to the Imaging Extension. The status message changes from *Disconnected* to *Connected* and the button changes to **Disconnect**. Clicking **Disconnect** then terminates the connection.

Data Selection Section	Selects the data to be displayed in the Raster Scan display (see section 5.2.7).
Channel 1, Channel 2	Selection of the channel to which the receiver antenna of the Imaging Extension is connected.
Peak Amplitude, Peak Position	Selection of the contrast parameter of the xy map.
Manual Position Control Section	The four arrow keys enable a manual movement of the translation stages of the THz Imaging Extension.

NOTE !	The translation stages can also be moved via the rotary knobs at the Scanner Control Unit shown in Figure 56.

Current Position X,Y (mm) Section	The two boxes display the actual position of the stages in X and Y, respectively. The position is also displayed by a red cross in the xy map of the central panel.
Move to Position Section	Click the respective button to execute the desired function.
Home	Moves the stages to the "home" position, as defined in Preferences > Scanner tab (see Figure 53).
Center	Moves the stages to the center of the image. The image size is defined in the Imaging Area Control section (see below).
Start Position, End Position	Moves the stages to the selected start or end position of the image, as defined in the Imaging Area Control section.
Transmission, Reflection	Moves the stages to the transmission and reflection positions used for aligning the optics of the Imaging Extension. These positions are defined in Preferences > Scanner tab (see Figure 53).

Imaging Area Control Section

Store as Start Position/	Defines the actual position of the stages as **Start Position** or **End Position** of the image, respectively.

NOTE !	With respect to the representation on the central panel, the **Start Position** has to be in the bottom left corner of the area to be imaged. The **End Position** is in the top right corner. Otherwise, the imaging measurement may not start.

Start Position (mm)	Manual input fields of the X/Y coordinates of the THz image.
Size (mm)	The two input fields define the size of the image in X (left-right) and Y (up-down) directions, in units of mm. Values can be entered manually, or are computed automatically from the selected **Start Position** and **End Position**.
Step Size (mm)	The two input fields define the step size in X and Y, in units of mm. Values between 0.01 mm and 5 mm are selected from a drop-down list.

Data Control Section

Number of Steps	Indicates the number of pixels in the image.
Est. File Size (MB)	Estimated file size of the THz image, in MB.
Est. Time	Estimated remaining acquisition time, in hours : minutes : seconds.
Start Scan	Click to move the stages to the **Start Position** and initiate an imaging measurement.
Abort	Click to stop an ongoing imaging measurement.
Save	Click to open a dialog window to select a file name and destination.
Progress bar	Shows the status of the image as it is being completed (0 % to 100 %).

NOTE !	You can choose to record an image with or without trace averaging. The number of trace averages selected in the System Control Panel, Acquisition section (see section 5.2.2) is applied to each image pixel.

NOTE !	The RAM usage during the recording of an image depends on the number of pixels the image will consist of. If the estimated RAM usage is higher than the recommended RAM usage of 6 GB, the warning shown in Figure 45 is displayed before starting the measurement.

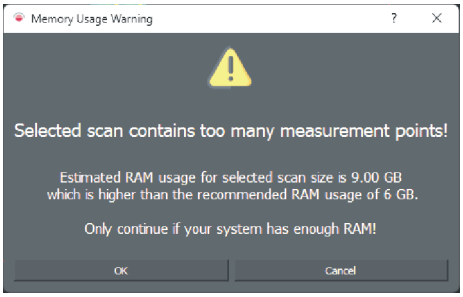


Figure 45 Memory usage warning

5.2.11 Menu System

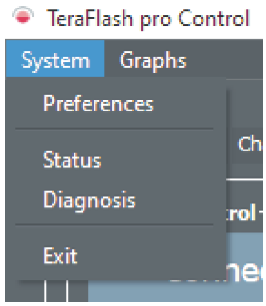


Figure 46 Menu System

System > Preferences Opens the Preferences window with various tabs to configure basic system settings, such as IP address, transimpedance amplifier gain and scanner settings of an Imaging Extension. You can also change software settings, e.g. default file paths.

System > Preferences > System Tab

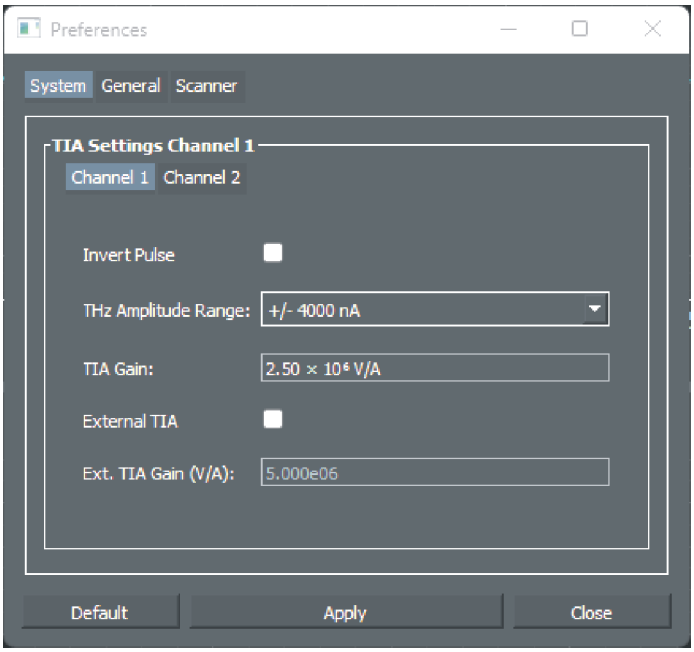


Figure 47 Preferences window, System tab

TIA Settings Channel 1/2 Section	Settings of the transimpedance amplifier (TIA) integrated in the TERAFLASH pro.
Channel 1/Channel 2	Select settings for channel 1 or channel 2, respectively.
Invert Pulse	Check the box to invert the signal in the pulse-trace display.
THz Amplitude Range	Drop-down menu to select the maximum allowed amplitude range of the THz pulse.You can select three different settings between +/- 140 nA and +/- 4000 nA.
TIA Gain	Displays the gain factor of the TIA, in V/A. Values vary from 2.50×10^6 V/A to 7.14×10^7 V/A, depending on the selected **THz Amplitude Range**.
External TIA	Check the box to enable the use of an external TIA, shortcutting the internal one.
Ext. TIA Gain (V/A)	Input field for the gain factor of an external TIA, if used (see checkbox **External TIA**).
Default	Click to restore the default settings.
Apply	Click to save any changes made in this tab.
Close	Click to close this menu.

System > Preferences > General Tab

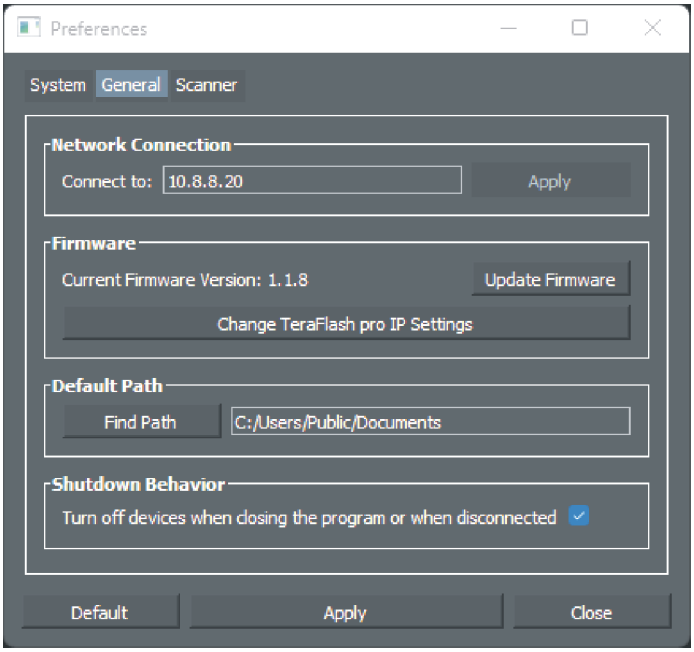


Figure 48 Preferences window, General tab

The General tab allows to change some of the fundamental system settings, including the IP address configuration of the TERAFLASH pro and the firmware version.

Network Connection Section

NOTE !	The default IP address of the TERAFLASH pro Main Unit is 10.8.8.20. The subnet mask is 255.255.255.0 and the Gateway setting is 10.8.8.1. The recommended setting for the USB-Ethernet adapter is 10.8.8.10.

Connect to — Input field for the IP address of the TERAFLASH pro Main Unit. You only need to modify this setting in case you change the IP address of the system (see **Change TeraFlash pro IP Settings** below).

Apply — Applies the new settings.

Firmware Section

Current Firmware Version — Display of the firmware currently installed on the system.

Update Firmware — Opens a dialog window for executing a firmware update. For details, please refer to section 8.7.

NOTE !	A firmware update will restore the default network connection settings.

Change TeraFlash pro IP Settings	Option to assign a different IP address to the TERAFLASH pro Main Unit. Afterwards, you need to restart the system.

CAUTION ! If you use an invalid IP address, you may not be able to connect to the system any longer. Should this have happened, please contact TOPTICA Photonics.

In order to change the IP address, please follow the steps below:

1. Click **Change TeraFlash pro IP Settings.**

2. A Caution window opens (see Figure 49), warning you that an incorrect setting can transfer the TERAFLASH pro system into an "unreachable" state. Confirm by clicking **OK** or **Cancel** to abort.

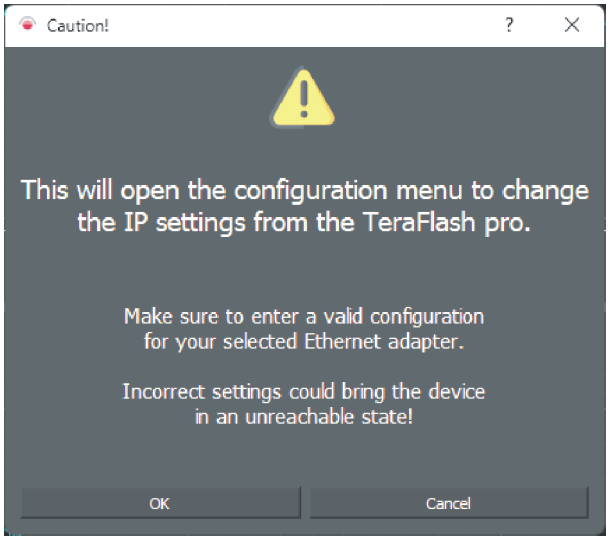


Figure 49 Caution window before change of IP address

3. In the next window, enter the new network settings. Confirm by clicking **Apply** or **Cancel** to abort.

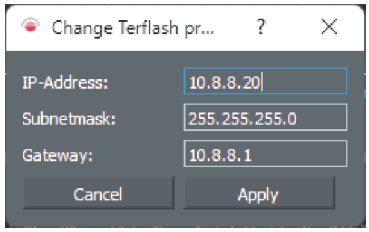


Figure 50 Window to enter the new network settings

4. You are once again warned that you are about to change the network settings. Confirm by clicking **OK** or **Cancel** to abort.

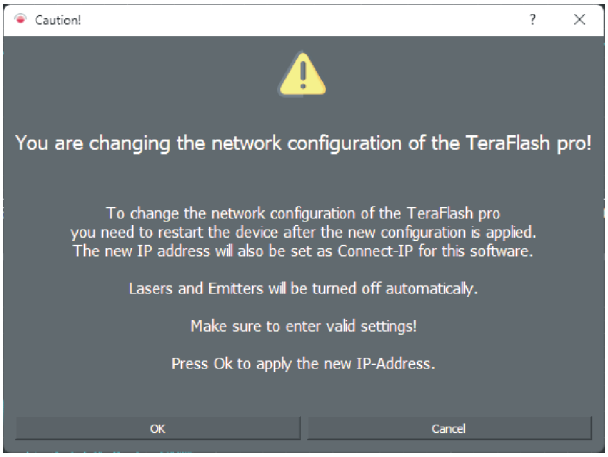


Figure 51 Caution window before change of IP address

4. The next window confirms that the configuration was changed successfully. Please restart the system to apply the new settings. Confirm by clicking **OK**.

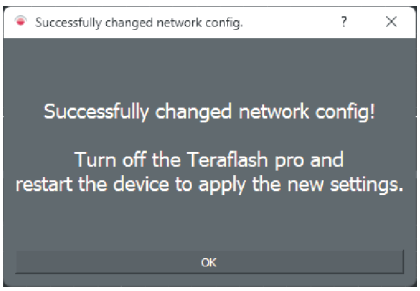


Figure 52 Confirmation window

Default Path Section

Find Path	Opens a dialog window to select the default path for storing measurement data.

Shutdown Behavior Section

Turn off devices when closing the program...	Check the box to shut bias voltage(s) and laser down when the software is closed.
Default	Click to restore the default settings.
Apply	Click to save any changes made in this tab.
Close	Click to close this menu.

System > Preferences > Scanner Tab

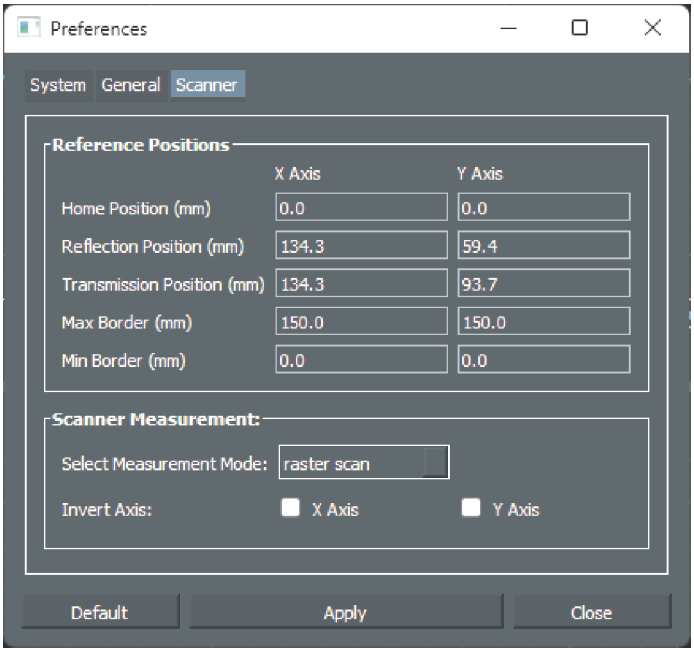


Figure 53 Preferences window, Scanner tab

Reference Positions Section For TERAFLASH pro systems with Imaging Extension, the settings in this tab define the default positions of the linear X/Y stages.

Home Position (mm) X/Y coordinates of the home position in THz imaging.

Reflection Position (mm) (Approximate) X/Y coordinates of the reflection target, i.e. the feature of the structured aluminum plate used for aligning the optics of the Imaging Extension in reflection (see section 6.1.2).

Transmission Position (mm) (Approximate) X/Y coordinates of the pinhole of the structured aluminum plate that is used for aligning the optics of the Imaging Extension in transmission (see section 6.1.2).

Max Border (mm) Maximum X/Y coordinates of the field of view.

Min Border (mm) Minimum X/Y coordinates of the field of view.

Scanner Measurement Section

Select Measurement Mode Defines the movement of the X/Y stages. Via the drop-down menu, you can select the options "raster scan" (unidirectional scan) and "traversal scan" (bidirectional or "zigzag" scan).

Invert Axis Checkboxes to invert the X and Y axis, respectively.

NOTE !	This option should only be used for customized imaging setups, but not for the Complete Imaging Extension as described in section 6.1.

Default Click to restore the default settings.

Apply Click to save any changes made in this tab.

Close Click to close this menu.

System > Status

Opens the System Status window that displays information about the connected TERAFLASH pro. If no system is connected and this line is clicked, an error message appears.

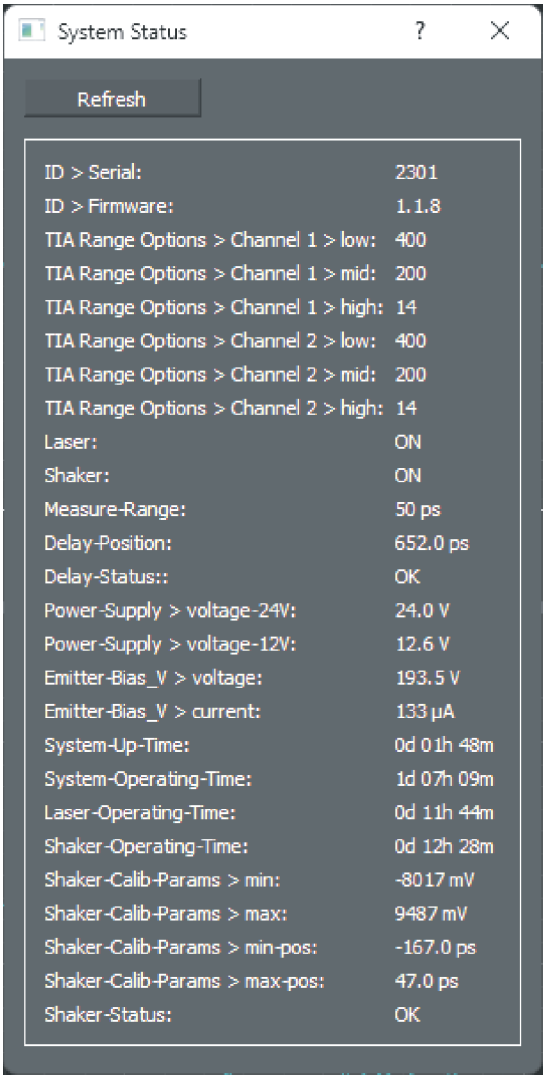


Figure 54 System Status window

Refresh

When activated, the displayed parameters are measured continuously.

System > Diagnosis

Opens the Diagnosis window to facilitate troubleshooting. Via this window, a self-test of the system can be initiated, checking the delay stages and the terahertz antennas.
Please refer to section 8.8 for a detailed description.

System > Exit

Closes the TERAFLASH pro Control Software.

5.2.12 Menu Graphs

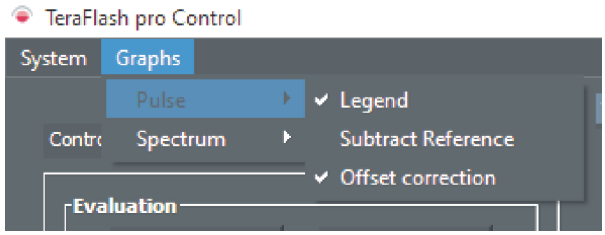


Figure 55 Menu Graphs (example)

This menu defines the display options of the two central panels, the pulse trace (Pulse settings) and the spectrum (Spectrum settings).

Graphs > Pulse > Legend	Activates or deactivates the display of the legend in the pulse trace display.
Graphs > Pulse > Subtract Reference	Subtracts a reference measurement (see section 5.2.4) from the displayed pulse trace.
Graphs > Pulse > Offset Correction	Applies an offset shift to the pulse trace, excluding low-frequency components from the signal. When activated, the offset is subtracted automatically, so that the signal level is zero at the beginning of the pulse trace.
Graphs > Spectrum > Legend	Activates or deactivates the display of the legend in the spectrum display.
Graphs > Spectrum > Phase	Activates of deactivates the display of the phase spectrum.
Graphs > Spectrum > Unwrap Phase	Activates of deactivates the display of the unwrapped phase spectrum, i.e. corrected for 2π repetitions of the phase.
Graphs > Spectrum > Divide by Reference	Activates/deactivates the display of a normalized spectrum, i.e. the ratio of the signal spectrum divided by the reference spectrum.
Graphs > Spectrum > Log Scale	When activated, the vertical axis changes from a linear scale to a logarithmic scale.
Graphs > Spectrum > Waterlines	Enables a visualization of the rotational absorption lines of water vapor. This feature is useful when conducting measurements in air, and can be activated in the spectrum graph.
Graphs > Spectrum > Normalize	With normalization activated, the peak of the spectrum is set to 0 dB.

6 Imaging Extension

The Imaging Extensions are available in two different versions. The Complete Imaging Extension scans a sample through the terahertz beam. It comprises two precise linear stages with motion controllers, mounts for the terahertz antennas and four parabolic mirrors for beam shaping and focusing. The optical elements can be arranged for terahertz imaging in transmission or reflection; adjustment pins enable a quick and easy reconfiguration from one geometry to the other.

The Flexible Imaging Extension consists of a compact reflection head that scans across static samples. This assembly is mounted on an aluminum frame, which can be positioned horizontally, vertically or at any oblique angle.

Electric connections as well as the control software are identical for both versions. Specifications are listed in section 8.2.

6.1 Complete Imaging Extension

6.1.1 Connections

NOTE !	In the following installation guidelines, it is assumed that the TERAFLASH pro Main Unit is already set up and operational. If this is not the case, please follow the instructions as described in sections 4 and 5 first.

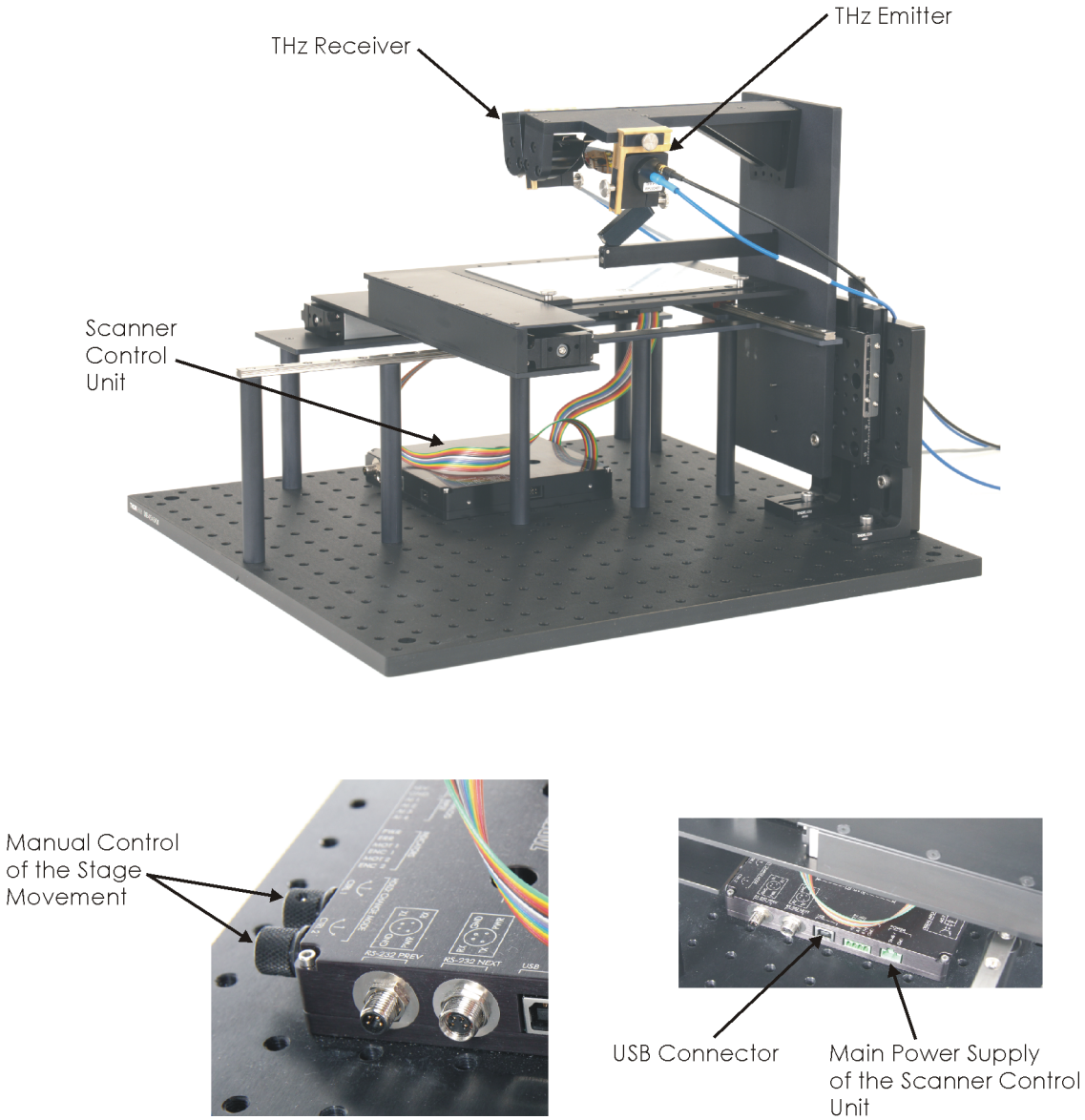


Figure 56 Top view of the Complete Imaging Extension, configured for reflection measurements.

1. Make sure that the TERAFLASH pro Main Unit is switched off.

2. Connect the supplied USB cable to the controller of the Imaging Extension, and to a free USB interface of the control PC.

3. Connect the supplied AC/DC adapter to the Main Power Supply connector, as shown in Figure 56.

4. **Optional**: Mount the fiber/cable clips.

4.1 Remove the two screws (1) indicated in Figure 57.

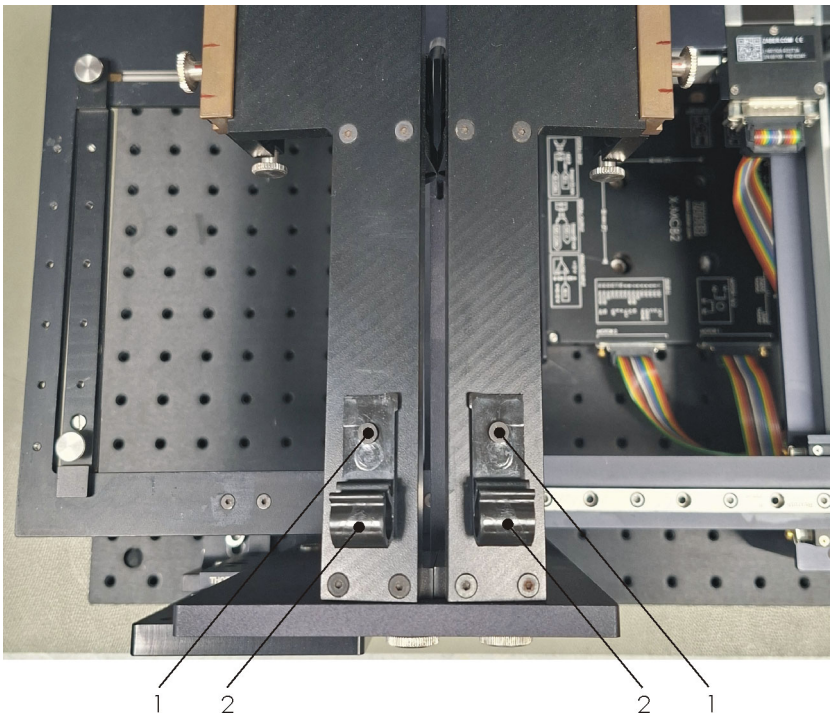


Figure 57 Mounting the optional fiber/cable clips

4.2 Mount the two cable clips (2) using the longer screws provided with the cable clips (M3 x 16 mm) as shown in Figure 57.

4.3 Carefully secure the antenna fibers and cables in the clips as shown in Figure 58 and make sure they cannot get in contact with the moving sample table.

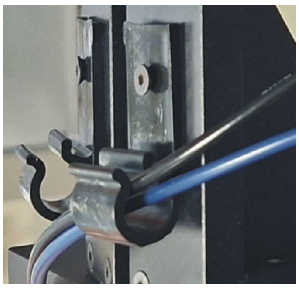


Figure 58 Securing the antenna fiber and cable

5. Start the TERAFLASH pro Main Unit and the measurement as described in section 4.4.

6.1.2 Optical Alignment

In the Complete Imaging Extension, the optical assembly contains four mirrors and two antenna mounts. Two mirrors and one antenna mount each are arranged as a rigid unit. The units can be fixed in two positions to enable imaging in transmission or reflection, respectively (see section 6.1.3). The parabolic mirrors are pre-aligned and fixed. The position of the antenna mounts can be adjusted manually.

A tapered pointer indicates the approximate position of the terahertz focus, when the optics are correctly aligned. To aid the alignment, TOPTICA provides a structured aluminum plate that fits on the 2D translation stage.

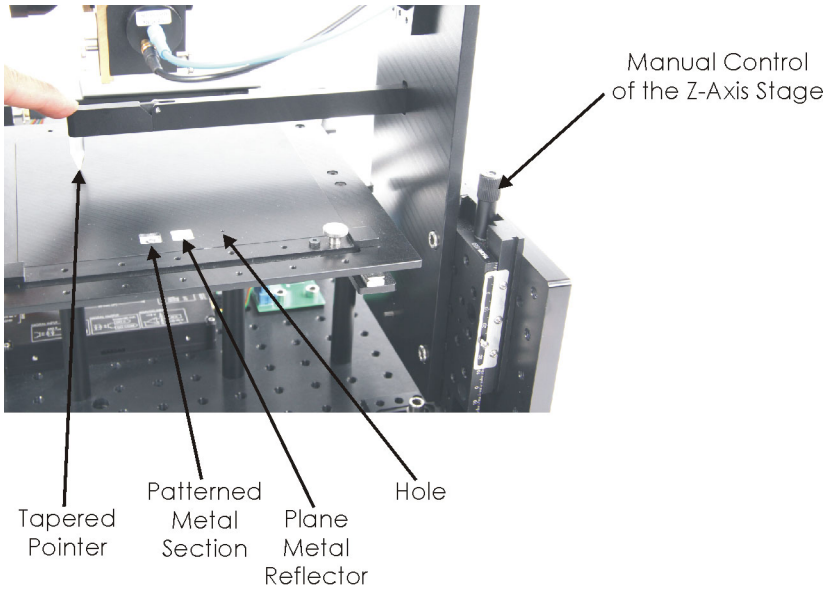


Figure 59 Structured aluminum plate and tapered pointer

The structured aluminum plate comprises three features:

- a hole for adjusting the optics in transmission,
- a patterned metal section for adjusting the optics in reflection,
- a plane metal reflector for reference measurements.

Preparation

1. If the antennas are not mounted, place the emitter and receiver modules in the corresponding mounts (see Figure 60) and fix them with screw C.

2. Place the structured aluminum plate on the 2D translation stage so that the three features are located on the right-hand side (see Figure 59).

3. Switch to the imaging mode by selecting the Raster Scan tab in the TERAFLASH pro Control Software. Connect to the Imaging Extension by clicking the **Connect** button in the Connection section of the Raster Scan tab (see section 5.2.10).

4. Use the buttons **Transmission** or **Reflection** in the Move to Position section of the Raster Scan tab (see section 5.2.10) to set the stage to the approximate position. Fine-tune the stage position in x/y until the relevant feature – either the hole or the center of the patterned metal section – sits under the tip of the tapered pointer (when pushed down).

5. Adjust the height of the optics assembly relative to the aluminum plate using the manual Z-axis control (see Figure 59). The aluminum plate should slightly touch the tapered pointer.

Antenna Alignment

The emitter/receiver antennas are fastened with the help of three locking screws A, B, C (see Figure 60). The antenna modules have to be oriented in the same direction (axis of polarization, refer to Figure 14) and their rear side should protrude approx. 5 mm from the respective mount.

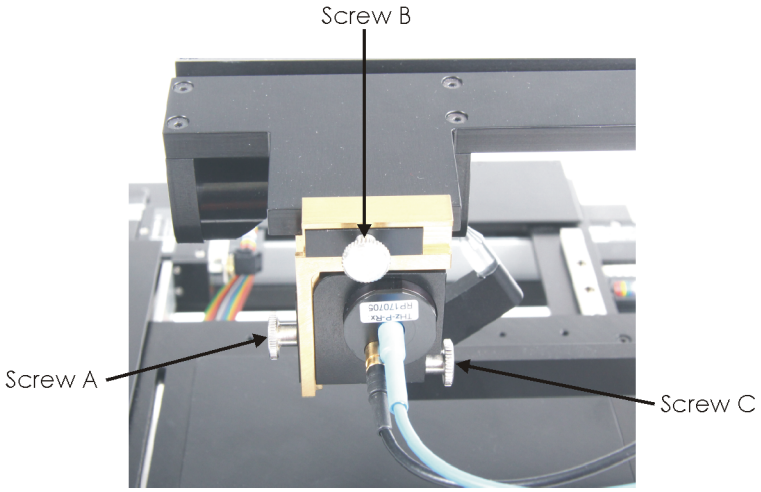


Figure 60 Emitter/receiver antenna mounting

To optimize the alignment, make sure the terahertz signal is displayed in the pulse trace and spectrum graphs.

1. Loosen screw A. Carefully shift the TX antenna up and down in its mount and try optimizing the pulse amplitude. Tighten screw A again.

2. Loosen screw B. Shift the TX antenna horizontally. Tighten screw B again.

3. Loosen screw C to vary the distance of the antenna to the mirror. Tighten screw C again.

4. Repeat steps 1 - 3 for the RX antenna.

It might be necessary to repeat the process for both antennas several times to find the correct alignement.

For transmission measurements, maximize the amplitude of the pulses transmitted through the hole in the structured aluminum plate.

To align the optics in reflection mode, observe the pulse echoes generated by the patterned metal section of the aluminum plate. The pattern consists of 5 elements with a size of 1.5 x 1.5 mm^2 but different heights each. The individual pulse echoes appear at different times in the pulse trace.

The first echo comes from the central section. When the terahertz beam is correctly aligned, this is the largest pulse. The following 4 reflexes result from off-centered light. All of these pulses should exhibit similar amplitudes, smaller than that of the central pulse. The sixth, somewhat broader pulse originates from the large background area (see Figure 61).

NOTE !	The individual pulses can show up as peaks or dips in the terahertz pulse trace depending on the orientation of the antenna (dips in Figure 61).

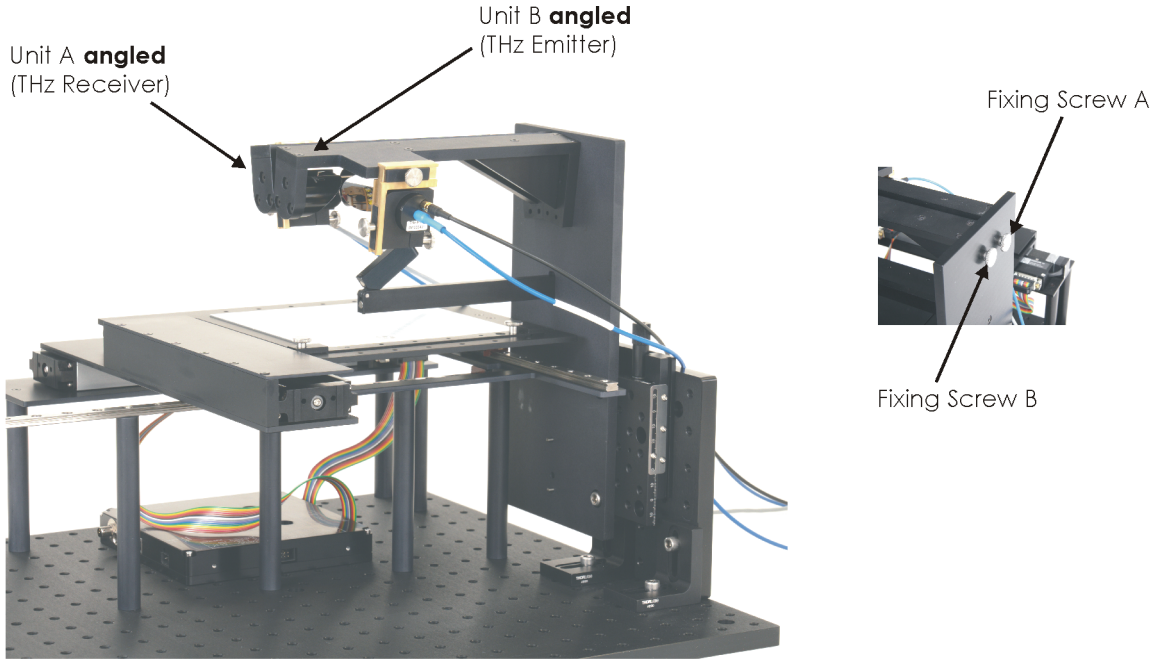


Figure 61 Reflected pulse trace of the patterned metal section of the structured aluminum plate. The arrows indicate the pulse echoes.

NOTE !	Depending on the type of antenna used, the focus can be so small that only the first echo is observed.

6.1.3 Configuration in Transmission and Reflection

The following instructions describe the reconfiguration of the imaging setup from a reflection geometry to a transmission setup. In order to change from transmission to reflection, simply follow the steps in reverse order. The optics are mounted on two rigid units for an easy configuration. Each unit contains one antenna and two mirrors. The unit is positioned with the help of alignment pins, and is fixed with one screw.



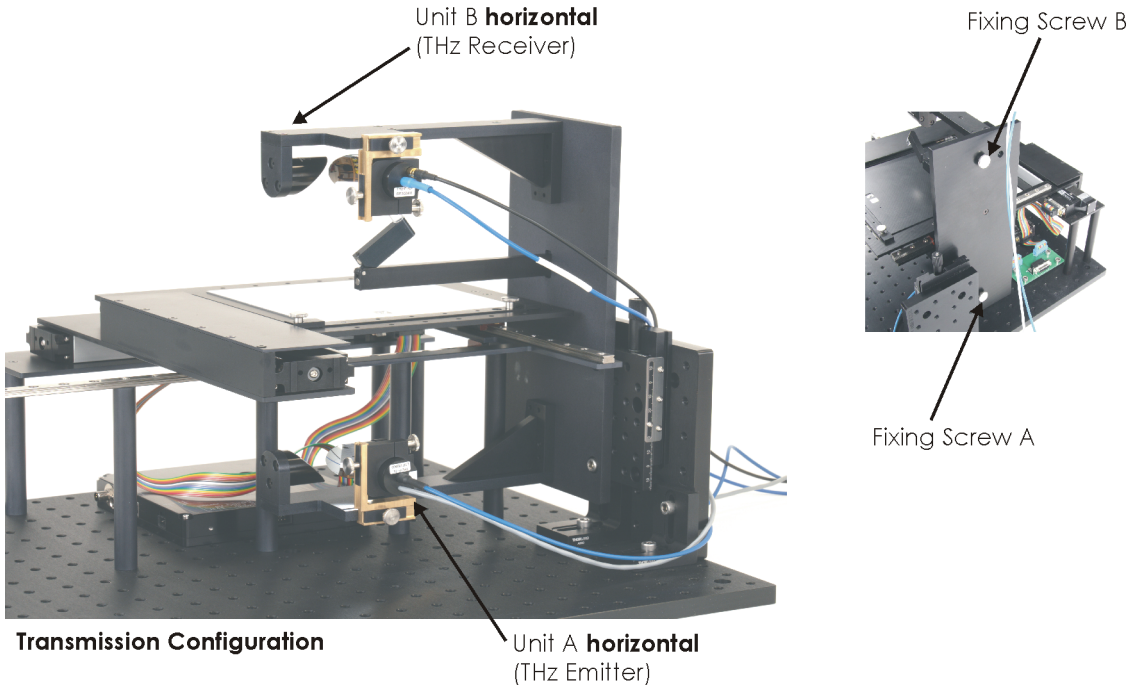


Figure 62 Imaging setup in reflection configuration (top) and transmission configuration (bottom)

To change the imaging setup to transmission configuration, please follow these steps:

1. Remove the fixing screw A and remove unit A (see Figure 62, top).

2. Place unit A underneath the sample area. Rotate the block so that the long metal bar is **horizontal** and the terahertz beam is guided vertically (see Figure 62, bottom). Two alignment pins indicate the correct position. Fix the unit with fixing screw A (see Figure 63).

3. Remove the fixing screw B. Remove and rotate unit B so that the long metal bar is **horizontal** and the terahertz beam propagates vertically (see Figure 62, bottom). Four alignment pins indicate the correct position. Fix the unit with fixing screw B (see Figure 63).

4. If required, fine-adjust the positions of the antennas as described in section 6.1.2.

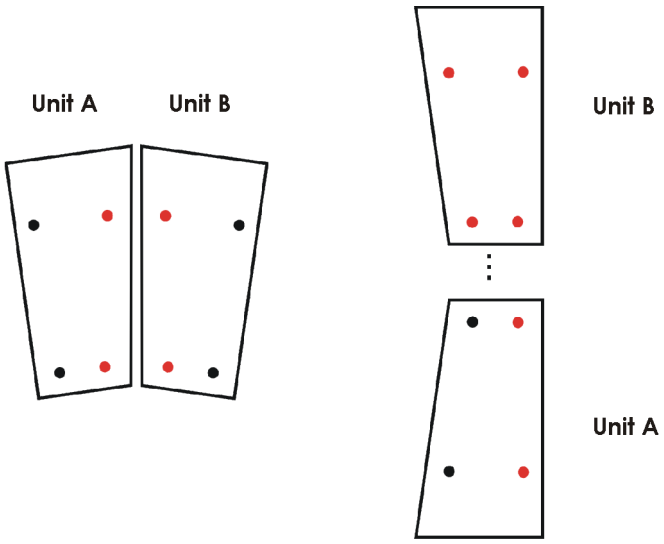


Figure 63 Schematic of the alignment pins (red) used for correct orientation of the optical blocks in reflection configuration (left) and transmission configuration (right).

6.2 Data File Format

Data acquired with the Imaging Extension are saved to a .csv file ([chosen file name].tim.csv). It comprises three parts: the header, a comment section and the data section.

The header consists of two lines with information about the measurement (refer to the table below). In the data section, each line represents one pixel and contains the amplitude values of the measured terahertz pulse trace (in nA).

The header specifies the necessary information to assign the pixels to rows and columns (e.g. number of pixels in X and Y direction and scanning profile).

Name	Meaning
No_Y, No_X	Number of pixels in Y and X direction, respectively
dy, dx	Step size of the translation stages in X and Y direction in mm (i.e., spatial resolution)
No_com-lines	Number of comment lines used at the beginning of the file
No_values	Number of measurement points in each pulse trace (e.g., 70 ps scan range / 0.05 ps resolution + 1 = 1401 points)
dt/ps	Time resolution of the pulse trace measurement in ps
Begin/ps	Start of the pulse trace with respect to the position of the long delay in ps

Table 2 Information given in the header: names and meaning.

NOTE !	The number of pulse traces is simply No_Y x No_X. The total number of data points is given by the number of pulse traces multiplied with the number of measurement points per trace (No_values).

6.3 THz-MAP Image Analysis Software

Once a scan is completed, the result can be evaluated with the THz-MAP Image Analysis Software.

6.3.1 Software Installation

The THz-MAP Image Analysis Software is not part of the TERAFLASH pro Control Software, but an additional software program. It requires Windows 10 (or higher) and administrator rights for installation.

NOTE !	The numeric format of your PC needs to be set to points (.) as decimal separator. Otherwise, data will not be displayed correctly.

1. Copy the installation files from the provided flash drive to hard disk.

2. Use Windows Explorer to navigate to the selected directory on the hard disk. Right-click on Setup.exe and select "Run as administrator".

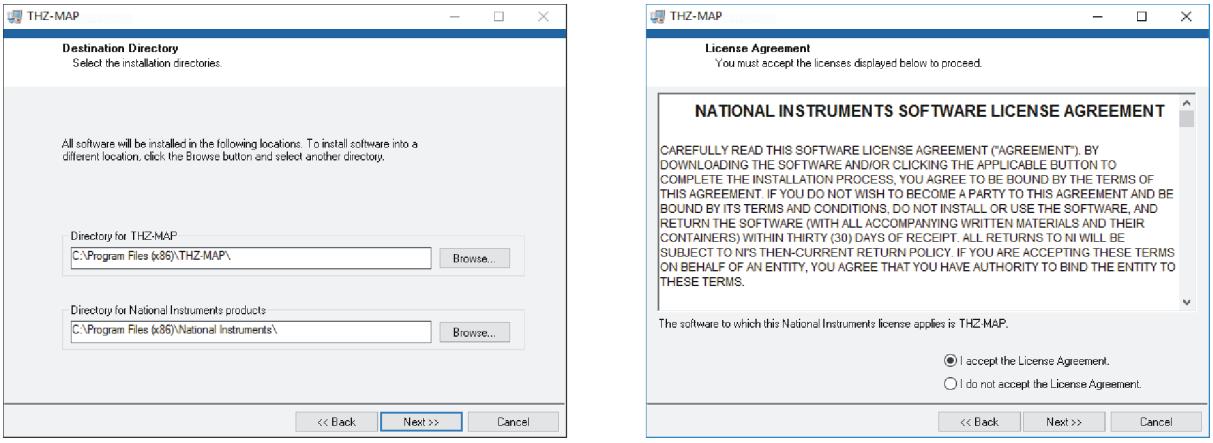


Figure 64 Selection of directory and acceptance of license agreement

3. Select the installation directory and if asked confirm the license agreement. Click **Next** to continue.

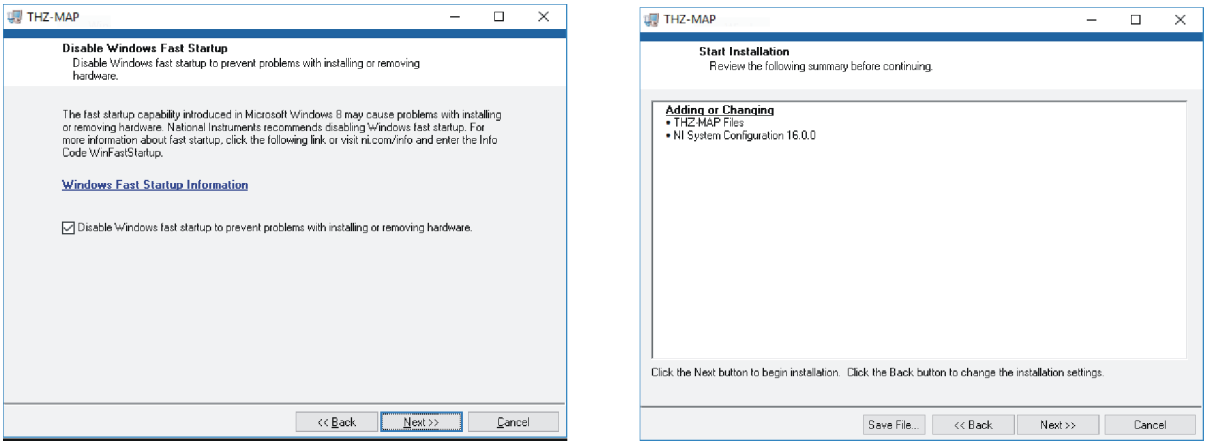


Figure 65 Disable the "Fast startup" option and Installation Overview

4. If asked, disable the "Fast startup" option (Figure 65 left). Click **Next** to continue. On the Start Installation screen (Figure 65 right) click **Next** to continue. The installation progress is indicated by green progress bars.

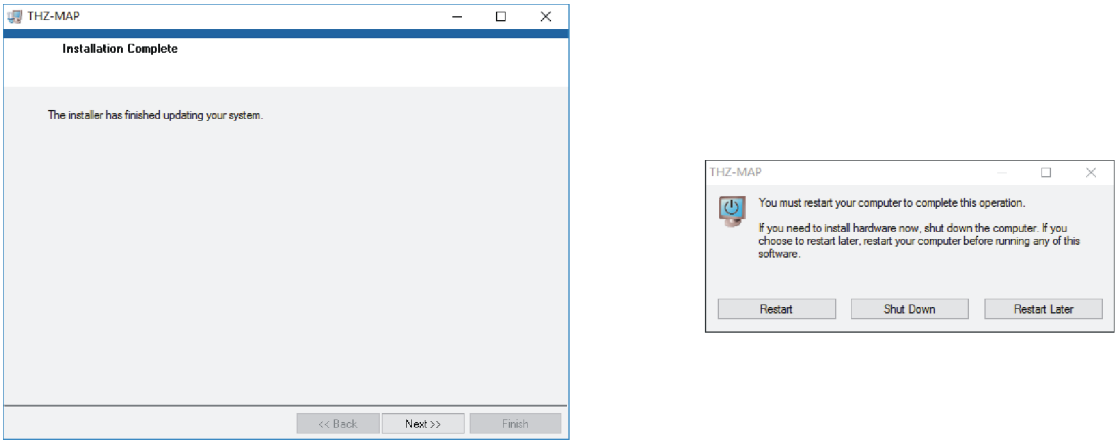


Figure 66 Completion of Installation and restart prompt

5. Click **Finish** to complete the installation. When prompted, restart your computer to complete the installation.

6.3.2 User Interface

After starting the application, a file dialog appears, allowing you to select the data file created with the Imaging Extension Control Software. A progress bar is displayed at the bottom of the THz-MAP Main window while the file is being loaded.

NOTE !	After loading a new dataset please move the cursors in the Pulse Trace Graph to the outer edges of the trace, so that no pulses are inadvertently excluded from the analysis. Confirm by clicking **Apply Changes** (green cursor: start; black cursor: end).

Overview

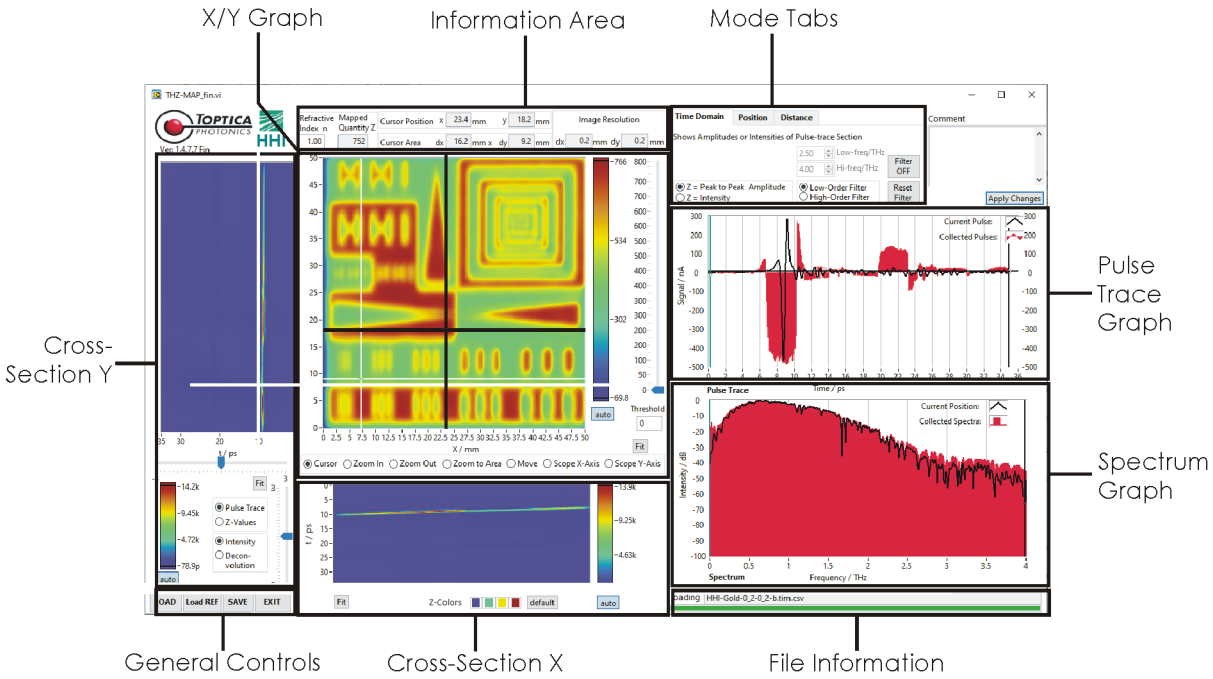


Figure 67 THz-MAP Image Analysis Software, overview of the Main window

The Main window visualizes the measured data in three different ways: (i) in a two-dimensional parameter map (X/Y Graph, often called "terahertz image"), (ii) in 2 cross-sections along the black cursor lines of the X/Y Graph (Cross-Section X and Cross-Section Y), (iii) specific data of the pixel selected by the point of intersection of the two cursor lines (black crosshair cursor) is shown in the Pulse Trace Graph and Spectrum Graph.

The X/Y Graph and the cross-sections display parameter values, which are selected via the Mode tabs in the upper right corner.

The **Time Domain** mode displays and analyses data based on the original terahertz pulse traces. The modes **Position** and **Distance** localize the peaks within the time traces, and display the absolute position of the highest peaks, or the relative distance between the two strongest peaks, respectively.

General Features

All graphs and controls share common LabView features. The ranges displayed in the graphs can be edited manually by a double click on the first and last value of the abscissa, ordinate or color scale: Simply type in the numbers of choice. Keep in mind that any cursors that are not located within the displayed range might no longer be accessible.

For all of the controls and numerical input fields, a right mouse click opens a further context menu, allowing to copy a parameter value or change a control to its default setting.

X/Y Graph

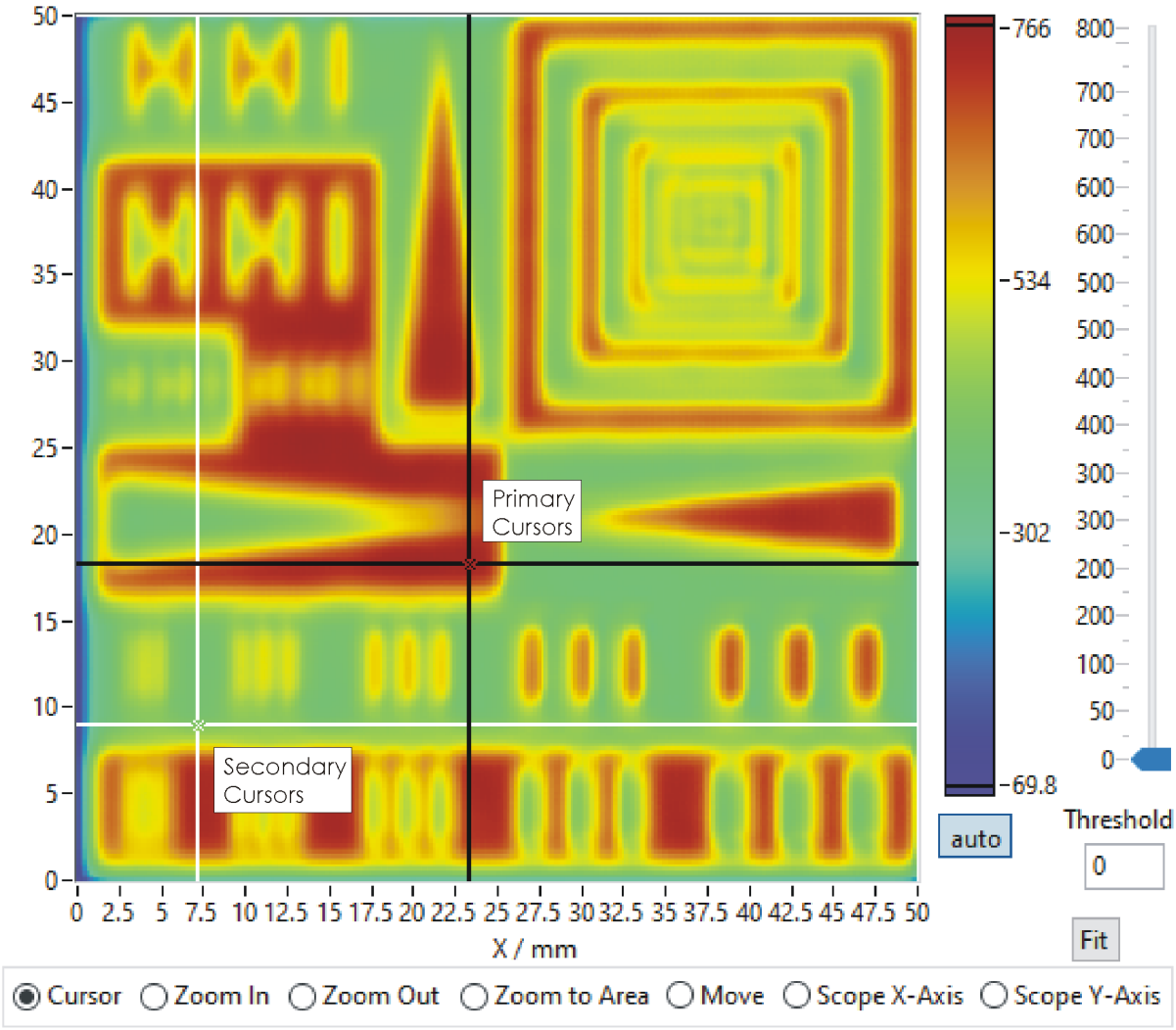


Figure 68 X/Y Graph

This graph maps a single quantity (Z) of each pulse trace in false colors, as function of the 2D position of the translation stage ("terahertz image"). The default parameter displayed is the peak-to-peak amplitude of the pulse trace of each image pixel. Alternative parameters are selected via the Mode tabs (see below).

The color bar to the right of the X/Y Graph indicates the range of values shown in the image. The **auto** button below the color bar activates or disables the auto scaling of the color map. A threshold control on the right permits a manual input of the lowest value used in the color map. Setting this threshold is only possible when the automatic color scaling is deactivated. The threshold value can either be entered via the numeric input field next to the **auto** button, or adjusted with the **Threshold** slide bar above. Alternatively, the color scale can be optimized by directly changing the range of the color bar (double click on values next to it and type in desired number). The color code itself can be modified at the bottom of the user interface (**Z Colors**). Clicking **default** restores the original color settings.

The X/Y Graph features two crosshair cursors. By default, the cursors are located in the lower left corner of the graph, and can be dragged with the mouse to the desired position within the image. The primary (black) cursor selects a pixel for inspection in the Pulse Trace Graph and Spectrum Graph. The black crosshair also defines cross-sections of the image along the horizontal and vertical axis, respectively (see description "Cross-Sections along x and y" below). The secondary (white) cursor measures the distance to the primary one. The corresponding value is displayed in the information area above the X/Y Graph.

A set of buttons below the X/Y Graph modifies the display:

Cursor	Reactivates the cursors after the use of one of the Zoom or Move buttons.
Zoom In	Zooms into a section of the image. After clicking **Zoom In**, move the mouse to the center of the section that you want to magnify. Click or hold the left mouse button to zoom in.
Zoom Out	Reverses the zoom.
Zoom to Area	Zooms into a rectangular image section, which is selected with the mouse.
Move	Moves the part of the image displayed in the X/Y Graph (Drag the image with the mouse, useful in conjunction with one of the Zoom functions).
Scope X-Axis	Enables the selection of an image section with the mouse and stretches its horizontal dimension across the full image.
Scope Y-Axis	Enables the selection of an image section with the mouse and stretches its vertical dimension across the full image.
Fit	Fits the entire image into the X/Y Graph, undoing any previous zoom.

Mode Tabs

The Mode tabs define the parameter (z-value) that is displayed in the X/Y Graph in false colors, as well as in the "Z-Value" mode of the corresponding Cross-Section Graphs. They also provide specific options for data analysis, including frequency filters and amplitude thresholds. A mode is activated by opening the corresponding tab.

Time Domain Mode

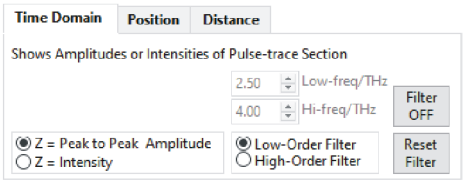


Figure 69 Time Domain mode

The Time Domain mode selects either the peak-to-peak pulse amplitude or the intensity of the strongest peak in the pulse trace as Z value for the color-coded display of the X/Y Graph (depending on the control **Z = Peak to Peak Amplitude/Z = Intensity** on the left).

Spectral filtering can be applied to the pulse traces by enabling the filter controls (**Filter ON/OFF** button). Two filter options with different frequency roll-offs are available (**Low-Order Filter/High-Order Filter**).

The **Low-freq/THz** and **Hi-freq/THz** fields specify the low- and high-frequency cut-offs, respectively. The effects of spectral filtering on the pulse trace and spectrum are shown in the respective graphs (black curves). The two cursors in the Spectrum Graph indicate the selected frequency range (see section Spectrum Graph below). In the terahertz image (X/Y Graph), low-pass filtering is helpful to reduce noise contributions from the high-frequency content. Vice versa, high-pass filtering increases the spatial resolution, which comes at a cost in amplitude reduction. Turning the filter on or off has an immediate effect; however changing the filter settings requires pressing the **Apply Changes** button to update all graphs. The **Reset Filter** button switches off the spectral filtering and restores the default settings.

NOTE !	The peak analysis is restricted to the time interval selected by the two cursors in the Pulse Trace Graph (see section Pulse Trace Graph). This time domain filtering enables, for example, a selective analysis of different layers in a sample. When two filters are nested, spectral filtering is applied to the entire pulse trace first, and then the time domain filter selects the specified range for evaluation.

Position Mode

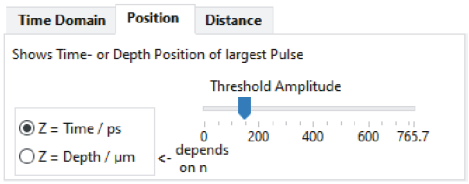


Figure 70 Position mode

In Position mode, the highest amplitude within a pulse trace is located, and its absolute position is displayed as Z-value in the X/Y Graph.

Depending on the control setting **Z = Time/ps** or **Z = Depth/µm** on the left, the displayed value is either the time delay in ps or the corresponding distance in µm, relative to the start of the time trace (t = 0). Note that the distance calculation requires an input for the refractive index n of the material (input field in the "Information Area", see Figure 77).

A low-amplitude threshold can be selected with the help of the **Threshold Amplitude** slide bar on the right. This excludes low-signal pulse traces from the peak localization. The **Apply Changes** button (top-right of the THz-MAP Image Analysis Main window) recalculates the image after a change of the threshold.

NOTE !	The position analysis only includes the time interval selected by the cursors in the Pulse Trace Graph (see section Pulse Trace Graph below). This time domain filtering enables for example the selective analysis of different layers in a sample. The spectral filtering selected in the Time Domain Mode tab is also applied prior to the position analysis.

NOTE !	The position values displayed in the X/Y Graph and the Cross-Section Graphs are changed by the tilt correction, which serves to compensate for any misalignment of the sample (see section Leveling below). In contrast, the Pulse Trace Graph remains unchanged.

Distance Mode

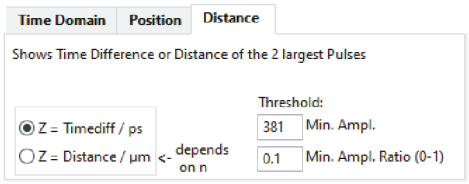


Figure 71 Distance mode

In Distance mode, the interval between the two strongest peaks in the pulse trace is determined and displayed as Z-value in the X/Y Graph.

Depending on the control setting **Z = Timediff/ps** or **Z = Distance/µm**, the displayed value is either the time difference in ps or distance in µm. Akin to the Position mode analysis, the distance calculation requires an input for the refractive index of the material (input field in the "Information Area", see Figure 77).

A lower threshold for the pulse amplitude (**Min. Ampl.**) and the amplitude ratio of the pulses (**Min. Ampl. Ratio (0-1)**) helps discard unwanted artifacts of noisy pulse traces. Clicking **Apply Changes** recalculates the X/Y Graph accordingly.

NOTE !	Peak localization and the distance measurement analysis only includes the time interval selected by the cursors in the Pulse Trace Graph (see section Pulse Trace Graph below). In contrast, spectral filtering is not applied to the peak analysis.

Pulse Trace Graph

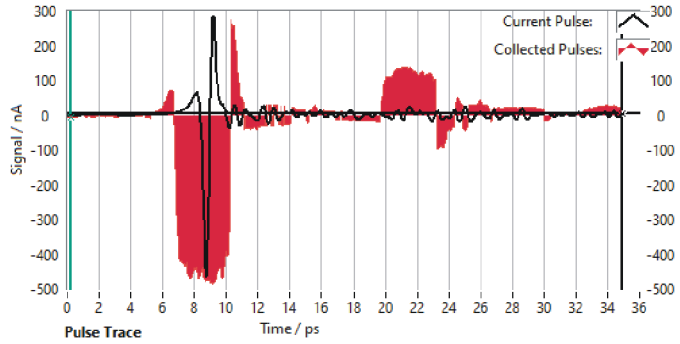


Figure 72 Pulse Trace Graph

The Pulse Trace Graph shows the terahertz waveform of the pixel selected by the black crosshair cursor in the X/Y Graph (black line, right ordinate). Additionally, the red shaded area (left ordinate) represents a superposition of the pulse traces of all image pixels, providing an overview of the range of values across the image. Note that the two ordinates may have different scales and that the zero lines may differ.

Time Domain Filtering

Two cursors in the pulse trace graph enable the selection of a time interval within the pulse trace. Only the enclosed interval is evaluated according to the analysis mode chosen (see Mode tabs above). This time domain filter applies both to the calculation of the X/Y Graph, and to the display of the z-value as cross-sections.

The horizontal positions of the green and black cursors mark the beginning and end of the interval, respectively. Both cursors can be moved horizontally by mouse dragging. Clicking the **Apply Changes** button in the top-right corner updates the analysis and the display in the X/Y Graph.

NOTE !	After loading a new dataset please move the cursors in the Pulse Trace Graph to the outer edges of the trace, so that no pulses are inadvertently excluded from the analysis. Confirm by clicking **Apply Changes** (green cursor: start; black cursor: end).

Spectrum Graph

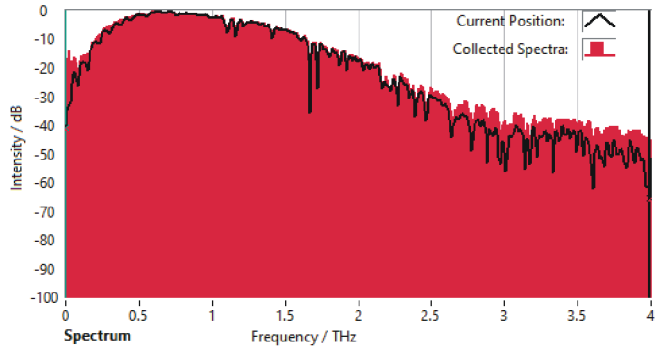


Figure 73 Spectrum Graph

Akin to the Pulse Trace Graph, the Spectrum Graph represents the spectrum, i.e. the Fourier transform of the pulse trace, of the pixel selected by the crosshair cursor in the X/Y Graph. This spectrum is displayed as a black line. In addition, a superposition of all of the spectra in the image is shown in red.

NOTE !	The calculation of the spectrum (single pixel) takes the spectral filters set in the Time Domain Mode tab (see above) into account. The two cursors mark the frequency boundaries selected for filtering.

Cross-Sections along X and Y

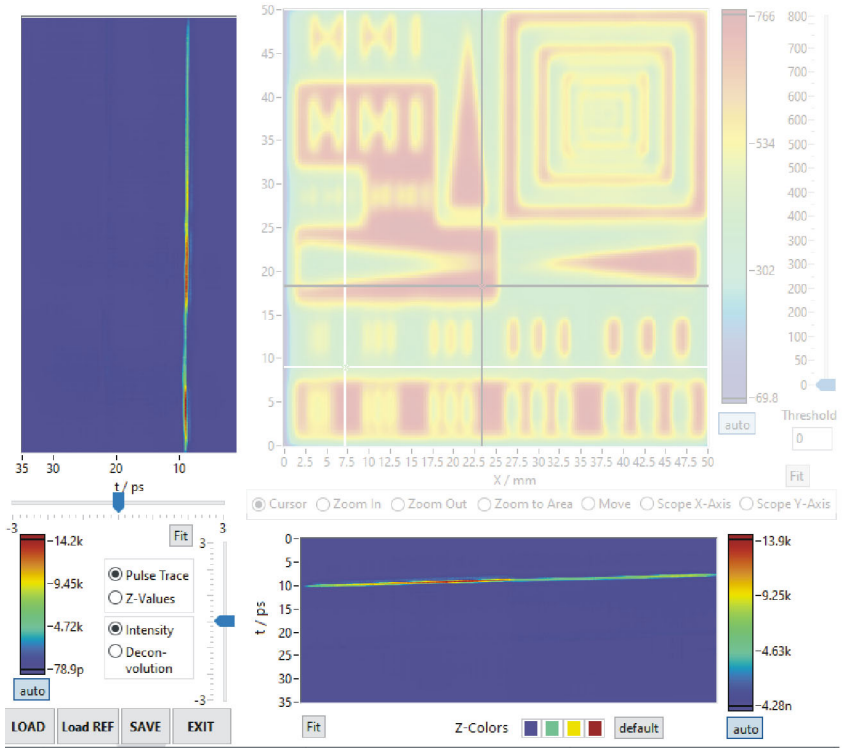


Figure 74 Cross-Sections along X and Y showing pulse trace properties in false colors.

The black crosshair cursor in the X/Y Graph defines two cross-sections, one along the x-axis and one along the y-axis. Parameter values along these cross-sections are displayed in separate graphs. The x and y cross-section are located below and to the left of the X/Y Graph, and share the dimensions of the respective axis. The cross-section graphs offer two general representation options selected via the control in the bottom-left corner: **Pulse Trace** and **Z-Values**.

For the **Z-Values** option, the cross-section graphs show the z-value of the X/Y Graph (i.e., the value represented in false colors in the image), as measured along the lines of the crosshair cursor. An example of this representation is shown in Figure 75.

NOTE !	The value shown in the z-value cross-sections depends on the options selected in the Mode tabs, including the filters applied to the Pulse Trace Graph and the tilt compensation.

The **Pulse Trace** option is a sectional view similar to B-scans in ultrasound or radar imaging. The graphs display properties of the original terahertz pulse traces in false colors (see Figure 72). Depending on the **Intensity/Deconvolution** setting (bottom-left corner), either the intensity or the deconvoluted pulse trace is shown. For the x cross-section, the horizontal axis corresponds to the position along the cross-section. The vertical axis shows the time in the pulse trace. The color scale represents the pulse trace signal. For the y cross-section the axes are flipped.

The calculation of the deconvolution requires a single reference measurement, which is loaded via the button **Load REF** on the bottom-left corner of the Main window. For the analysis, the spectrum of each image pixel is divided by the reference spectrum, and back-transformed into the time domain. The result is displayed in the cross-section graphs.

For reflection measurements, the cross-sections show information about the internal structure of the sample. Layer interfaces at different depths create reflections at different times and appear as lines in the graphs. This deconvolution option helps to identify such interfaces, as differences e.g. compared to a plane metal reflector are highlighted.

The range of the time axis in the cross-sections can be edited by either selecting an area with the mouse or by a double click on the first and/or last value of the axis. Clicking **Fit** retrieves the original settings.

For the false-color representation, each cross-section graph has an individual color bar, located below the bottom-right corner of the X/Y Graph (Cross-Section X) and in the bottom-left corner of the User Interface (Cross-Section Y). By default, the colors are autoscaled. If the respective auto option is switched off, the scales can be modified by a double-click on the first and/or last value. Reactivating the respective auto button restores the original color scale.

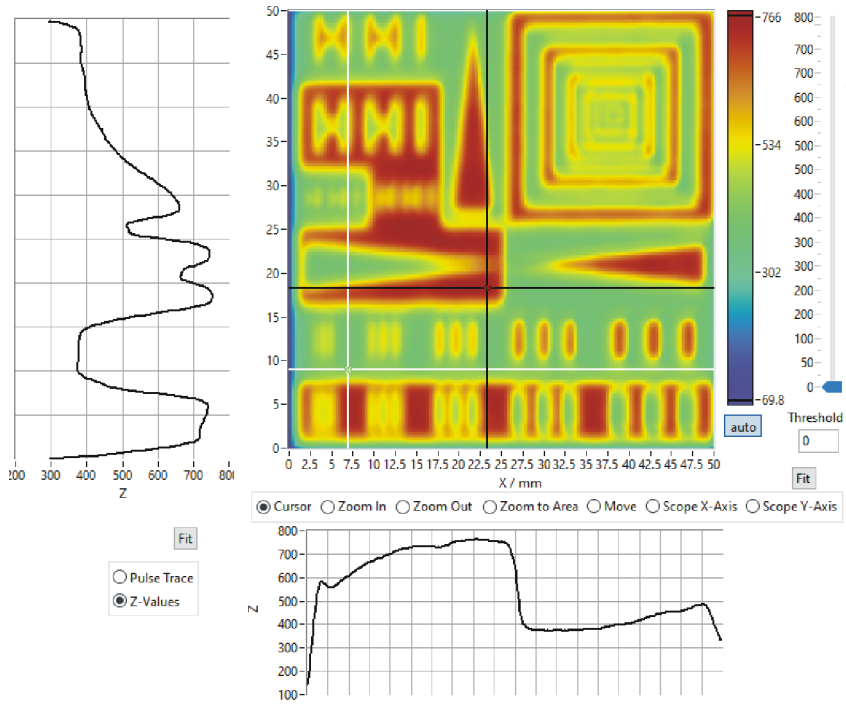


Figure 75 X and y cross-section plots, showing the z-values along the lines of the black crosshair cursor in the X/Y Graph.

Leveling

Due to the alignment of the sample surface with respect to the translation stage, the display in the Cross-Section Graph might appear tilted. This tilt can be compensated with the leveling controls.

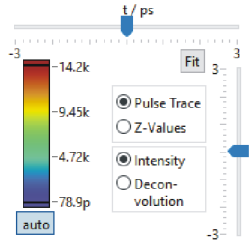


Figure 76 Leveling controls

Use the leveling controls to apply a vertical and/or horizontal tilt correction (upper and lower slider respectively). The changes are directly applied to the Cross-Section Graphs in Pulse Trace Mode. In order to recalculate the X/Y Graph after a change of the leveling controls, click the **Apply Changes** button at the top-right of the window. By default, minimum and maximum values of the controls are set to -3 and 3 ps, respectively, however these settings can be modified by double-clicking the number and typing a new value.

NOTE !	The tilt correction only applies to the Cross-Section Graphs and the X/Y Graph (Analysis mode "Position"), whereas the Pulse Trace Graph remains unaffected.

Information Area

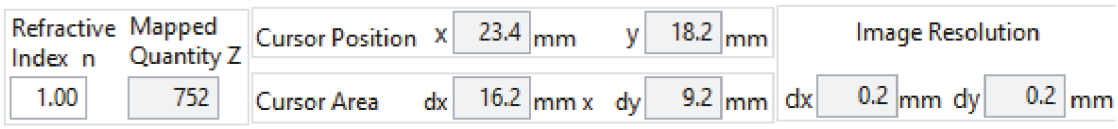


Figure 77 Information Area

The Information Area in the upper-left corner displays general information about the image, the sample, and distance/size measurements in the X/Y Graph.

Refractive index n	Input field to enter a (frequency-independent) value of the refractive index of the material. This is used to calculate the peak position and distance (please refer to the Mode tabs Position and Distance above).
Mapped Quantity Z	Displays the z-value at the momentary position of the black crosshair cursor in the X/Y Graph.
Cursor Position (x, y)	Indicates the momentary horizontal and vertical position of the black crosshair cursor in the X/Y Graph, relative to the bottom-left corner, in mm.
Cursor Area (dx, dy)	Tool to measure distances in the X/Y Graph. The values displayed show the difference in the horizontal and vertical positions, respectively, of the black and white crosshair cursors in the X/Y Graph, in mm.
Image Resolution (dx, dy)	Denotes the horizontal and vertical step size of the translation stage for the displayed image, in mm.

Comment Field, Apply Changes Button (Top-right)

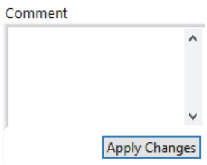


Figure 78 Comment field and **Apply Changes** button

Comment	In case the data file was saved with a comment, the statement is displayed in the **Comment** field in the top-right corner of the User Interface.
Apply Changes	Updates all displays and applies any changes, e.g. threshold or filter settings, to the terahertz image on display.

General Control Buttons (Bottom-left)



Figure 79 General control buttons

LOAD Opens a file dialog to load a data file created with the TERAFLASH pro Control Software.

NOTE !	After loading a new dataset, please move the cursors in the Pulse Trace Graph to the outer edges of the trace, so that no pulses are inadvertently excluded from the analysis. Confirm by clicking **Apply Changes** (green cursor: start; black cursor: end).

Load REF Opens a file dialog to load a single reference pulse trace (not a complete image!) for cross-section analysis (see description "Cross-Sections along x and y" below).

SAVE This button opens a dialog box (see Figure 80) allowing you to select the data and graphs you want to export. You can choose to save the maps and graphs as images (png – portable network graphic), as plane text files (csv – comma separated values), or both by highlighting the respective boxes.

EXIT Terminates the THz-MAP Image Analysis Software.

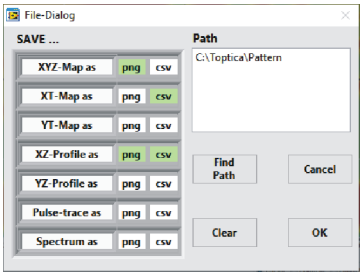


Figure 80 Dialog Box

Save... Selection matrix for images and datasets to save. Highlight the respective boxes (png and/or csv) to select the datasets you wish to export.

Clear Resets the options of the "Save" matrix.

Path Displays the path and general file name for saving. Each dataset selected is saved to a separate file with a specific file name extension and data format (see table below).

Find Path Opens a file dialog to select the desired path and filename.

OK Initiates the export procedure and closes the dialog box.

Cancel Closes the dialog box without saving any data.

Dataset	File Name Extension	Data Format of csv files
	Format: FileName.Extension.png FileName.Extension.csv	
XYZ-Map	xyz	Matrix with z-values for each pixel as displayed in the XY-Graph. Each line in the data set corresponds to one line along the x-axis in the graph, starting at the bottom-left corner of the graph.
XT-Map	xt	Matrices representing the cross-section images. Each line corresponds to one pixel along the cross-section, containing the pulse trace values (intensity or deconvoluted pulse trace), starting at the left/bottom side.
YT-Map	yt	
XZ-Profile	xz	Single line: Selected z-value for each pixel along the cross-section, starting at the left/bottom side.
YZ-Profile	yz	
Pulse-Trace	pls	First line: pulse trace/spectrum of selected pixel (including filtering options as applied). Second line: superposition of the pulse traces/ spectra of all image pixels
Spectrum	spc	

Table 3 Names and formats of data files created by export option.

File Information



Figure 81 File Information

Filename The name of the selected data file is displayed in the bottom-right corner.

Loading Progress bar while an image file is being loaded.

7 Alignment of the Terahertz Optomechanics

CAUTION !	**Photoconductive antennas are ESD-sensitive devices. We recommend wearing a high-impedance grounding strap around the wrist when handling photoconductive antennas. Take care not to touch the inner pins of the connectors.**

CAUTION !	When handling the photoconductive antennas, it is necessary to protect them from any inadvertent electrical short-circuits and to avoid contact with current-carrying parts. The antennas are particularly sensitive to voltage peaks. These might considerably affect the service life of the antennas, even if a disturbance is not immediately noticeable.

7.1 Compact Rail System (THZ OPTICS / TR COMP)

The optomechanical assembly THZ OPTICS / TR COMP is a simple, compact setup for terahertz transmission measurements. 2 off-axis parabolic mirrors collimate and re-focus the terahertz beam.

The photoconductive antennas and the mirrors are fixed in cages, which reside on a rail. Moving the cages along the rail changes the length of the collimated terahertz beam path. To retrieve the terahertz pulse after a change of the beam length, use the stationary delay stage of the TERAFLASH pro (see description of Delay (ps) in section 5.2.1).

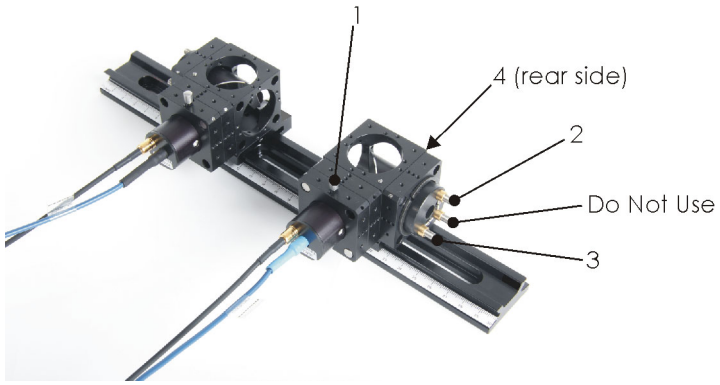


Figure 82 Compact terahertz optomechanics with two parabolic mirrors (TOPTICA article code THZ OPTICS / TR COMP)

Function of the screws:

(1) Locking screw for fixing the photoconductive antennas
(2) Vertical and
(3) Horizontal tip/tilt alignment of the mirrors,
(4) Locking screw for the optical cage

7.1.1 Terahertz Beam Alignment

1. Preparation

1.1 Place the photoconductive antennas in their mounts, so that the rear side of the aluminum housing is approx. 6 cm away from the center of the mirror. When correctly positioned, the full length of the flattened part of the cylindrical housing just about protrudes out of the mount (see Figure 83). Make sure that the angular orientation of both photoconductive antennas is the same, i.e. the polarization is either vertical or horizontal. The polarization axis is indicated by two small grooves on the front side of the housing (see Figure 14). Fix the photoconductive antennas with the locking screws (1).

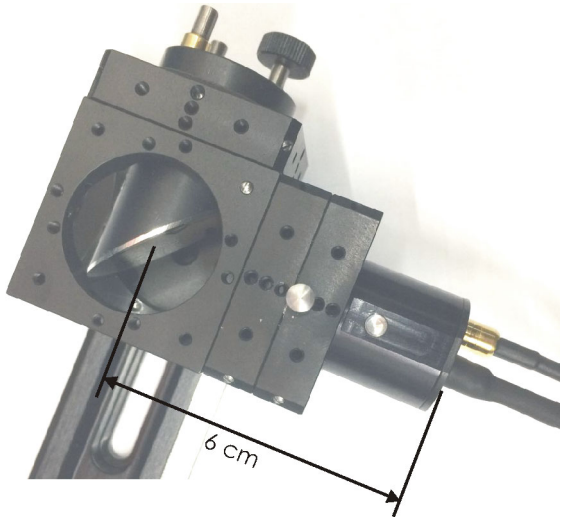


Figure 83 Positioning the photoconductive antennas

1.2 Start the system as described in section 4.4.
Move the stationary delay stage (see description of Delay (ps) in section 5.2.1) until the terahertz pulse is displayed in the upper screen of the TERAFLASH pro Control Software (see Figure 16).

2. Mirror Alignment

2.1 Adjust the tip/tilt screws (2, 3) of each mirror separately.

2.2 Next, "walk" the beam:
Change the "tip" screw of mirror 1. The signal should now decline a bit, as the position has already been optimized. Try to compensate the signal loss by adjusting the "tip" screw of mirror 2.
Repeat the previous step a couple of times, until you see a trend to either higher or lower signal values. If the signal increases, continue. If not, try the opposite direction.

2.3 Repeat the same procedure with the "tilt" screws of both mirrors.

3. Focus Alignment

3.1 Loosen the locking screws (1). Carefully move the antennas in the direction of the beam, i.e. towards the mirrors or further out of the cages. Make sure the angular orientation of the antennas is not changed, to keep the polarization axis correct.
If the signal increases, continue the movement; if not, reverse the direction. Fix the antennas by tightening the locking screws (1) again.
Should the terahertz pulse move out of the display, then adjust the stationary delay stage.

3.2 Repeat steps 2.1 to 2.3.

7.2 Two-Mirror Optomechanics (THZ OPTICS / TR 2 M)

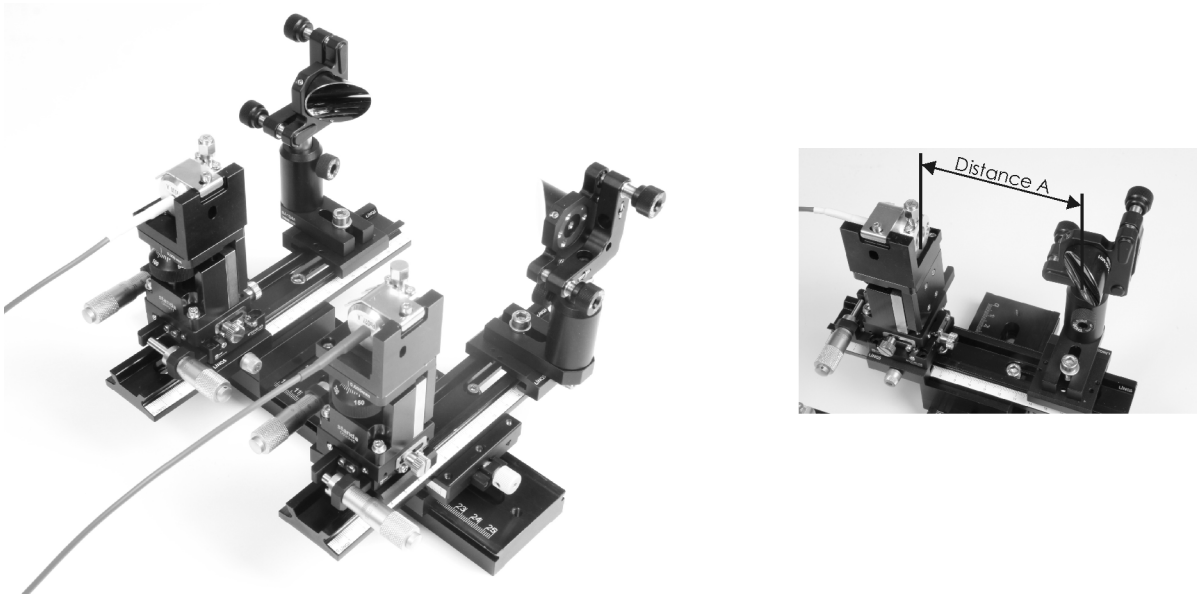


Figure 84 Two-mirror optomechanics (TOPTICA article code THZ OPTICS / TR 2M)
The distance A between the front side of the antenna module and the center of the mirror shown on the right is approx. 5.5 cm for mirrors with 3" focal length.

The optomechanical assembly THZ OPTICS / TR 2M consists of two 3-axis stages (one for each photoconductive antenna), 2 off-axis parabolic mirrors to collimate and re-focus the terahertz beam, kinematic mirror mounts and a manual delay stage. The entire setup is mounted on rails, with two shorter rails – carrying one antenna mount and one mirror mount each – residing on a longer system rail (see Figure 84).

The user can easily change the length of the collimated beam path by moving the slides that carry the short rails. When the system is correctly aligned, such a length shift should only have a minimal impact on the terahertz amplitude. To retrieve the terahertz pulse after a change of the beam length, use the stationary delay stage of the TERAFLASH pro (see description of Delay (ps) in section 5.2.1).

7.2.1 Antenna and Mirror Mounts

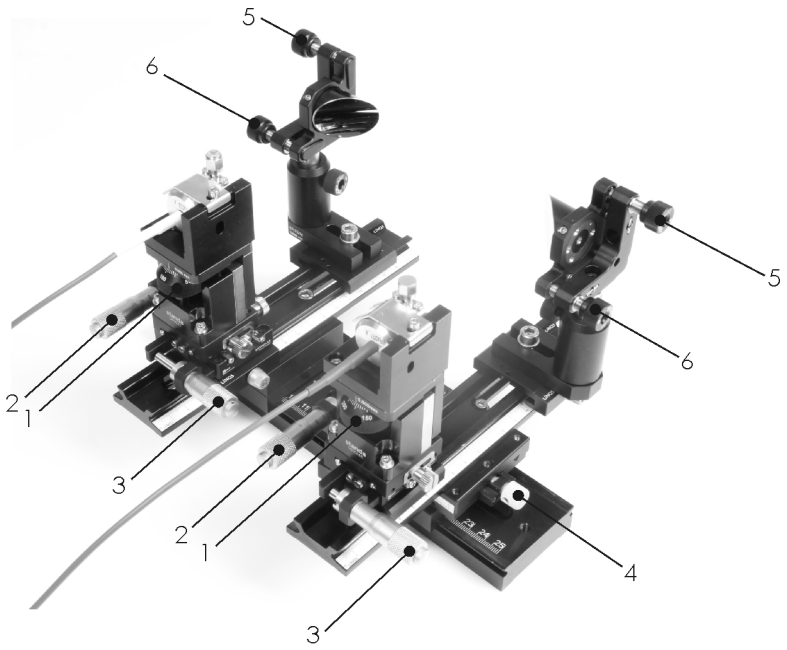


Figure 85 Adjustment screws of antenna and mirror mounts

Each antenna mount features three orthogonal translation stages. Their functions are (from top to bottom, as viewed from the side):

(1) Vertical alignment
(2) Focus alignment
(3) Horizontal alignment

In addition, a manual delay stage (4) adjusts the distance between the two short rails. This stage does not change the beam alignment.

Each mirror mount features two screws, for vertical (5) and horizontal (6) tip/tilt alignment, respectively.

7.2.2 Terahertz Beam Alignment

NOTE !	For location of the adjustment screws, please refer to Figure 85.

1. Preparation

1.1 Make sure that the photoconductive antennas are securely fixed on their mounts.

1.2 Make sure that the angular orientation of both photoconductive antennas is the same, i.e. the polarization is either vertical or horizontal. The polarization axis is indicated by two small grooves on the front side of the housing (see Figure 14).

1.3 Adjust the vertical alignment screws so that both antennas have the same height. Use a ruler to check the alignment. The center of the photoconductive antenna should be level with the center of the parabolic mirrors.

1.4 Measure the distance between the center of the mirror, and the front facet of the antenna modules. In the case of 3" mirrors with 75 mm focal length (the default mirrors used), adjust the distance to 55 mm.

1.5 Start the system as described in section 4.4.
1.6 Move the stationary delay stage (see description of Delay (ps) in section 5.2.1) until the terahertz pulse is displayed in the upper screen of the TERAFLASH pro Control Software (see Figure 32).

2. Antenna Position

2.1 Use the vertical-alignment screw (1) to shift the first antenna module up and down. Try to increase the signal. If the pulse amplitude rises – good. If not, try the opposite direction.
2.2 Repeat with the second antenna module.
2.3 If no further improvement is possible, try moving both antennas up or down in parallel.
2.4 Repeat these steps using the horizontal-alignment screws (3) of both antennas.

3. Mirror Alignment

3.1 Adjust the tip/tilt screws (5, 6) of each mirror separately.
3.2 Next, "walk" the beam:
Change the "tip" screw of mirror 1. Now, the signal should now decline a bit, as the position has already been optimized. Try to compensate the signal loss by adjusting the "tip" screw of mirror 2.
Repeat the previous step a couple of times, until you see a trend to either higher or lower pulse amplitudes. If the signal increases, continue. If not, try the opposite direction.
3.3 Repeat the same procedure now with the "tilt" screws of both mirrors.

4. Focus Alignment

Turning the focus-alignment screw (2) changes the focal distance between mirror and photoconductive antenna.

4.1 Continue turning the screw (2) of the first antenna in one direction and observe the pulse amplitude. If the pulse moves out of the display, use the stationary delay stage to change the offset of the scan range.
4.2 You will finally see an effect of the focus position, i.e. the pulse amplitudes become smaller or larger. Try to maximize the amplitude.
4.3 Repeat with the second antenna.

5. Iteration

Repeat steps 2 - 4.

7.3 Four-Mirror Optomechanics (THZ OPTICS / TR 4M)

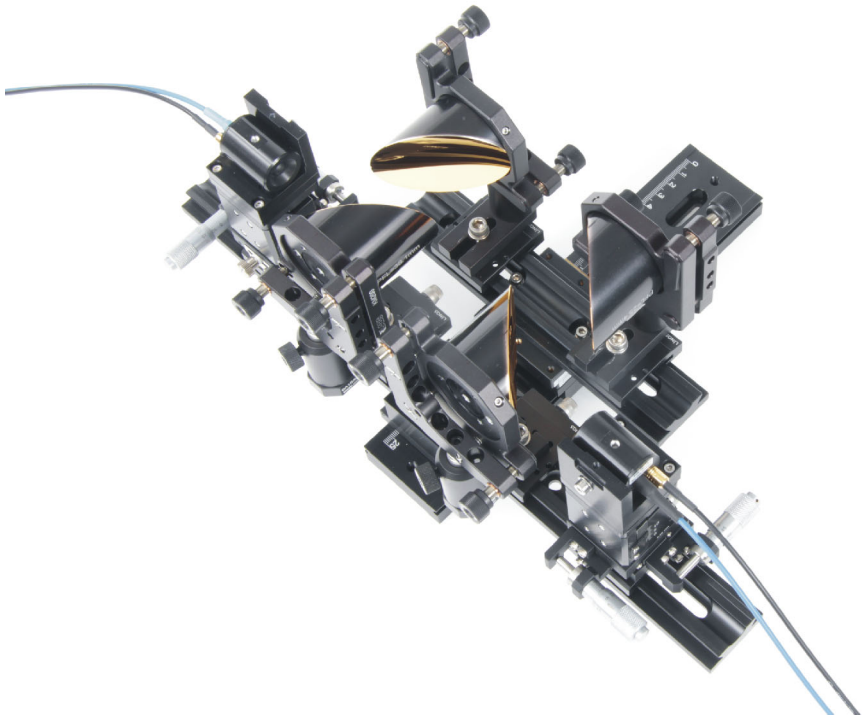


Figure 86 Terahertz optomechanics with four parabolic mirrors (TOPTICA article code THZ OPTICS / TR 4M)

The optomechanical assembly THZ OPTICS / TR 4M consists of two 3-axis stages (one for each photoconductive antenna), 4 off-axis parabolic mirrors to collimate and re-focus the terahertz beam, kinematic mirror mounts and a manual delay stage. The entire setup is mounted on rails (see Figure 86).

In addition to providing collimated beam paths between mirrors #1 and #2, as well as between mirrors #3 and #4, this assembly generates a beam focus halfway between mirrors #2 and #3. The focus size is approx. 1 mm at 1 THz.

NOTE !	For alignment of the four-mirror optomechanics, please refer to section 7.2 and follow the procedure accordingly. Before starting with the alignment, adjust the position and height of all of the mirrors, as well as the antenna mounts, as accurately as possible with the help of a ruler and a bubble level.

7.4 Reflection Head (THz Optics/Ref 4M)

The compact optical head THz Optics/Ref 4M enables terahertz measurements in reflection mode. The device comprises four parabolic mirrors, which generate a focus at the location of the sample and re-focus the reflected beam onto the receiver antenna. The sample should be positioned directly at the output of the measurement head. The incident angle of the terahertz beam on the sample is 8 degrees. The focus generated has a diameter of approx. 1 mm at 1 THz. The compact head features a connector for tubes with an outer diameter of 6 mm (FESTO - CK-M5-PK-4) for purging the setup with dry air or nitrogen, which helps to remove absorption due to ambient water vapor.

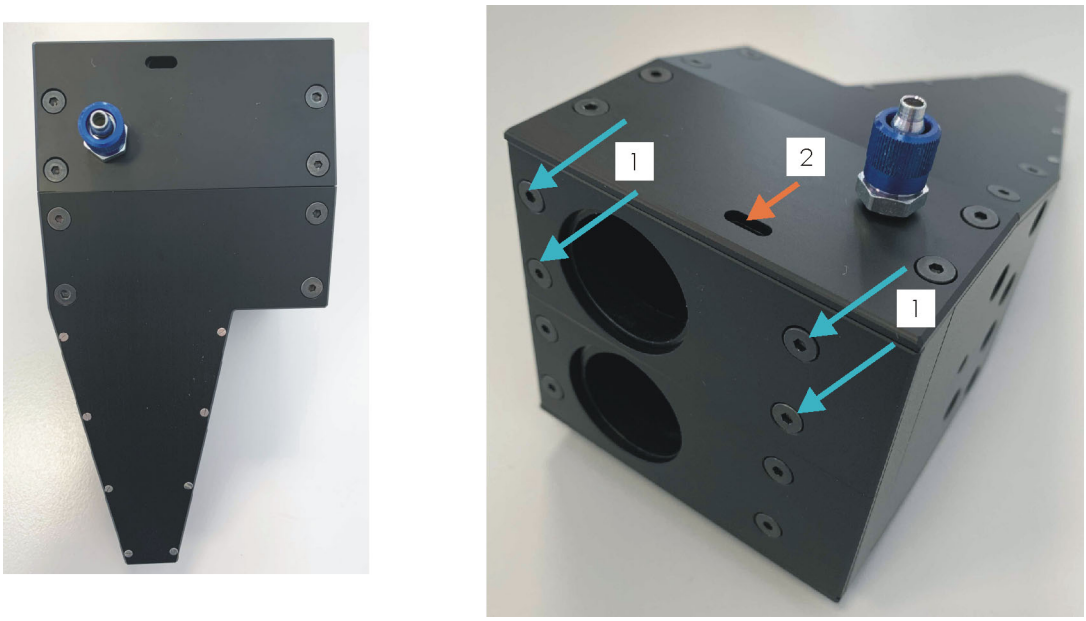


Figure 87 Left: Side view of the measurement head with connector for purging.
Right: Top view. The screws that adjust the antenna position are indicated with arrows.

7.4.1 Optical Setup

A schematic of the optical setup is shown in Figure 88. Four off-axis parabolic mirrors with a diameter of 1" are used for beam shaping. The two mirrors next to the antennas have a focal length of 2" and the two central mirrors have a focal length of 4". They are slightly tilted, resulting in an incident angle of the terahertz beam on the sample of 8 degrees.

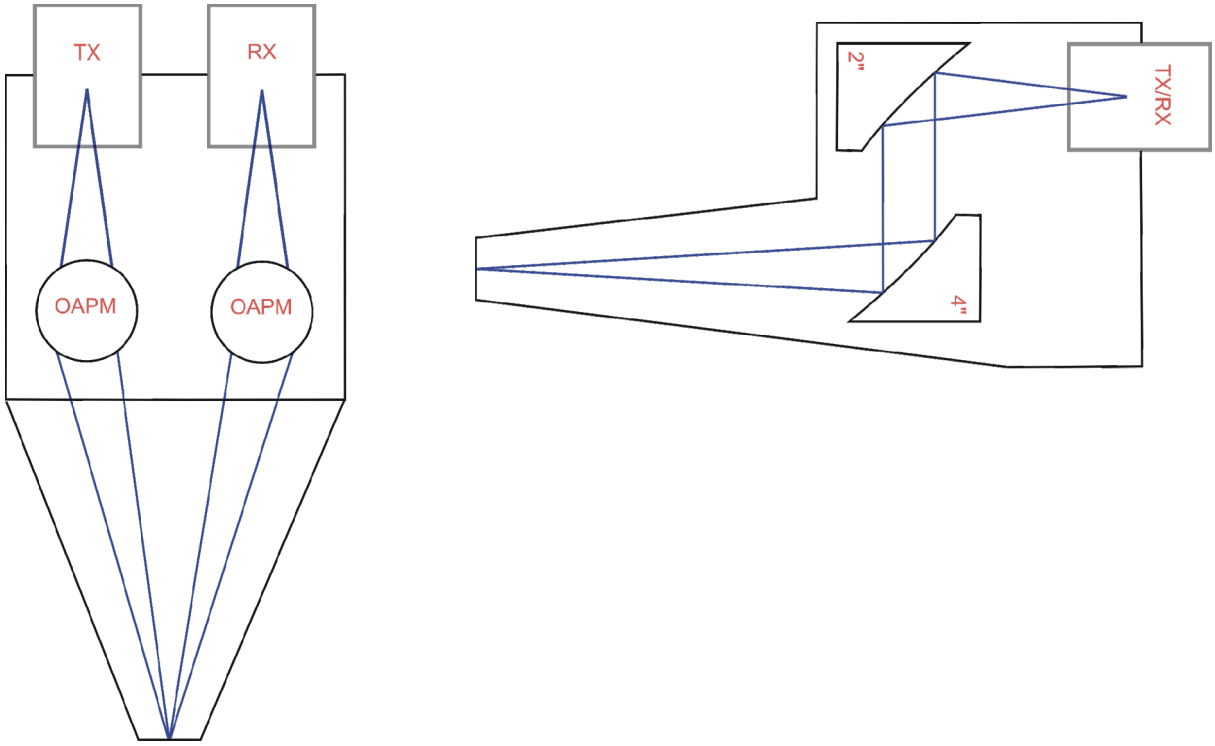


Figure 88 Schematic of the optical setup (left: front view, right: side view). The focal length of the off-axis parabolic mirrors (OAPMs) is indicated.

7.4.2 Installation of the Antennas and Optical Alignment

Place the antennas in the mounts. The rear-side of the emitter and receiver module should protrude approx. 12 mm from the reflection head and they have to be rotated outwards by ± 8 degrees, according to the arrangement of the terahertz beam path. The rotation is easily observable when looking from the module side onto a bright background, see Figure 89. Fix the antennas by tightening the locking screws. The screws are accessible through a small hole on each side, marked with arrow (2) in Figure 87.

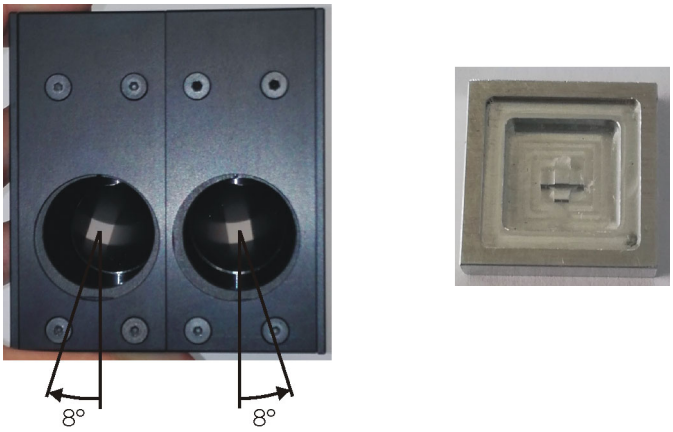


Figure 89 Left: Polarization rotation. Right: Alignment tool, patterned aluminum reflector.

The mirrors within the reflection measurement head are pre-aligned and fixed. Once the antennas are inserted, the focus position needs to be adjusted. This is done by carefully moving the antennas up and down in their mounts. In addition, the horizontal position of the antennas can be adjusted by releasing the four screws around each antenna on the top side of the measurement head (arrows (1) in Figure 87) and moving the antenna manually. Take care not to change the angular orientation of the antenna modules.

To facilitate the alignment in conjunction with a pulsed terahertz system, TOPTICA provides an alignment tool (patterned aluminum reflector, see Figure 89). It consists of 5 sections of the same size, but different heights. Placed underneath the measurement head, the reflector generates several pulse "echoes", visible at different times in the pulse trace. The first peak comes from the central section and has to be the largest one. The following 4 reflexes result from off-centered light along the x- and y-axis. They should exhibit similar amplitudes, but be small compared to the central pulse. The sixth, somewhat broader pulse originates from the large background area. Figure 90 shows an example of a pulse trace for a well-aligned measurement head.



Figure 90 Pulse trace acquired with reflection head and aluminum adjustment tool. The 6 pulses generated are marked by red arrows, their shape indicates a good central adjustment.

NOTE !	Depending on the type of antenna used, the focus can be so small that only the first echo is observed.

8 Appendix

8.1 Specifications of the TERAFLASH pro HP System

8.1.1 System Specifications

Specifications	**TERAFLASH pro HP**
Spectral range	0.1 - 6 THz (typ. 6.5 THz)
Spectral peak dynamic range	120 dB
Components	One femtosecond laser SM/PM fiber delivery 2 mechanical delay stages (stationary/moving) 2 (optionally 4) InGaAs photoconductive switches Electronics for data acquisition
Laser wavelength	1560 nm
Laser pulse width	typ. 50 .. 80 fs
External fiber length	2.5 m (other lengths u.r.)
Terahertz emitter	PCA-TD-250-TX-4: High-power InGaAs/InP photoconductive switch with strip-line antenna, 2.5 m fiber pigtail
Terahertz receiver	PCA-TD-250-RX-4: InGaAs/InP photoconductive switch, 2.5 m fiber pigtail
Antenna package	Cylindrical, Ø 25 mm, integrated Si lens and SM/PM fiber pigtail
Number of channels (antenna pairs)	2
Scan range	5 .. 3000 ps, software-controlled
Scan speed	166 traces/s (5 ps) 95 traces/s (20 ps) 60 traces/s (50 ps) 6 traces/s (200 ps) Intermediate settings possible
Average terahertz power	typ. 300 .. 400 µW
Useable terahertz path length	15 - 110 cm, continuously adjustable via software (stationary delay)
Frequency resolution @ max. scan range	0.3 GHz
Computer interface	Ethernet
Computer software	GUI or command set
Size (H x W x D)	180 x 450 x 560 mm³
System weight	20 kg
Operating voltage	110/220 V AC
Accessories	Transmission optomechanics, Reflection head, Imaging extension

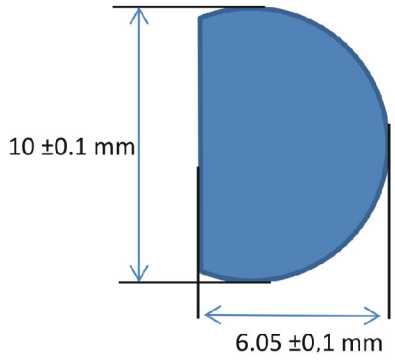
8.1.2 Environmental Specifications

Environmental requirements	Indoors, altitude < 2000 m
Temperature range (operating)	15 - 30 °C, relative humidity = 60 % (climatized lab recommended)
Temperature range (transport/storage)	0 - 40 °C, humidity non-condensing

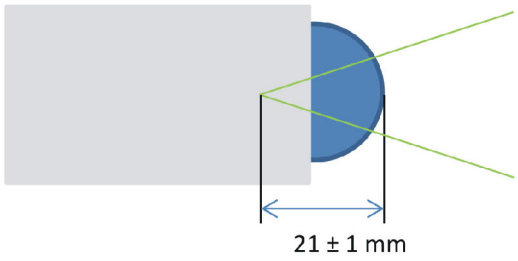
8.1.3 Lens Geometry and Properties of the Photoconductive Antennas

Hyper-Hemispherical lens geometry:

Height: 6.05 ± 0.1 mm
Diameter: 10 ± 0.1 mm



Back focus distance for pulsed Thz modules:



Beam Profile as measured with golay cell:

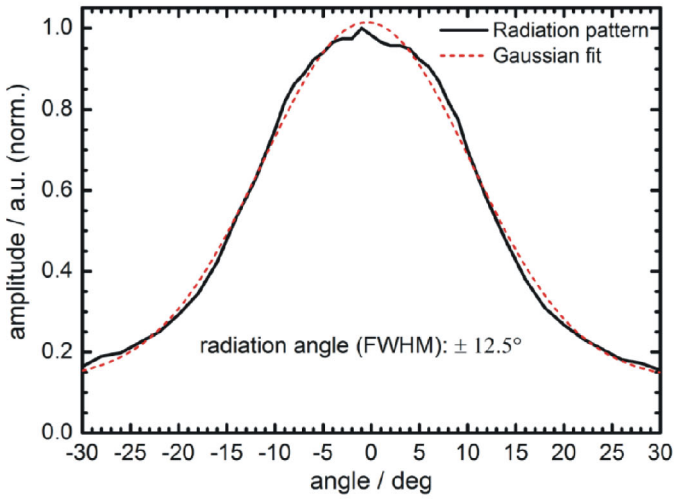


Figure 91 Lens specifications and beam divergence

8.2 Specifications of the Imaging Extensions

	Complete Imaging Extension	Flexible Imaging Extension
Concept	Moving sample, static TX/RX antennas	Static sample, moving reflection head
Linear stages	2 stages + motion controller included, positioning accuracy < 0.2 mm	
Terahertz optics	easy configuration in transmission and reflection, 4 mirrors: focal length: 2“ (collimating mirrors) and 4” (focusing mirrors)	reflection-mode only, 4 mirrors: focal length 2” (collimating mirrors) and 4” (focusing mirrors)
Mounts for THz antennas	Included for TX and RX	
Measurement speed	Up to 40 pixel/sec (@ 20 ps scan range)	
Positioning accuracy	0.1 mm	
Maximum field of view	15 x 15 cm	30 x 30 cm
Sample weight	Max. 2 kg	Any
Angle of incidence (reflection setup)	± 8 deg.	
Data analysis	Data post processing software offers multiple image representations, cross-section views and single-pixel analysis	

8.3 Cleaning

DANGER !	Before cleaning, the TERAFLASH pro system must be switched off and disconnected from mains.

Clean the outside of the TERAFLASH pro Main Unit with a cloth slightly moistened with a mild detergent solvent. Then dry with a clean cloth. **Do not clean the inside of the TERAFLASH pro Main Unit**.

CAUTION !	If any materials or liquids get into the TERAFLASH pro Main Unit during cleaning, contact TOPTICA. Never use strong solvents such as thinners, benzine, acidic- or alkaline solvents, spray-type cleaners or abrasive cleaners as they may destroy the surface.

8.4 Cleaning of the Photoconductive Antenna Modules

Clean the outside of the antenna modules with a cloth slightly moistened with a mild detergent solvent. Then dry with a clean cloth. Do not attempt to clean the inside of the antenna modules. The protective coating on the silicon lens must not get in touch with aggressive chemicals.

DANGER !	The antenna modules feature a silicon lens with a protective coating, which suppresses residual laser radiation. The antenna modules must not be used if the protective coating on the silicon lens is scratched or damaged.

8.5 Battery Recycling



NOTE !	**Batteries inside !** The return and disposal of batteries must be complied with and practised in accordance with the current applicable legal requirements and local regulations.

8.6 Data saved by Save Pulse/Save Spectrum

The data is stored as ASCII values as one or two comma-separated files (.csv), depending on whether the **Save Pulse**, **Save Spectrum** or **Save Pulse and Spectrum** option is selected.

The saved datasets match the displays of pulse trace and spectrum. Data for both Channel 1 and Channel 2 are saved into the same file.

The pulse file is called [chosen file name].pulse.csv and contains the terahertz pulse trace with the following columns:

- Time (ps)
- THz Signal Channel 1 (nA)
- THz Signal Channel 2 (nA)
- Reference Terahertz Signal Channel 1 (nA)
- Reference Terahertz Signal Channel 2 (nA)

The spectrum file is called [chosen file name].spectr.csv and contains power spectra and phase spectra with the following columns:

- Frequency Channel 1 (THz)
- Power Spectrum Channel 1 (dB)
- Frequency Channel 2 (THz)
- Power Spectrum Channel 2 (dB)
- Reference Spectrum Channel 1 (dB)
- Reference Spectrum Channel 2 (dB)
- Phase unwrapped Channel 1 (rad)
- Phase unwrapped Channel 2 (rad)
- Reference Phase unwrapped Channel 1 (rad)
- Reference Phase unwrapped Channel 2 (rad)

Both files contain a header line with column identifiers and channel numbers.

8.7 Firmware Update

The firmware of the TERAFLASH pro can be updated via **Menu System > Preferences > General Tab** (Firmware section).

CAUTION ! Only use this function when instructed to do so by TOPTICA Photonics. Only use files provided by TOPTICA.

Please perform the following steps to update the TERAFLASH pro firmware:

1. Make sure that the photoconductive antennas are electrically disconnected.

2. Click **Update Firmware** in the Firmware section of the **Menu System > Preferences > General Tab** (see Figure 48). This opens the Update Firmware window where you are prompted to drag the new firmware file into the input field. Please use the .zip archive as provided by TOPTICA.

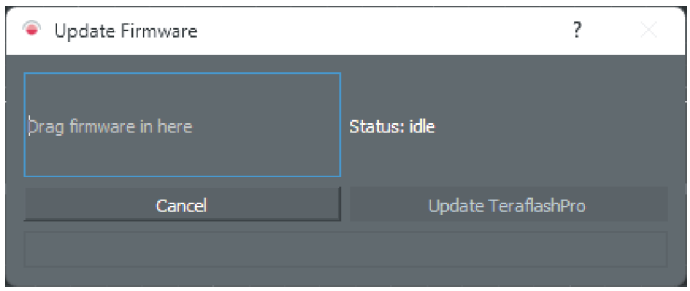


Figure 92 Update Firmware window

3. Click **Update TeraFlash Pro** to initiate the update process, or **Cancel** to abort.

4. A window opens (see Figure 93), reminding you not to turn off the TERAFLASH pro system during the firmware update.

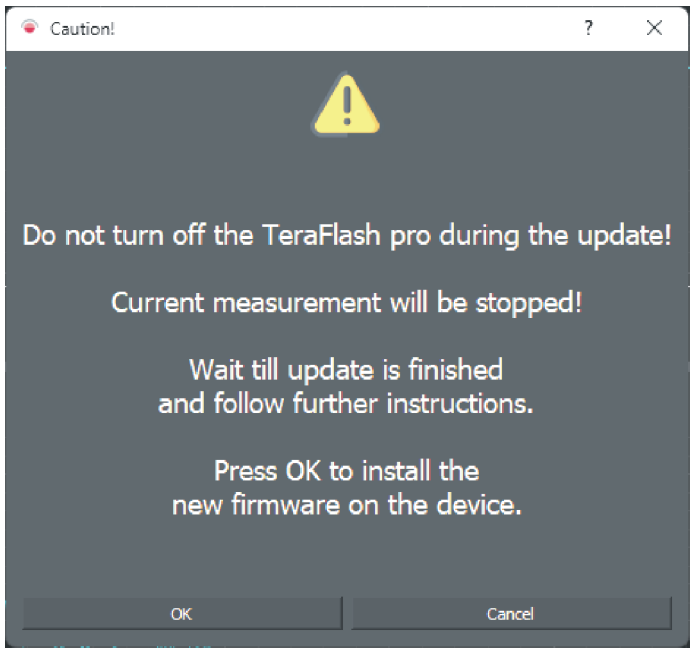


Figure 93 Caution window before performing firmware update

5. Confirm by clicking **OK** to start the firmware update. The status of the update is displayed by a progress bar.

6. After a successful firmware update, the window shown in Figure 94 opens.

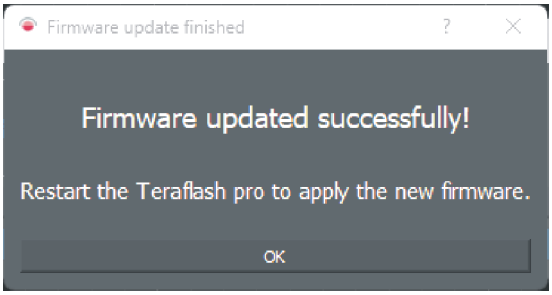


Figure 94 Firmware update finished window

Confirm by clicking **OK**. The connection to the TERAFLASH pro system is disconnected automatically.

7. Restart the TERAFLASH pro Main Unit at the red Power Switch at the front panel.

8. Connect the emitter and receiver photoconductive antennas electrically again.

8.8 System Self-Test (Diagnosis Option)

To facilitate troubleshooting, the TERAFLASH pro Control Software package includes a "diagnosis" routine that performs a self-test of the delay stages and the terahertz antennas, by measuring DC photocurrent values for both the emitter and the receiver.

1. Make sure that the emitter and the receiver under test are connected - both electrically and optically - to the same channel at the TERAFLASH pro Main Unit, and that the TERAFLASH pro Main Unit is able to communicate with the control PC.

2. Select **Menu System > Diagnosis** (see 5.2.11). A window opens (see Figure 95) stating that the laser and the THz emitters will be switched off.

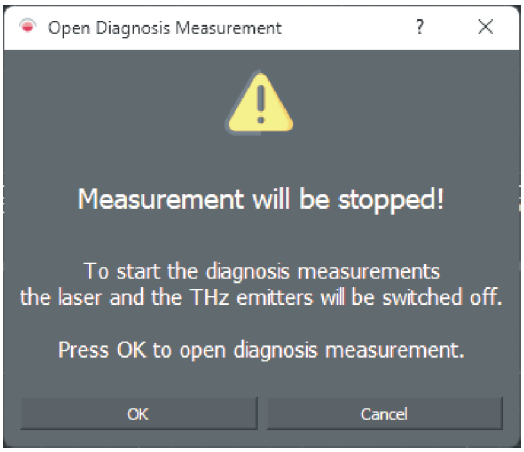


Figure 95 Open Diagnosis Measurement window

Confirm by clicking **OK**.

3. The main window of the Diagnosis routine opens (see Figure 96).

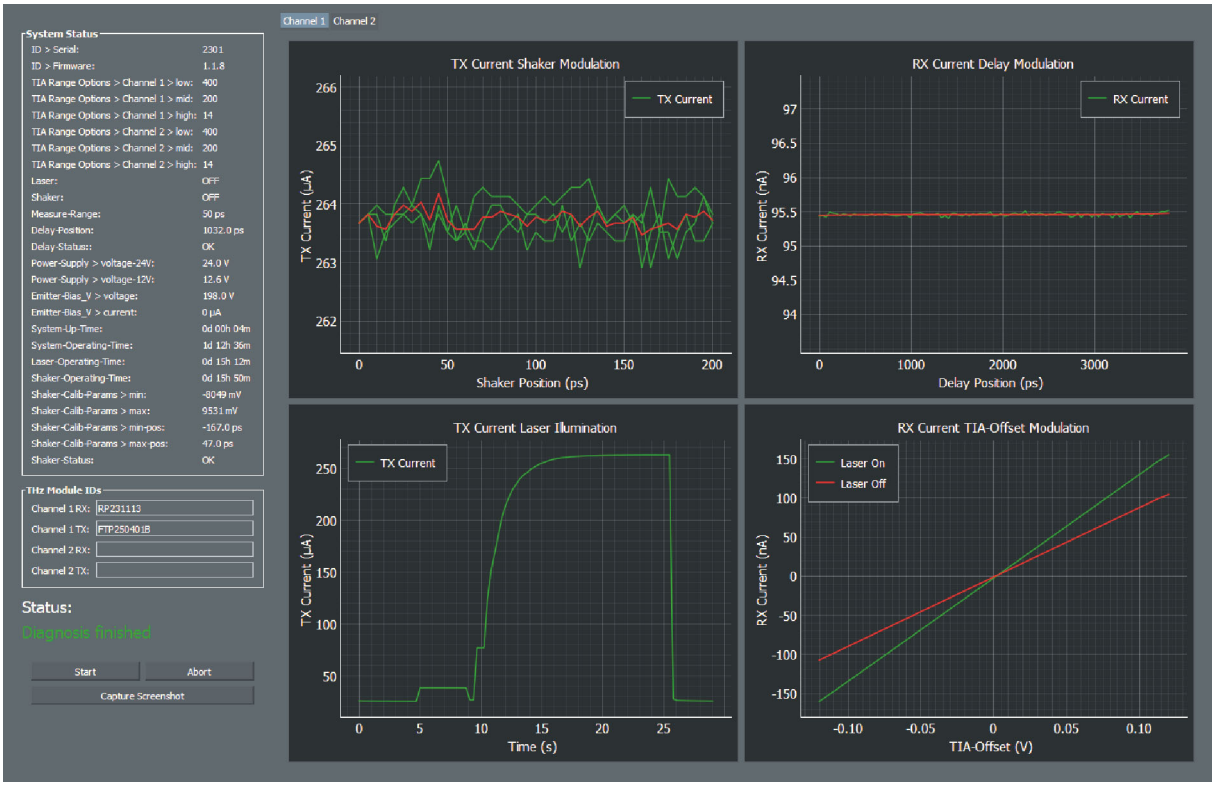


Figure 96 Main window of the Diagnosis routine

4. Fill in the serial numbers of the emitter and receiver modules that are connected to the TERAFLASH pro Main Unit. Please pay attention to use the correct serial numbers, as this will facilitate communication with TOPTICA service in case of a malfunctioning device.

Examples:
Channel 1 RX: RP231114
Channel 1 TX: FTP250402B

5. Click **Start** to initiate the self-test. The progress of the measurement is displayed by a flashing status message. If you need to terminate the measurement, click **Abort**.

NOTE !	The self-test takes approximately 3 minutes.

When completed, the four panels display the following information:
Top left: TX photocurrent vs. shaker position
Bottom left: TX photocurrent vs. time, laser is turned on and off again
Top right: RX photocurrent vs. delay line position
Bottom right: RX photocurrent vs. bias

6. Click **Capture Screenshot** to save the results. Enter a file name when prompted. The software saves two .png files, one for each channel. Please email these files to TOPTICA service for further analysis.

NOTE !	For the Diagnosis routine the optical setup is not important, as the self-test only records photocurrent data, but does not perform any actual terahertz measurements.

8.9 Troubleshooting

8.9.1 TERAFLASH pro Main Unit

Problem	Cause	Solution
No communication between PC and TERAFLASH pro Main Unit.	TERAFLASH pro Main Unit switched OFF.	Check if TERAFLASH pro Main Unit is switched ON.
	Ethernet cable missing, or USB-Ethernet adapter not properly installed.	Check ethernet connector cable. For installing the USB-Ethernet adapter, follow the instructions in section 5.1.1.
	Incorrect IP configuration.	Refer to 5.1.1 to configure the IP settings.
	Communication blocked by firewall.	Add the TERAFLASH pro Control Software to the list of allowed programs in the firewall (section 5.1.4).
Terahertz measurement does not start.	System not connected.	Check if "Connect" indicator in the GUI is green. If not, please refer to solutions to "No communication between PC and TERAFLASH pro Main Unit" above.
	Measurement not started.	Check if **Measurement** has been clicked in the System Control panel, Control tab (see section 5.2.1).
Strong background oscillations in pulse trace. Diagnosis file shows ripples in photocurrent measurements.	Possible cause: Ground loops, e.g. between the two photoconductive antennas, or between the antennas and the Mains ground of the TERAFLASH pro Main Unit.	If possible, fix the photoconductive antennas electrically insulated (e.g., by using plastic screws). Make sure the optical table is properly grounded, and uses the same ground as the TERAFLASH pro Main Unit.

Problem	Cause	Solution
Terahertz signal does not show a pulse, but only noise.	Laser switched OFF.	Check if laser is switched ON (laser status is "On", laser warning LEDs on TERAFLASH pro Main Unit lit up).
	Incorrect position of the stationary delay.	Change delay position. Refer to the Production and Quality Control Data Sheet for reference; modify the delay position according to the length of your terahertz beam path.
	Incorrect beam alignment.	Adjust terahertz beam (position of antennas, orientation of mirrors etc.).
	Incorrect channel selected.	Check to which channel the TX and RX antennas are connected. Check in the TERAFLASH pro Control Software, that the correct bias voltage channel is activated and this channel is selected for the measurement.
	TX or RX antenna not connected or defective.	Check the electrical and optical connections of the antennas. Use the Diagnosis option to check the status of the antennas (see section 8.8). If no photocurrent is detected, contact TOPTICA Photonics.
Terahertz signal does not show a pulse, Diagnosis option (see section 8.8) shows photocurrent levels as expected, but hardly any change when the two delays move (both upper graphs).	Optical connections of TX and RX antennas are interchanged.	Connect the optical fibers of the TX and RX antennas to the correct output ports.
Terahertz pulse shape is unusual, peaks are truncated.	**TIA** range too small.	Adjust **TIA** settings, see section 5.2.11. **(Menu System > Preferences > System tab).**
Display of terahertz signals is slow or appears frozen.	Not enough CPU power.	Make sure to use a suitable PC, with CPU of generation 14 or higher.
TERAFLASH pro Main Unit does not start when Mains Switch is turned ON.	Defective fuse.	Check the two fuses next to the Mains Connector of the TERAFLASH pro Main Unit (see Figure 13). Replace if necessary (5 A, slow blow).

8.9.2 Imaging Extension

8.9.2.1 Imaging Extension Control Software

Problem	Cause	Solution
Measurement can't be started.	Numeric settings on the PC do not use "." (point) as decimal separator.	Set numeric format of your PC to use "." (point) as decimal separator.
Stage controls and image size settings show unexpected behavior, e.g., delay stages move to unexpected position or can't be moved.	Numeric settings on the PC do not use "." (point) as decimal separator.	Set numeric format of your PC to use "." (point) as decimal separator.

8.9.2.2 THz-MAP Image Analysis Software

Problem	Cause	Solution
After loading a data file, no content is being displayed.	Numeric settings on the PC do not use "." (point) as decimal separator.	Set numeric format of your PC to use "." (point) as decimal separator.
2D Image displayed does not show features expected from measured pulse traces.	Time domain or spectral filtering might obscure the expected features.	Reset the filtering options - click Reset Filter, then move the cursors in the Pulse Trace Graph to the beginning and end of the trace (green cursor to 0 and black cursor to maximum value, so that no features of the pulse trace are excluded), then click **Apply Changes**.

8.10 Main Dimensions of the TERAFLASH pro Main Unit

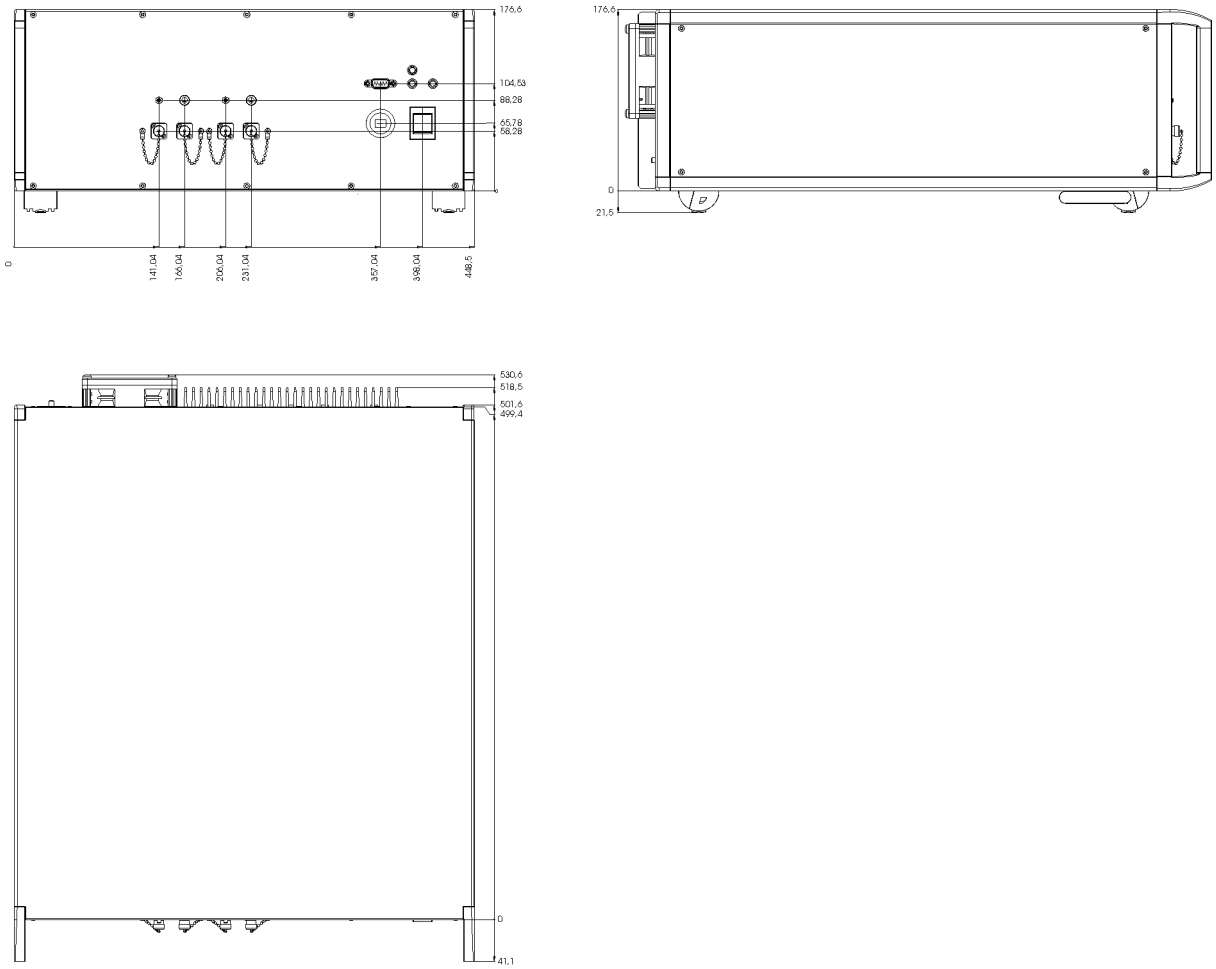


Figure 97 Main dimensions of the TERAFLASH pro Main Unit

8.11 Declaration of CE Conformity

Konformitätserklärung **Declaration of Conformity / Declaration de Conformitè**						TOPTICA	
QM-Formular:	**F-173**	Stand Formular:	**12.08.2025**	Version Formular:	**17**	Seite:	**1 von 1**

Wir / *We / Nous* **TOPTICA Photonics SE** CE

Anschrift / *Address / Adresse* **Lochhamer Schlag 19**
82166 Graefelfing
Germany

erklären in alleiniger Verantwortung, daß das Produkt / *declare under our sole responsibility, that the product / declarons sous notre seule responsibilité, que le produit*

Bezeichnung / *Name / Nom*	TeraFlash pro HP
Artikelnr. / Article No. / No. d'Article	TeraFlash pro HP
Beschreibung / Description / Description	Time-Domain Terahertz-Platform

mit den grundlegenden Anforderungen der Richtlinien / *fulfills the requirements of the standard and regulations of the directives / satisfait aux exigences des normes et directives*

2014/30/EU (Elektromagnetische Verträglichkeit), 2014/35/EU (Niederspannungsrichtlinie)
2011/65/EU and the amendment of Directive 2015/863 (RoHS-3 Richtlinie)

übereinstimmt und damit den Bestimmungen entspricht. / *and therefore corresponds to the regulations of the directive. / et, ainsi, correspond au règlements de la directive.*

Angewendete harmonisierte Normen / *Applied harmonized standards: / Normes harmonisées appliquées*:

DIN EN 61326-1	VDE 0843-20-1:2022-11 Elektrische Mess-, Steuer-, Regel- und Laborgeräte - EMV-Anforderungen - Teil 1: Allgemeine Anforderungen (IEC 61326-1:2020); Deutsche Fassung EN 61326-1:2021 *Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 1: General requirements (IEC 61326-1:2020); German version EN 61326-1:2021*
DIN EN 61010-1	Sicherheitsbestimmungen für elektrische Mess-, Steuer-, Regel- und Laborgeräte - Teil 1: Allgemeine Anforderungen (IEC 61010-1:2010 + COR:2011 + A1:2016, modifiziert + A1:2016/COR1:2019); Deutsche Fassung EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019 *Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements (IEC 61010-1:2010 + COR:2011 + A1:2016, modified + A1:2016/COR1:2019); German version EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019*
DIN EN 60825-1	Sicherheit von Lasereinrichtungen - Teil 1: Klassifizierung von Anlagen und Anforderungen (IEC 60825-1:2014); Deutsche Fassung EN 60825-1:2014 + A11:2021 *Safety of laser products - Part 1: Equipment classification and requirements (IEC 60825-1:2014); German version EN 60825-1:2014/A11_2021-02*
DIN EN IEC 63000	Technische Dokumentation zur Beurteilung von Elektro- und Elektronikgeräten hinsichtlich der Beschränkung gefährlicher Stoffe (IEC 63000:2016); Deutsche Fassung EN IEC 63000:2018 *Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances (IEC 63000:2016); German version EN IEC 63000:2018*

Durch nicht von uns autorisierte Änderungen am Produkt verliert diese Erklärung ihre Gültigkeit. / *Non authorized changes at the product result in the invalidity of this declaration. / Changements au produit, qui ne sont pas autorisées par nous, ont pour conséquence l'invalidité de cette declaration.*

Gräfelfing, den 26.02.2026

Dr. Wilhelm Kaenders
Vorstand

Ort und Datum der Ausstellung
Place and Date of Issue
Lieu et date d'établissement

Name und Unterschrift des Befugten
Name and Signature of authorized person
Nom et signature de la personne autorisée

8.12 License and Copyright Information associated with Third Party Software

This product incorporates certain third party software. The license and copyright information associated with this software is available in the folder Software License and Copyright Information on the supplied USB flash drive.
Please address your request to TOPTICA Photonics SE, Head of Development, Lochhamer Schlag 19, 82166 Graefelfing, Germany. This offer is valid during a 3-years-period beginning at the purchase date.

8.13 EU Legislation for Electrical and Electronic Equipment (EEE)

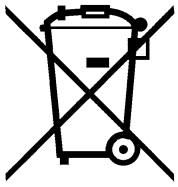
Companies selling electrical and electronic goods in the European Union must conform to the EU legislation for electrical and electronic equipment (EEE), which includes the Waste Electrical and Electronic Equipment Directive (WEEE). Assigned duties affect product design of the equipment, disposal of used appliances as well as organizational responsibilities, i.e. product registration.

There are different requirements for household WEEE and that which is sold business to business (B2B). All equipment TOPTICA Photonics SE handles is classed as B2B. TOPTICA is registered at the Competent Authority (Stiftung Elektro-Altgeräte Register EAR) under No. DE70442884.

At end-of life return your product back to TOPTICA. TOPTICA will dispose used equipment in such a manner as to meet all relevant local, country and EU requirements and guideline.

To return products please mark them clearly with "used device". Contact TOPTICA prior to shipping and send them to the following address:

TOPTICA Photonics SE
Lochhamer Schlag 19
D-82166 Graefelfing



The crossed out wheeled dust bin symbol indicates that products must be collected and disposed of separately from household waste.

TOPTICA reminds all end users of WEEE that you are responsible for deleting personal data on the waste equipment to be disposed of.

9 Guarantee and Service

As a first step toward obtaining technical support, please contact your local distributor or visit the support pages on our web site: https://www.toptica.com/support/.

In case you wish to return a product for diagnosis and/or repair, please contact us prior to sending it so we can issue a **Return Material Authorization** (RMA) number for you.

You can always contact TOPTICA service on the Internet by

- Email: service@toptica.com
- Web form: service.toptica.com

The service department at our headquarters in Munich can be reached by phone

- Tel: +49 89 85837150

For up-to-date phone numbers and addresses of our regional offices please refer to our web site: www.toptica.com/contact-us/